

# MATHEMATICS GRADE 10: SIMILARITY AND ENLARGEMENT

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.1 (12 Lessons)*

| SUB-STRAND OVERVIEW |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| Grade Level         | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Subject             | Mathematics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Strand              | Strand 2.0: Measurements and Geometry                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Sub-Strand          | Sub-Strand 2.1: Similarity and Enlargement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Total Duration      | 12 lessons × 40 minutes = 480 minutes total                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Sub-Strand Content  | <ul style="list-style-type: none"> <li>– Similarity: similar figures and their properties</li> <li>– Linear scale factor of similar figures</li> <li>– Area scale factor of similar figures</li> <li>– Volume scale factor of similar solids</li> <li>– Enlargement: centre of enlargement and scale factor</li> <li>– Constructing images under enlargement (positive, negative and fractional scale factors)</li> <li>– Finding the centre and scale factor of an enlargement from an object and its image</li> <li>– Application of similarity and enlargement in real life (scale drawings, maps, photographs, models)</li> </ul>                                                                                                          |
| Learning Outcomes   | <p>By the end of the sub-strand, the learner should be able to:</p> <ol style="list-style-type: none"> <li>identify similar figures in the environment,</li> <li>state the properties of similar figures,</li> <li>determine the linear scale factor of similar figures,</li> <li>determine the area scale factor of similar figures,</li> <li>determine the volume scale factor of similar solids,</li> <li>construct the image of an object under enlargement given the centre and the scale factor,</li> <li>determine the centre and the scale factor of an enlargement,</li> <li>apply similarity and enlargement in real-life situations,</li> <li>appreciate the use of similarity and enlargement in day-to-day activities.</li> </ol> |
| Core Competencies   | <ul style="list-style-type: none"> <li>– Communication and Collaboration: learners discuss with peers to identify similar figures, justify scale factors and share enlargement constructions in groups.</li> <li>– Critical Thinking and Problem Solving: learners reason about why a photo resizes without distortion, and use scale factors to solve length, area and volume problems.</li> <li>– Creativity and Imagination: learners design scale drawings and enlarged or reduced images of familiar objects, such as school logos and passport photos.</li> <li>– Digital Literacy: learners use digital devices or geometry software to enlarge and reduce images and to check scale factors where available.</li> </ul>                |

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| <b>Core Values</b>                                | <ul style="list-style-type: none"> <li>– Unity: learners work together in groups to construct and compare enlargements.</li> <li>– Responsibility: learners take care of geometrical instruments and use them accurately.</li> <li>– Integrity: learners report measurements and scale factors honestly, even when they differ from expected values.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>Science &amp; Engineering Practices</b>        | <ul style="list-style-type: none"> <li>– Asking questions and defining problems: learners ask why a passport photo can be much smaller than the person without being distorted.</li> <li>– Developing and using models: learners build and revise a cumulative model of similarity and enlargement (a scale factor model linking lengths, areas and volumes).</li> <li>– Analyzing and interpreting data: learners compare measured ratios of corresponding sides and angles in original and resized photographs.</li> <li>– Using mathematics and computational thinking: learners calculate scale factors and apply them to length, area and volume.</li> <li>– Constructing explanations: learners use evidence to explain why equal ratios and equal angles preserve shape.</li> </ul> |
| <b>Pertinent &amp; Contemporary Issues (PCIs)</b> | <ul style="list-style-type: none"> <li>– Life Skills (self-esteem and critical thinking): learners participate confidently in discussions and defend their reasoning.</li> <li>– Financial Literacy: learners use scale and area factors to estimate the cost of printing photographs and painting enlarged signs.</li> <li>– Citizenship (national identity): learners use maps of Kenya and photographs for IDs and passports, linking enlargement to official documents.</li> </ul>                                                                                                                                                                                                                                                                                                     |
| <b>Career Connections</b>                         | <ul style="list-style-type: none"> <li>– Architect and Civil Engineer: use scale drawings and models to plan buildings and roads in Nairobi and Mombasa.</li> <li>– Surveyor and Cartographer: use scale factors to produce maps, such as the Survey of Kenya maps.</li> <li>– Graphic Designer and Photographer: resize images for passports, posters and billboards without distortion.</li> <li>– Tailor and Fashion Designer: scale patterns for different sizes of kitenge and school uniform garments.</li> <li>– Sign Writer: enlarges logos and painted shop signs in market centres.</li> </ul>                                                                                                                                                                                   |
| <b>Focus for Lessons</b>                          | This unit uses the passport photograph puzzle as an anchor for investigating why figures can change size but keep the same shape. Learners measure, predict, construct and explain how corresponding sides and angles, scale factors and the centre of enlargement keep shapes undistorted. They extend this to areas and volumes and apply it in Kenyan contexts such as maps, scale models and posters. Through the lessons they revise a cumulative model and a Driving Question Board.                                                                                                                                                                                                                                                                                                 |
| <b>Driving Question / Key Inquiry Questions</b>   | <p>DRIVING QUESTION: How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?</p> <p>KICD KEY INQUIRY QUESTIONS:</p> <ol style="list-style-type: none"> <li>1. Why do we need to enlarge or reduce objects?</li> <li>2. How do we use similarity and enlargement in day-to-day activities?</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Anchoring Phenomenon</b>                       | A Kenyan student has her full-size photograph taken at a studio in Kisumu for her national ID and passport application. The photographer prints a passport photo of only 3.5 cm × 4.5 cm. Her whole face and shoulders fit into this tiny box, yet she still looks exactly like herself: her face is not stretched, squashed or skewed. The same picture can also be enlarged to a large poster for her school's prize-giving day, and it still looks like her. Yet if the photographer forces the picture into the box by changing only its width, the face looks stretched and distorted. This is puzzling because the size changes by a huge amount, yet the shape is retained. What must stay the same, and what is allowed to change?                                                 |
| <b>Storyline Thread</b>                           | <p>Lesson 1: Anchoring phenomenon. Learners predict how a full-size photo fits a 3.5 cm × 4.5 cm box, examine undistorted and distorted passport photos, and build the first Driving Question Board.</p> <p>Lesson 2: Identifying similar figures in the environment (photos, maps, models, school logos) and sorting similar from non-similar</p>                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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|                             | <p>figures.</p> <p>Lesson 3: Properties of similar figures: measuring corresponding angles and ratios of corresponding sides on original and resized photos.</p> <p>Lesson 4: Linear scale factor: calculating scale factors for enlargements and reductions, such as a full-size photo to a passport photo, and finding missing lengths.</p> <p>Lesson 5: Solving problems involving similar triangles and other figures, such as heights of trees or school buildings using shadows.</p> <p>Lesson 6: Area scale factor: investigating how the area changes when lengths are enlarged by <math>k</math>, using grid paper and cut-outs, and relating it to <math>k^2</math>.</p> <p>Lesson 7: Volume scale factor: comparing small and large similar solids such as cartons or containers, and relating volume to <math>k^3</math>.</p> <p>Lesson 8: Introduction to enlargement: centre of enlargement and scale factor, and constructing images with positive scale factors using rays from the centre.</p> <p>Lesson 9: Enlargement with fractional and negative scale factors, so learners see reduction and inverted images, as in a projector.</p> <p>Lesson 10: Finding the centre and scale factor of an enlargement from an object and its image by joining corresponding points.</p> <p>Lesson 11: Applications: scale drawings, maps of Kenya, scale models and costing painted signs, using area and volume factors.</p> <p>Lesson 12: Consolidation: learners complete the cumulative model, return to the passport photo phenomenon, explain it using ratios, angles and scale factors, and resolve the Driving Question Board.</p> |
| <b>Supporting Phenomena</b> | <ul style="list-style-type: none"> <li>– A map of Kenya in an atlas and a large wall map of Kenya show Lake Victoria and Mombasa with the same shape, though the sizes differ greatly.</li> <li>– A matchbox-size model of a matatu looks just like the real matatu, but its paint and material needs are very different.</li> <li>– A tailor in Gikomba market cuts small and large versions of the same dress pattern, and the large version needs much more cloth than just twice as much when the length is doubled.</li> <li>– A projector in a classroom enlarges a small slide into a large image on the wall, keeping the same shape but flipped when the lens is reversed.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

**Production & Layout:** Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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## LESSON 1 (40 minutes): Wanjiru's Passport Photo: What Stays the Same When Size Changes?

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Purpose</b>              | To launch the anchoring phenomenon: Wanjiru Atieno's photo can shrink to a 3.5 cm × 4.5 cm passport print or grow to a 70 cm × 90 cm poster and still look like her, yet a 4.5 cm × 4.5 cm print looks distorted. Learners record their first ideas about what must stay the same and what is allowed to change.                                                                                              |
| <b>Knowledge</b>            | Learners notice that the original (14 cm × 18 cm), passport (3.5 cm × 4.5 cm) and poster (70 cm × 90 cm) prints differ greatly in size but look the same in shape, while the distorted 4.5 cm × 4.5 cm print has a different shape. They name the sizes and say that some feature of the shape must be kept (the idea of similar figures, ~, is developed in later lessons). No calculations beyond noticing. |
| <b>Skills</b>               | Making predictions; observing and comparing images; asking questions; developing initial ideas and a first model; communicating reasoning in pairs and groups.                                                                                                                                                                                                                                                |
| <b>Attitudes</b>            | Curiosity about how everyday objects such as photographs are resized; willingness to share tentative ideas without fear of being wrong; respect for peers' ideas; collaboration and unity in group work.                                                                                                                                                                                                      |
| <b>Key Inquiry Question</b> | 1. Why do we need to enlarge or reduce objects? 2. How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                                         |
| <b>Purpose in Storyline</b> | Lesson 1 is the launch lesson and builds on nothing earlier. It creates the puzzle (size changes a lot, shape stays the same) and opens the Driving Question Board. Lesson 2 will use the same four prints as sorting cards to spot similar figures by shape. Later lessons develop the properties of similar figures, scale factor, area and volume factors, and enlargement with a centre.                  |
| <b>Safety Notes</b>         | Use rulers and scissors carefully; cut paper only at desks, pass scissors handle-first, and keep walkways clear when learners move to the Driving Question Board. No electricity or internet is needed.                                                                                                                                                                                                       |

### B. LESSON OVERVIEW

Mrs Achieng opens by telling the story of Wanjiru Atieno, a Grade 10 learner at Nyalenda Secondary School in Kisumu, who visits Mr Otieno at Lakeview Photo Studio for her national ID and passport application. Learners first predict, individually, how a full-size photo (14 cm × 18 cm) can fit a 3.5 cm × 4.5 cm box without her face looking wrong. Groups then examine four prepared prints: the original 14 × 18, the passport 3.5 × 4.5, the prize-giving poster 70 × 90 and the distorted 4.5 × 4.5 print, made by keeping the height at 4.5 and stretching only the width. They measure each side with a ruler, record the sizes, and notice that the sizes change by a huge amount while the shape of the face stays the same, except for the distorted print. In the Explain phase learners give first explanations of what must stay the same and what may change. The class builds the first Driving Question Board with the driving question 'How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?' and learners post their own questions. Finally, learners sketch an initial model of their current thinking, which will be revised through Lesson 12.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience                                               | Teacher Moves                                                     | Sensemaking Strategy                                         | Formative Assessment Strategy                                 |
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| <b>Predict Phase</b> | Learners listen to the story of Wanjiru Atieno at Lakeview Photo | Draw a 3.5 cm × 4.5 cm rectangle and a 14 cm × 18 cm rectangle on | Predict-and-record (public prediction). Writing a prediction | Check that at least 80% of learners have written a prediction |

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| <p> <b>VIDEO:</b><br/> <a href="#">Scale drawings</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Corresponding Parts of Similar Figures (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p>                                                       | <p>Studio in Kisumu. Mr Otieno must print her 14 cm × 18 cm photo into a 3.5 cm × 4.5 cm passport box. Individually, in their notebooks, learners write a prediction answering: 'How can her whole face and shoulders fit into this tiny box and still look like her? What will go wrong if the photographer changes only the width?' They sketch the box and the picture inside it, then share with a partner for 2 minutes.</p>                                                                                                                                                                                                                                                                                                                                                                                                 | <p>the board (width first, then height). Say: 'Wanjiru has her photo taken at a studio in Kisumu. Mr Otieno prints a tiny passport photo, only 3.5 cm by 4.5 cm. Her whole face still fits. How is this possible without her looking different?' Pose the prediction prompt and give 3 minutes of silent writing. Then say: 'Turn to your partner and compare predictions.' Circulate and listen without correcting. Cold-call 4 learners: 'What did you predict would happen to her face?' After each answer, WAIT TIME of 10-15 seconds before calling the next learner so that others can think. Record 3-4 predictions on the board without saying which is right. Ask: 'What word would you use for what changes? What word for what stays?'</p>                                                                                                            | <p>before any instruction reveals learners' preconceptions, such as 'just squeeze it in' or 'cut off the edges', and commits them to an idea they will test and revise through the unit. Comparing with a partner turns private guesses into ideas that can be questioned.</p>                                    | <p>and a sketch. Look for observable indicators: (a) predictions that mention 'smaller' or 'reduce'; (b) predictions that mention 'stretched', 'squashed' or 'same shape'; (c) learners who only say 'crop' or 'cut'. Note the ideas heard in partner talk on a class list to revisit in the Explain phase and in Lesson 2.</p>                                                                                                                                      |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Dilating shapes: shrinking by 1/2</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Corresponding Parts of Similar Figures Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>In groups of 4, learners receive a set of four prepared prints of Wanjiru: the original (14 cm × 18 cm), the passport photo (3.5 cm × 4.5 cm), the prize-giving poster (70 cm × 90 cm; shown as a scaled sketch or a taped outline on the wall or floor) and the distorted trial print (4.5 cm × 4.5 cm). They use a ruler to measure width and height of each print that fits on a desk, record the sizes in a table (width first, then height), and write a note on how Wanjiru's face looks in each: 'looks like her' or 'looks stretched/squashed'. They place the passport print over the original and the distorted print over the passport and describe what they notice. They also pick one feature, such as the distance between the eyes or the face shape, and discuss whether it looks the same in each print.</p> | <p>Give each group a ruler, the four prints and an observation table with the columns Print, Width (cm), Height (cm), Looks like Wanjiru? Say: 'Measure carefully and write width first, then height. Do not calculate anything yet; just notice.' Explain the distorted print: 'Mr Otieno kept the height at 4.5 cm and stretched only the width.' Circulate and ask each group: 'Which prints look like the same picture, just bigger or smaller? Which one looks different? What do you see on her face that tells you?' WAIT TIME of 10-15 seconds after each question. Prompt without giving the rule: 'Compare the shape of the box for the passport and the original. What do you notice?' After 12 minutes, cold-call 3 groups to report one observation each. Record on the board: 14 × 18, 3.5 × 4.5, 70 × 90, 4.5 × 4.5. Ask: 'The poster is much</p> | <p>Structured observation with a data table. Recording measured sizes next to a judgement of how the face looks lets learners separate what changes (size) from what appears to stay (shape), which creates the puzzle. Overlaying the prints makes the differences visible before any formula is introduced.</p> | <p>Observable indicators: learners measure with the ruler starting at 0 and record widths and heights in the correct order; they correctly label the 4.5 × 4.5 print as looking stretched or squashed; they say the passport and poster look like the original even though the sizes are different. Note groups that focus only on size rather than shape, and ask them a prompt such as 'Is the picture shape the same?'. No numerical ratios are expected yet.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | bigger than the passport photo, yet both look like Wanjiru. What does that tell you?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Identifying scale factors</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Scale drawings</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to their predictions and discuss in their groups: 'What must stay the same, and what is allowed to change when the photo is resized?' Each group writes one 'what changes' statement and one 'what stays the same' statement on a strip of paper, using words and a sketch. They use the sentence frame: 'The size changes because _____, but the shape stays the same because _____.' They also say why the <math>4.5 \times 4.5</math> print looks distorted. Groups share in a whole-class talk and compare with their earlier predictions, marking which predictions they now want to test.</p> | <p>Say: 'Look back at your prediction. Talk in your group for 5 minutes. What must stay the same and what is allowed to change?' Hand out the sentence frames. Circulate and record on a clipboard which ideas appear: 'sides get smaller by the same amount', 'looks the same', 'the box shape', 'angles', 'ratio'. Do not confirm or reject. Ask: 'Why do you think the <math>4.5</math> by <math>4.5</math> print looks stretched? What was changed and what was not?' Then say: 'I will cold-call 5 groups. Listen for ideas that are the same as yours and ideas that are different.' After each group shares, WAIT TIME of 10-15 seconds, then ask another learner: 'Can you add to or challenge that idea?' Post strips on the board in two columns: 'Changes' and 'Stays the same'. Tell learners: 'We do not yet have the full explanation. That is what this unit will help us find. Some of these ideas will be tested in later lessons.'</p> | <p>Claim-and-evidence talk with a sentence frame. Asking learners to separate 'what changes' from 'what stays' focuses their attention on the key mathematical contrast, while the frame requires a reason, not just an opinion. Posting ideas without judgement keeps them open for testing in later lessons.</p> | <p>Listen for: (a) 'size/length changes' appearing in the 'changes' column; (b) 'shape/look/box proportions' appearing in the 'stays' column; (c) an explanation of the distorted print such as 'only the width was changed'. Count how many groups wrote both statements. Record misconceptions, for example 'all sides must grow by adding the same number of centimetres', to address in Lesson 3 and Lesson 4. Do not correct them now.</p>                           |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Scale drawings</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Scale Factor of Similar Polygons (Flexbook)</a><br/> Source: CK-12</p>                                                                                                                              | <p>Learners view the driving question written by Mrs Achieng on a large chart: 'How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?' Individually, each learner writes at least two questions on sticky notes or paper slips, such as 'Why does changing only the width spoil the picture?', 'Is there a number that tells how much smaller the passport photo is?', 'Do maps and posters work the same way?' They also connect to the key inquiry questions: 'Why do we need to enlarge or reduce</p>                                                         | <p>Place the driving question at the top of the board. Say: 'Write at least two questions you have about Wanjiru's photo. Think about what you wonder, not what you already know.' Give 4 minutes of silent writing. Invite learners to post: 'Who will bring up one question to place on the board?' Cold-call 6 learners to read and place their questions; after each, WAIT TIME of 10-15 seconds, then ask: 'Who has a question that is similar or connected to this one?' Group the slips with the class into 3-4 categories and label</p>                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Driving Question Board. Learner-generated questions give ownership of the learning path and make the unit's purpose visible. Sorting questions into themes shows connections among ideas, and the board becomes a living record of growing understanding to which learners return in each lesson.</p>           | <p>Observable indicators: each learner posts at least 2 questions; questions refer to the phenomenon (the photo, the box, the stretched print) rather than being off-topic; some questions go beyond description to ask why or how (for example, 'what number tells the change?'). Tally how many questions fall into each theme. Learners whose questions are only factual (such as 'What is the size?') are encouraged to rewrite them as 'why' or 'how' questions.</p> |

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| <a href="#">Search ARES for similar readings</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>objects?' and 'How do we use similarity and enlargement in day-to-day activities?' Learners post their questions on the DQB and, in groups, sort them into themes (for example, size, shape, measurement, real-life uses).</p>                                                                                                                                                                                                                                                                                                                                                                     | <p>them. Add the two KICD key inquiry questions to the board. Say: 'Over the next lessons we will come back to this board, cross off questions we can answer, and add new ones.' Take a photo or copy of the board for the record, and keep it displayed in the classroom.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Dilating shapes: shrinking by 1/2</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Scale Factor of Similar Polygons Practice</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p> | <p>In groups, learners draw an initial model on a sheet of paper: the original photo (14 cm × 18 cm), the passport photo (3.5 cm × 4.5 cm) and the poster (70 cm × 90 cm) drawn as rectangles with labelled sizes, with Wanjiru's face outlined in each, and the distorted print (4.5 cm × 4.5 cm) beside them. They use arrows and words to show 'what changes' and 'what stays the same' and add a box labelled 'We still need to find out ...'. Groups display their model on the wall for a short gallery walk and each learner adds one sticky comment or question to another group's model.</p> | <p>Say: 'Draw a model that explains your current best thinking about why the passport photo and the poster still look like Wanjiru, but the 4.5 by 4.5 print does not. It is fine if you are not sure. Show what you know and what you still need to find out.' Provide a rubric of three things to include: sizes labelled, what changes, what stays the same. Circulate and ask: 'What does this arrow show? What part of your model are you least sure of?' WAIT TIME of 10-15 seconds after each question. After 10 minutes run a 5-minute gallery walk and then cold-call 3 learners: 'What is one idea you saw that you want to test?' Close by saying: 'Keep your model safe. In Lesson 2 we will use these same prints to sort and identify figures that have the same shape. We will revise this model several times before Lesson 12.' Collect the models and store them.</p> | <p>Initial model building and revision cycle. Drawing a first model forces learners to organise their ideas into a representation that can be tested, shows gaps in their reasoning, and creates a baseline against which later revisions in Lessons 2-12 can be compared.</p> | <p>Look for models that: label <math>14 \times 18</math>, <math>3.5 \times 4.5</math>, <math>70 \times 90</math> and <math>4.5 \times 4.5</math> correctly; show size as changing and some feature of shape as staying the same; and name at least one open question. Record which groups include the idea of 'same shape' versus 'only the size' as a baseline for Lesson 2. Accept incomplete models; they are a starting point rather than a final answer.</p> |

## D. TEACHER REFLECTION

Which learners connected 'same shape' with the passport and poster prints and which focused only on size? What preconceptions (such as 'add the same number of cm to every side', 'just cut it') appeared, and how will I address them in Lessons 2-4 without giving the rule now? Was wait time of 10-15 seconds kept after each question, and did a wide range of learners, not just the usual few, contribute through cold-calls? Did every learner post at least two questions on the DQB? Which questions on the DQB can be linked to which later lessons, and which key inquiry question, 'Why do we need to enlarge or reduce objects?' or 'How do we use similarity and enlargement in day-to-day activities?', needs more exploration? How well did Kenya-based context (Wanjiru at Lakeview Photo Studio in Kisumu, Nyalenda Secondary School) motivate learners?

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                   |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What did I observe?                                          | We observed that Wanjiru's original photo (14 cm × 18 cm), the passport photo (3.5 cm × 4.5 cm) and the poster (70 cm × 90 cm) are very different in size but her face looks the same in each, while the 4.5 cm × 4.5 cm print, made by changing only the width, looks stretched. |
| What did I learn?                                            | We learned that a picture can change size by a large amount and still keep its shape, and that changing only one direction (the width) spoils the shape. We do not yet know exactly what must stay the same.                                                                      |
| How does this explain the phenomenon?                        | Our first idea is that the size can change but something about the shape, such as how the sides compare, must stay the same in every direction. We recorded our questions on the Driving Question Board and drew an initial model to test in later lessons.                       |

## LESSON 2 (40 minutes (1 lesson)): Spotting Similar Figures Around Us: Sorting Photos, Maps, Models and Logos

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners use visual judgement of 'same shape' to identify and sort similar and non-similar figures in photos, maps, models and school logos, building the idea that size may change while shape stays the same. This advances the driving question by giving learners a working sorting rule to test in later lessons.                                                                       |
| <b>Knowledge</b>            | Figures are similar (~) when one looks like an exact enlargement or reduction of the other: same shape, possibly different size. Wanjiru's passport print (3.5 cm × 4.5 cm) and poster (70 cm × 90 cm) look like the original (14 cm × 18 cm), but the 4.5 cm × 4.5 cm distorted print does not. Numeric ratios and measurement are NOT used yet.                                            |
| <b>Skills</b>               | Sorting and classifying figures; making and justifying visual judgements using words such as 'stretched', 'squashed' and 'skewed'; identifying corresponding parts (width, height, eyes, chin); communicating reasoning in groups (NGSS SEP: Analysing and Interpreting Data; Constructing Explanations).                                                                                    |
| <b>Attitudes</b>            | Curiosity about shapes in the environment; unity and cooperation in group sorting; respect for peers' reasoning; appreciation of how similarity is used in day-to-day activities such as photography, maps, models and signs.                                                                                                                                                                |
| <b>Key Inquiry Question</b> | 1. Why do we need to enlarge or reduce objects? 2. How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                        |
| <b>Purpose in Storyline</b> | Lesson 1 launched the puzzle: the passport photo and poster look like the original but the distorted print does not. Lesson 2 turns that feeling into a sorting activity with many figures and leaves learners with a question: how can we be sure, and not just guess by eye? This prepares Lesson 3, where properties of similar figures (angles and side ratios) are tested by measuring. |
| <b>Safety Notes</b>         | Scissors: cut away from the body, pass handles-first, and keep them on the desk when not cutting. Keep walkways clear during the gallery walk. No electricity or internet is needed.                                                                                                                                                                                                         |

### B. LESSON OVERVIEW

Mrs Achieng returns to Wanjiru Atieno's photographs from Lakeview Photo Studio in Kisumu. Groups receive four sorting cards (the original 14 cm × 18 cm print, the 3.5 cm × 4.5 cm passport print, the 70 cm × 90 cm poster and the 4.5 cm × 4.5 cm distorted print), plus cards of other figures: a Kenya map and a smaller copy, a toy matatu model and a picture of a real matatu, school logo copies in two sizes (one squashed), and a pair of triangles. Learners predict which look like 'the same shape', sort them, justify with everyday words, and build a class sorting rule. They add questions to the Driving Question Board and revise their initial model: size may change, but shape must be kept. No numeric ratios are used; measuring and ratios come in Lesson 3.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                       | Learner Experience                                                                                                                | Teacher Moves                                                                                                                       | Sensemaking Strategy                                                                                                                   | Formative Assessment Strategy                                                                                                             |
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| <b>Predict Phase</b><br> <a href="#">VIDEO: Dilations and properties</a> | Learners recall Lesson 1: Wanjiru's passport photo and poster still look like her, but the 4.5 cm × 4.5 cm print looks stretched. | Display the pictures and say: 'Yesterday we saw that Wanjiru's passport photo still looks like her. Today, before I teach anything, | Predict-Observe-Explain (Predict step) with a Think-Pair-Share: committing to a prediction makes learners' prior ideas visible so they | Look for: (1) learners writing a prediction AND a reason, not just a choice; (2) use of informal shape language ('stretched', 'squashed', |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>HTML:</b><br/> <a href="#">Corresponding Parts of Similar Figures (Flexbook)</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                     | <p>Each learner silently looks at 6 mixed pictures on the board (a small and large school logo, a squashed logo, a map of Kenya and a smaller copy, a matatu model and a real matatu). In their notebooks they write a prediction: 'Which pairs have the same shape even though the sizes differ?' and one reason. They compare with a partner.</p>                                                                                                                                                                                                                                                                                                                    | <p>predict: which of these pairs have the SAME SHAPE, even if one is bigger?' Give WAIT TIME of 15 seconds in silence before writing. Walk the room and note words used. Cold-call 4 learners: 'Which pair did you choose, and what made it look right or wrong?' Record their words (e.g. 'stretched', 'just smaller') on the board without confirming correctness. Say: 'We will test these ideas with evidence today.'</p>                                                                                                                                                                        | <p>can be tested and revised against evidence later in the lesson.</p>                                                                                                                                                                         | <p>'just smaller'); (3) whether learners treat 'same size' as required for 'same shape'. Tally on a class list how many say size must match. Misconception to note: 'different size means different shape'.</p>                                                                                                                                                                                                           |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Corresponding points and sides of scaled shapes</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Corresponding Parts of Similar Figures Practice</a><br/> Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p> | <p>In groups of 4, learners receive an envelope of sorting cards: the four photo prints (original 14 cm × 18 cm, passport 3.5 cm × 4.5 cm, poster 70 cm × 90 cm, distorted 4.5 cm × 4.5 cm), plus card pairs for the Kenya map, matatu model, school logo (normal and squashed) and two triangles. They lay out the original as the reference, place each other print beside it, and sort every card into 'looks the same shape as the original/partner' and 'does not'. They may cut out and lay prints on top of one another to compare, but may NOT measure with a ruler yet. They record each decision in a table: Figure pair   Same shape?   What I noticed.</p> | <p>Distribute envelopes and say: 'Use only your eyes and your hands today. Lay one figure over another and ask: does it look like a true enlargement or reduction, or is it stretched, squashed or skewed?' Remind of scissor safety. Circulate and ask: 'Which part of the face looks wrong in the 4.5 by 4.5 print? Where do you see it?' Ask groups who finish early: 'Can you find a similar pair and a non-similar pair in the classroom, such as the school notice board and its picture?' Allow WAIT TIME of 10 seconds after each question. Do NOT tell learners about ratios or angles.</p> | <p>Sorting and classifying (pattern-seeking) with a data table: handling many contrasting examples helps learners notice the common feature of similar figures (same shape, different size) and what breaks it (uneven stretching).</p>        | <p>Observable indicators: (1) group sorts the passport and poster as the same shape as the original and the 4.5 cm × 4.5 cm print as different; (2) the table has a reason for each decision; (3) learners use correct vocabulary 'enlarged', 'reduced', 'stretched'; (4) learners overlay prints to compare corresponding parts (eyes, chin, corners). Intervene if a group sorts by size alone.</p>                     |
| <p><b>Explain Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Dilations and properties</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Similarity &amp; transformations</a><br/> Source: Khan</p>                                                                                              | <p>Groups share their sorts in a gallery walk, then in a whole-class discussion explain their rule for deciding whether two figures are the same shape. Learners attempt to answer: 'Why do we need to enlarge or reduce objects?' and 'Where do we use similar figures in day-to-day life?' using Kenyan examples (passport photos in Kisumu, maps of Kenya, a matatu model, a school logo on a signboard, a sketch of a farm plot).</p>                                                                                                                                                                                                                              | <p>Ask: 'Which print did every group sort as different? Why?' Cold-call 3 groups to share, then 2 learners to restate another group's reasoning: 'Say in your own words what Group 2 noticed.' WAIT TIME of 10-15 seconds after each question. Press: 'You said it looks stretched. Stretched in which direction? Wide only, or tall as well?' Introduce the symbol ~ and the terms 'corresponding parts' and 'enlargement/reduction' using</p>                                                                                                                                                      | <p>Claim-Evidence-Reasoning with a gallery walk and peer restating: learners justify each sorting claim with observed evidence and compare reasoning across groups, so the class constructs a shared definition rather than receiving one.</p> | <p>Listen for: (1) a rule that does not rely on size ('same shape, different size'); (2) identification of the non-uniform stretching (width changed but height did not) in the 4.5 cm × 4.5 cm print; (3) at least two real-life examples of similar figures; (4) recognition that visual judgement may be unreliable. Use a 3-question exit check: 'Name a similar pair. Name a non-similar pair. Give one reason.'</p> |

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| <p>Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                      | <p>They write the class rule: 'Similar figures (~) have the same shape; one is an enlargement or reduction of the other. A figure stretched or squashed in one direction only is NOT similar.' They also identify corresponding parts (width, height, eyes, chin).</p>                                                                                                                                                                                                                                               | <p>learners' own words. Say: 'Our eyes can be fooled. What could we do to be sure?' Take 3 answers, noting 'measure' and 'compare angles' for Lesson 3. Do not teach scale factors or ratios.</p>                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                     |
| <p><b>Driving Question Board (DQB) Creation</b><br/> <b>VIDEO:</b><br/> <a href="#">Corresponding points and sides of scaled shapes</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Dilations and properties</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Learners write questions on sticky notes and add them to the DQB beside the driving question: 'How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?' Examples: 'How can we check shape without just looking?', 'Is there a number that shows the picture was enlarged evenly?', 'Why does the 4.5 by 4.5 print look wrong?', 'How do map makers shrink Kenya?' They cluster the notes under headings such as Checking shape, Numbers, Maps and models.</p> | <p>Say: 'Write one question that today's sorting left you wondering about.' Give WAIT TIME of 15 seconds before writing. Cold-call 4 learners to read a question and say where on the board it belongs. Ask: 'Which question could we answer by measuring?' Highlight questions that point to Lesson 3 and later work (angles, ratios, scale factor), and mark them with a star. Do not answer them now.</p>                                                                        | <p>Driving Question Board with question clustering: turning observations into questions gives learners ownership of the inquiry and shows how today's evidence links to coming lessons.</p> | <p>Check that questions: (1) are linked to the phenomenon; (2) move beyond 'what' to 'how' or 'why'; (3) show recognition that visual judgement needs a test. Count how many learners wrote a question pointing to measurement. Note learners who need sentence starters such as 'How can we be sure that...'</p>                   |
| <p><b>Model Building Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Dilations: scale factor</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>HTML:</b><br/> <a href="#">Corresponding Parts of Similar</a></p>                                                                                   | <p>Learners draw a first-draft model in their notebooks: three prints of Wanjiru (3.5 cm × 4.5 cm passport, 14 cm × 18 cm original, 70 cm × 90 cm poster) shown as the same shape in three sizes, with the 4.5 cm × 4.5 cm print beside them labelled 'not the same shape'. Under it they write: 'What may change: _____. What must stay the same: _____. What I still need to check: _____. They finish a summary table row for today.</p>                                                                          | <p>Say: 'Revise your model from Lesson 1 using today's evidence. Show what may change and what must stay the same. Also mark what you are still unsure about.' Give WAIT TIME of 10 seconds before drawing. Circulate and ask 3 learners: 'Where in your drawing can I see the difference between the passport print and the 4.5 by 4.5 print?' Select 2 learners to share. Close by saying: 'Next lesson we will measure to test your rule.' Keep drawings for later revision.</p> | <p>Cumulative model building with a 'still need to check' section: learners record what their evidence supports and openly mark gaps, so the model grows with each lesson.</p>              | <p>Check the model: (1) shows three similar prints and one non-similar print clearly; (2) states that size may change and shape must stay; (3) includes an honest 'still need to check' item such as measuring. Look for numeric ratios or angles appearing too early, and redirect gently. Collect the exit check as evidence.</p> |

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| <a href="#">Figures (Flexbook)</a><br>Source: CK-12<br><a href="#">Search ARES for similar readings</a> |  |  |  |  |
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D. TEACHER REFLECTION

Which learners judged similarity by size alone, and did the overlay of prints change their thinking? Did the gallery walk surface the idea that stretching in one direction only breaks the shape? Which questions on the DQB point to measuring in Lesson 3? Adjust grouping and examples (e.g. add local signboards or a farm plan) if some groups relied on guesswork.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What did I observe?                                          | We observed that Wanjiru's 3.5 cm × 4.5 cm passport print and the 70 cm × 90 cm poster look the same shape as the 14 cm × 18 cm original, but the 4.5 cm × 4.5 cm print looks stretched. Other pairs (map, matatu model, school logo) can also be sorted this way. |
| What did I learn?                                            | We learned that similar figures (~) have the same shape but can have different sizes, and that a figure stretched or squashed in one direction only is not similar to the original.                                                                                |
| How does this explain the phenomenon?                        | We can explain that size can change a lot while shape stays the same only when the figure is enlarged or reduced evenly. We still need a reliable test, not just our eyes, which we will build by measuring in Lesson 3.                                           |

## LESSON 3 (40 minutes (one lesson; double period recommended for the full measuring activity)): Properties of Similar Figures: Equal Angles and Equal Ratios in Wanjiru's Photos

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | To help learners discover, by measuring, that similar figures have equal corresponding angles and equal ratios of corresponding sides, and to use these two tests to explain why the passport print and poster keep Wanjiru's likeness while the 4.5 cm × 4.5 cm trial print does not.                                                                                                                                                                                                                                                                                       |
| <b>Knowledge</b>            | Similar figures (~) have all corresponding angles equal and all ratios of corresponding sides equal. Original 14 cm × 18 cm, passport 3.5 cm × 4.5 cm and poster 70 cm × 90 cm all have width : height = 7 : 9 and 90° corners. The distorted 4.5 cm × 4.5 cm print has width : height = 1 : 1. Face triangle (eye gap : chin distance) is 5.6 : 7.2, 1.4 : 1.8 and 28 : 36 (all 7 : 9) in the original, passport and poster, but 1.8 : 1.8 in the distorted print. The chin angle is about 42.5° in the original, passport and poster but about 53° in the distorted print. |
| <b>Skills</b>               | Measuring lengths with a ruler and angles with a protractor; calculating and simplifying ratios; comparing ratios and angles across figures; planning and carrying out an investigation (NGSS SEP: Planning and Carrying Out Investigations); analysing and interpreting data (NGSS SEP: Analyzing and Interpreting Data); constructing explanations from evidence; communicating and collaborating in groups.                                                                                                                                                               |
| <b>Attitudes</b>            | Appreciate the value of careful, honest measurement; show curiosity about why shape is kept; Unity and Responsibility when sharing instruments and recording data; confidence in justifying claims with evidence.                                                                                                                                                                                                                                                                                                                                                            |
| <b>Key Inquiry Question</b> | 1. Why do we need to enlarge or reduce objects? 2. How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Purpose in Storyline</b> | Lesson 1 launched the puzzle of Wanjiru's photos and Lesson 2 sorted figures by looking at shape. Lesson 3 replaces 'it looks the same' with measurable evidence: equal angles and equal ratios of corresponding sides. This gives learners the two tests they need before Lesson 4 introduces the linear scale factor k, and answers the first half of 'What must stay the same, and what is allowed to change?'                                                                                                                                                            |
| <b>Safety Notes</b>         | Scissors and protractor points are sharp: carry them point-down and use only at the desk. Keep bags and chairs out of walkways when groups move to the gallery walk. Do not use pencil points for pointing at faces; use the ruler edge. Everyone shares instruments politely and returns them to the group kit.                                                                                                                                                                                                                                                             |

### B. LESSON OVERVIEW

Mrs Achieng returns to Wanjiru's photographs from Lakeview Photo Studio, Kisumu. Groups receive four cut-out prints (original 14 cm × 18 cm, passport 3.5 cm × 4.5 cm, poster 70 cm × 90 cm represented by a scaled sheet or the stated measurements, and the distorted 4.5 cm × 4.5 cm print) with a marked face triangle (eyes and chin). Learners first predict what must be equal for two prints to be similar, then measure corner angles, widths and heights, the eye gap and chin distance, and the chin angle with a protractor. They compute width : height and eye gap : chin distance ratios, compare angles, and explain in groups why the passport and poster are similar to the original (~) while the distorted print is not. The class adds evidence to the Driving Question Board and revises the group model of 'what must stay the same'. No electricity or internet is required.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
|-------|--------------------|---------------|----------------------|-------------------------------|
|-------|--------------------|---------------|----------------------|-------------------------------|

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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Intro to triangle similarity</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Corresponding Sides of Similar Triangles (PLIX)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>                 | <p>Learners recall the four prints from Lessons 1 and 2 (the passport and poster looked like the original, the distorted 4.5 cm × 4.5 cm print did not). Individually they write in their books: 'If two figures are similar, which measurements must be equal and which can be different?' and choose from a list: all lengths, all angles, ratios of sides, areas, colour. They then sketch the expected result for the corner angle and for width : height of the passport print versus the distorted print, and share with a partner.</p>                                                                                                                                                                                                                                                                                                                                                                      | <p>Display the four print sizes on the board: 14 × 18, 3.5 × 4.5, 70 × 90 and 4.5 × 4.5 (cm). Say: "In Lesson 2 we sorted by looking. Today we want proof from measuring. Before we measure, what do you think must stay the same for Wanjiru to still look like Wanjiru?" Give 15 seconds of silent WAIT TIME for writing. Then cold-call 4 learners from different groups (not volunteers) to give one prediction each; record exactly what they say on the board under 'Our predictions' without judging. Ask: "Who can add a reason, not just an answer?" and WAIT TIME 10 seconds. Do not tell learners the correct properties.</p>                                                                                                                                                                                                         | <p>Prediction with justification (Think-Write-Pair-Share): committing to a written prediction before measuring exposes preconceptions (for example, that only the size or area must match), so measurement results later create productive conflict or confirmation linked to the anchoring phenomenon.</p>   | <p>Look for: learners who write that 'size' or 'area' must stay the same (misconception to revisit); learners who mention angles or 'proportion/ratio' unprompted; whether predictions include a reason. Tally on the board how many pairs predicted angles, ratios, both, or neither.</p>                                                                                                                                        |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Triangle similarity postulates/criteria</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Corresponding Parts of Similar Figures (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>In groups of four with a ruler, protractor, pencil and the four print cut-outs (face triangle marked with eye-gap line and chin point), learners complete a data table. Task A: measure the four corner angles of each print (expect 90°). Task B: measure width and height of each print and compute width : height (7 : 9 for original, passport and poster; 1 : 1 for the distorted print). Task C: measure the eye gap along the top and the chin distance down from the eye line (original 5.6 and 7.2; passport 1.4 and 1.8; poster 28 and 36; distorted 1.8 and 1.8) and compute eye gap : chin distance. Task D: measure the apex angle at the chin with a protractor (about 42.5° for the original, passport and poster; about 53° for the distorted print). Each group records results, allowing small measurement error of about 1° or 1 mm, and marks which measurements matched across prints.</p> | <p>Model one measurement under the document camera or on the board: place the protractor centre on the chin point and read from 0°. Say: "Measure twice, record once. Honest numbers, even if they are a little off." Circulate and ask each group: "Which two prints have the same width : height? How do you know?" and "What did you get for the chin angle on the distorted print?" Use WAIT TIME of 10 seconds after each question. Cold-call 3 groups to read their chin angle for the original, passport and poster so the class sees repeats of about 42.5°. If values differ by more than 2°, ask the group to re-measure with a partner group. Give support to groups struggling with simplifying ratios (divide both by 0.5 to get 7 : 9). Extension: ask quick finishers to compare eye gap : chin distance with width : height.</p> | <p>Structured data collection and pattern spotting: a common table lets groups compare columns across prints, so patterns (all 90°; 7 : 9 repeated; 42.5° repeated; the distorted print breaks every pattern except the 90° corners) emerge from the learners' own data rather than from teacher telling.</p> | <p>Observable indicators: protractor correctly centred on the vertex; angle read from the right scale; ratio simplified correctly to 7 : 9 or 1 : 1; table complete with units in cm; measurement within about 1° or 1 mm of expected values; groups re-measure when results disagree. Note groups that conclude the distorted print is 'different in everything' (it still has 90° corners) to address in the Explain Phase.</p> |

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| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Corresponding points and sides of scaled shapes</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Corresponding Parts of Similar Figures Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>                             | <p>Groups study their completed tables and write a claim-evidence-reasoning statement: 'The passport print and the poster are similar to the original (~) because ... The distorted 4.5 cm × 4.5 cm print is not similar because ...'. They must use the terms corresponding angles and corresponding sides. A spokesperson from each group shares; the class then agrees on two tests for similar figures: (1) all corresponding angles are equal; (2) ratios of corresponding sides are equal. Learners discuss why the distorted print passes test (1) for corners (90°) but fails test (2) and the chin-angle part of test (1), and therefore both tests are needed. They link this to the phenomenon: the size changed by a huge amount but the ratio 7 : 9 and the angle 42.5° stayed the same.</p> | <p>Say: "Use your numbers as evidence. Tell me what stayed the same and what changed." Give 2 minutes of group talk, then WAIT TIME 15 seconds before calling on anyone. Cold-call 5 learners to give one part of the explanation each (one on corner angles, one on width : height, one on the chin angle, one on the face triangle ratio, one on the distorted print). Press with: "Which number is your evidence?" and "The distorted print has 90° corners, so why is it still not similar?" Record the two properties on the board using the words 'corresponding angles' and 'corresponding sides' and the symbol ~. Do not introduce the scale factor k yet; say: "Next lesson we will find the single number that links the lengths." Correct language: use 'distorted' only for the 4.5 cm × 4.5 cm print.</p> | <p>Claim-Evidence-Reasoning with a class consensus discussion: requiring numerical evidence for each claim and then synthesising a shared definition builds a precise, evidence-based concept of similarity rather than a visual impression.</p> | <p>Look for: explanations that cite specific measured values (7 : 9, 42.5°, 53°, 1 : 1); correct use of 'corresponding'; recognition that equal corner angles alone do not prove similarity; avoiding the misconception that equal area or equal length is needed. Use thumbs up/sideways/down on 'Are equal angles enough?' (answer: no, ratios also needed) and note the count.</p> |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Intro to triangle similarity</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Corresponding sides and points</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to the Driving Question Board posed by Mrs Achieng: 'How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?' Each group writes on sticky notes (a) one thing they now know, with evidence from today's table (for example, 'Original, passport and poster all have width : height = 7 : 9'), and (b) one new question (for example, 'What number do we multiply 14 by to get 3.5?' or 'Do the areas follow the same pattern?'). They place notes on the DQB under 'Answered so far' or 'New questions' and link them to earlier notes from Lessons 1 and 2.</p>                                                                                                                                                                   | <p>Say: "Read the Driving Question again. What part have we answered today? What do we still need to find out?" Give 10 seconds WAIT TIME. Cold-call 3 groups to read and place a note, asking: "Where does this connect to an earlier note?" Group related questions (for example, the 'what number' notes) and tell learners the next lesson will take up the question about the number that links 14 to 3.5 without giving the answer. Keep notes that mention area and 3D shapes for Lessons 6 and 7.</p>                                                                                                                                                                                                                                                                                                           | <p>Driving Question Board questioning: learners see their questions drive the unit sequence, which keeps the anchoring phenomenon central and motivates later lessons on scale factor, area and volume.</p>                                      | <p>Check that notes: cite measured evidence; pose a testable question; are placed in the right category. Note how many groups ask about the linking number between lengths (readiness for Lesson 4). Quality indicator: a question that starts with 'How much?' or 'What number' rather than a yes/no question.</p>                                                                   |
| <p><b>Model</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Each group revises its class model</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Say: "Update your model with</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Cumulative model revision: adding</p>                                                                                                                                                                                                         | <p>Indicators: model correctly lists</p>                                                                                                                                                                                                                                                                                                                                              |

## Building Phase

 **VIDEO:**

## Triangle similarity postulates/criteria

Source: Khan Academy (English - US curriculum)

 [Search ARES](#)  
for similar videos

 **EXERCISE:**

## Unknown Measures of Similar Figures Practice

Source: CK-12  
 [Search ARES](#)  
[for similar](#)  
[readings](#)

of 'What must stay the same, and what is allowed to change?' on a poster or in their books. The model has two columns: 'Allowed to change' (size: width and height, area, length of the eye gap and chin distance) and 'Must stay the same' (corresponding angles such as  $90^\circ$  corners and about  $42.5^\circ$  at the chin; ratio of width to height  $7 : 9$ ; ratio of eye gap to chin distance  $7 : 9$ ). They add a labelled sketch of the original and passport prints with matching angles marked, and a sketch of the distorted  $4.5 \text{ cm} \times 4.5 \text{ cm}$  print with the broken ratio ( $1 : 1$ ) and the chin angle of about  $53^\circ$ . Finally they complete the exit-ticket question: 'A new print is  $10.5 \text{ cm} \times 13.5 \text{ cm}$ . Is it similar to the original  $14 \text{ cm} \times 18 \text{ cm}$ ? Show your test.' ( $10.5 : 13.5 = 7 : 9$ , so the width : height ratio matches; a full proof would also check angles, which are  $90^\circ$  for any rectangle).

today's evidence. Show me what stays the same, what changes, and which measurement proves it." Circulate with the prompt: "Where in your model is the chin angle?" and WAIT TIME 10 seconds. Select 2 groups to present for 1 minute each, then cold-call 2 other learners to say one thing they would add or change. Collect the exit tickets. Close with: "Size can change by a lot. Shape stays when angles and ratios of corresponding sides stay equal. Next time we find the number that links the lengths." Do not state the term scale factor definition yet beyond previewing it.

measured evidence to a visual model consolidates the two properties, makes the distorted-print contrast explicit, and prepares learners to quantify the change with a scale factor in Lesson 4.

angles and ratios under 'Must stay the same' and size under 'Allowed to change'; diagram labels  $42.5^\circ$  and  $53^\circ$  correctly; exit ticket applies the width : height test ( $10.5 : 13.5$  simplifies to  $7 : 9$ ) and states a conclusion. Sort tickets into three piles: complete, partly right (for example, ratio only), needs reteach, and plan follow-up in Lesson 4.

After the lesson, consider: Did learners rely on measured data rather than appearance to decide on similarity? Which groups mis-measured the chin angle, and was the cause the protractor placement, reading the wrong scale or a poorly marked vertex? How many learners recognised that the distorted print still has  $90^\circ$  corners yet is not similar? Did any learners hold on to the idea that equal area is needed? Which DQB questions show readiness for the scale factor in Lesson 4? Note what you would change in grouping, pacing or the data table for next time.

## What did I observe?

We measured the corner angles, widths, heights, eye gap, chin distance and chin angle of the original (14 cm × 18 cm), passport (3.5 cm × 4.5 cm), poster (70 cm × 90 cm) and the distorted (4.5 cm × 4.5 cm) prints. Record your values: corners \_\_\_\_°, width : height \_\_\_\_, eye gap : chin distance \_\_\_\_, chin angle \_\_\_\_°.

## What did I learn?

Similar figures ( $\sim$ ) have equal corresponding angles and equal ratios of corresponding sides. Write the two tests in your own words and give one number from today's data for each.

## How does this explain the phenomenon?

Explain why the passport print and poster keep Wanjiru's likeness but the 4.5 cm × 4.5 cm print does not, using the ratio 7 : 9 versus 1 : 1 and the chin angle of about 42.5° versus about 53°.



## LESSON 4 (40 minutes): Linear Scale Factor: How Many Times Bigger or Smaller? (Wanjiru's Photo Prints)

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <b>Purpose</b>              | Learners calculate the linear scale factor $k = \text{image length} \div \text{object length}$ for Wanjiru's prints, classify it as an enlargement ( $k > 1$ ) or reduction ( $0 < k < 1$ ), and use $k$ to find missing lengths. They also see that the distorted $4.5 \times 4.5$ cm print has two different factors ( $9/28$ and $1/4$ ), so no single $k$ exists for it.                                                                                                                                                                                                                                                                           |
| <b>Knowledge</b>            | Learners know that the linear scale factor is $k = \text{image length} \div \text{object length}$ , and they always state which print is the object and which is the image. For original ( $14 \times 18$ ) to passport ( $3.5 \times 4.5$ ), $k = 1/4$ (a reduction). For original to poster ( $70 \times 90$ ), $k = 5$ (an enlargement). For passport to poster, $k = 20$ . The distorted print has a width factor of $9/28$ (about 0.32) and a height factor of $1/4$ (0.25), which are not equal, so it is not similar to the original. Learners know that a missing length $= k \times \text{object length}$ , or $\text{image length} \div k$ . |
| <b>Skills</b>               | Learners calculate $k$ from corresponding sides and check that the same $k$ works for width and height. They use $k$ to find missing lengths in either direction, for example the face-triangle eye gap of 5.6 cm becoming 1.4 cm in the passport photo. They justify claims with calculations (NGSS SEP: Using Mathematics and Computational Thinking; Constructing Explanations). They also carry out the NGSS SEP of Analysing and Interpreting Data by comparing the two factors for the distorted print.                                                                                                                                          |
| <b>Attitudes</b>            | Learners show curiosity and precision when calculating and checking factors. They show teamwork and unity in groups, and responsibility by caring for rulers and sharing materials. They value the use of scale factors in day-to-day activities such as photo printing in Kisumu.                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Key Inquiry Question</b> | 1. Why do we need to enlarge or reduce objects? 2. How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Purpose in Storyline</b> | Lesson 3 showed that similar figures have equal corresponding angles and equal ratios of corresponding sides. Lesson 4 turns the idea of 'equal ratios' into a single number, the linear scale factor $k$ . This gives the precise answer to what must stay the same: one $k$ for all lengths. It prepares learners for similar triangles in Lesson 5, area and volume factors in Lessons 6 and 7, and enlargements with a centre in Lessons 8 to 10.                                                                                                                                                                                                  |
| <b>Safety Notes</b>         | Learners use rulers and scissors with care and walk, not run, when moving to the Driving Question Board (DQB). Keep bags and desks clear. No electricity or internet is needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

### B. LESSON OVERVIEW

Mrs Achieng revisits Wanjiru's photo from Lakeview Photo Studio, Kisumu. Learners first predict how many times smaller the  $3.5 \times 4.5$  cm passport photo is than the  $14 \times 18$  cm original, and how many times bigger the  $70 \times 90$  cm prize-giving poster is. In groups they then measure and calculate  $k = \text{image} \div \text{object}$  for width and for height on each print. They discover that the passport print gives  $1/4$  for both width and height, the poster gives 5 for both, and passport to poster gives 20. For the distorted  $4.5 \times 4.5$  cm print, the factors are  $9/28$  for width and  $1/4$  for height, which are not equal. In the Explain phase learners use  $k$  to find missing lengths such as the face-triangle eye gap and chin distance on each print. They then add to the DQB and revise their class model: what must stay the same is one scale factor  $k$  applied to every length. The lesson stays on linear lengths only. Angle properties belong to Lesson 3 and area and volume factors to Lessons 6 and 7.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Dilations: scale factor</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Dilations: scale factor</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners look at the three similar prints drawn on the board or shown as cut-outs: original <math>14 \times 18</math> cm, passport <math>3.5 \times 4.5</math> cm, poster <math>70 \times 90</math> cm. Individually they write a prediction in their books: 'The passport photo is ____ times smaller than the original, and the poster is ____ times bigger. I think this because ____.' They share with a partner. They then predict whether the same number will work for both width and height on the distorted <math>4.5 \times 4.5</math> cm print.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Hold up the three cut-outs and say: 'Wanjiru's face is the same in all three. Mr Otieno at Lakeview Photo Studio needs one number to tell his printer how much to shrink or stretch. What number would you give him?' Give WAIT TIME of 10-15 seconds of silent thinking before anyone writes. Then ask partners to share for 1 minute. Cold-call 4 learners with different predictions and record all of them on the board without judging. Ask: 'Who has a different number? What did you compare to get it?' Note any learners who compare only the width, or who subtract (<math>14 - 3.5</math>) instead of dividing.</p>                                                                  | <p>Prediction with Think-Pair-Share: committing to a prediction first creates a need to know and surfaces the common subtraction misconception ('10.5 cm smaller') so that later evidence can be compared with learners' own ideas. Learners see that they need a way to compare sizes as a multiplier rather than a difference.</p>           | <p>Listen for learners who use multiplication or division ('about 4 times smaller') rather than subtraction. Check that at least 3 of the 4 cold-called learners can say which print they compared from and which to. Tally on the board how many predicted 4, 5, or 'don't know'. Look for who expects the same factor for the distorted print.</p>                                                                                                                  |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Scale drawings</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Scale drawings</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar readings</p>                 | <p>In groups of 4, learners get a data card of the print sizes (original <math>14 \times 18</math>, passport <math>3.5 \times 4.5</math>, poster <math>70 \times 90</math>, distorted <math>4.5 \times 4.5</math> cm) and measure each print with a ruler to confirm the sizes. They complete a table with the columns: Object, Image, Image width <math>\div</math> Object width, Image height <math>\div</math> Object height, Same? Rows: (1) original to passport, (2) original to poster, (3) passport to poster, (4) original to distorted print. They calculate: row 1 gives <math>3.5/14 = 1/4</math> and <math>4.5/18 = 1/4</math>. Row 2 gives <math>70/14 = 5</math> and <math>90/18 = 5</math>. Row 3 gives <math>70/3.5 = 20</math> and <math>90/4.5 = 20</math>. Row 4 gives <math>4.5/14 = 9/28</math> (about 0.32) and <math>4.5/18 = 1/4</math> (0.25), which are not equal. Learners write <math>k = \text{image} \div \text{object}</math>, stating which is the object and which is the image. They then label each result as an enlargement (<math>k &gt; 1</math>) or a reduction (<math>0 &lt; k &lt; 1</math>).</p> | <p>Model the first calculation aloud, then release: 'k equals image length divided by object length. Say it with me. Before you divide, point to your object and point to your image.' Circulate and ask: 'Which print is your object? Which is your image?' Ask: 'Check: is the width factor equal to the height factor?' If a group reverses the division, ask: 'Is the passport bigger or smaller than the original? Should k be bigger or smaller than 1?' Give WAIT TIME of 10-15 seconds after each question. For row 4 ask: 'Your two numbers are different. What does that tell you about this print?' Stop the class after 12 minutes and cold-call 3 groups, each to report one row.</p> | <p>Structured data table with Claim-Evidence comparison: calculating width and height factors side by side makes the pattern visible. Equal factors mean one k, unequal factors mean no single k. Linking 'equal ratios of corresponding sides' from Lesson 3 to one number helps learners see k as a multiplier rather than a difference.</p> | <p>Check the table: each row should have both factors calculated with the object and image named. Look for correct values: <math>1/4</math>, 5, 20, and <math>9/28</math> versus <math>1/4</math>. Observe whether learners classify <math>k = 1/4</math> as a reduction and <math>k = 5</math> as an enlargement. Note groups who divide the wrong way round and prompt them with the bigger/smaller check. Ask 2 learners to explain why row 4 has no single k.</p> |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Dilations: scale</a></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Learners use k to find missing lengths on the face triangle (eye gap and chin distance). Given the</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Pose: 'Mr Otieno knows the original eye gap is 5.6 cm. How can he find the eye gap on the</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Claim-Evidence-Reasoning (CER) with Sentence Frames: learners state a claim (the passport print is</p>                                                                                                                                                                                                                                      | <p>Listen for: (1) correct missing lengths of 1.4, 1.8, 28 and 36 cm. (2) division by k used to find object</p>                                                                                                                                                                                                                                                                                                                                                       |

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| <p><a href="#">factor</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/><a href="#">Dilations: scale factor</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar readings</a></p>                                                                    | <p>original eye gap 5.6 cm and chin distance 7.2 cm, they calculate the passport values with <math>k = 1/4</math> (1.4 and 1.8 cm) and the poster values with <math>k = 5</math> (28 and 36 cm). They find an original length from an image length by dividing: an image eye gap of 28 cm on the poster, divided by <math>k = 5</math>, gives 5.6 cm. They try the distorted print, where the eye gap and chin distance are both 1.8 cm, and explain why that print has no single <math>k</math>. Each group writes a 2-3 sentence explanation: 'The passport photo keeps Wanjiru's likeness because ...'. They then compare how <math>k = 20</math> (passport to poster) relates to <math>1/4</math> and 5: <math>1/4 \times 5</math> would not give 20, but <math>5 \div (1/4) = 20</math>.</p> | <p>passport photo without measuring it?' Give WAIT TIME of 10-15 seconds, then collect 3 responses. Write the rule: 'missing image length = <math>k \times</math> object length'. Then ask: 'If you know the image length, how do you find the object length?' Cold-call 3 learners. Ask the groups to write the sentence frame '<math>k = \underline{\hspace{1cm}} \div \underline{\hspace{1cm}}</math> because <math>\underline{\hspace{1cm}}</math>'. Challenge: 'The poster is <math>k = 5</math> from the original. The passport is <math>k = 1/4</math>. Why is passport to poster 20 and not 5?' Allow 15 seconds of silence. Use the answer to restate that the object changes, so <math>k</math> changes. Ask one group to explain why the distorted print is not similar and state that the width and height factors are different. Do not re-teach angle measurement. Refer only to Lesson 3's finding that similar figures have equal ratios of corresponding sides.</p> | <p>similar to the original), support it with calculated evidence (<math>k = 1/4</math> for both width and height), and reason from the rule (one <math>k</math> for all lengths means the shape is kept). This links calculation to the driving question and supports the NGSS practice of Constructing Explanations.</p> | <p>lengths. (3) the explanation of why the distorted print has no single <math>k</math> (<math>9/28 \neq 1/4</math>). Check that at least 3 of the 4 groups state which print is the object. Mark with a tick when a learner uses the words 'corresponding sides'. Spot learners confusing <math>k = 5</math> with <math>k = 20</math> and prompt them to name the object and image.</p> |
| <p><b>Driving Question Board (DQB) Creation</b><br/> <b>VIDEO:</b><br/><a href="#">Scale drawings</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/><a href="#">Scale drawings</a><br/>Source: Khan Academy (English - US curriculum)<br/><a href="#">Search ARES for similar readings</a></p> | <p>Learners return to the DQB, where Mrs Achieng has posted the driving question: 'How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?' Individually they write on sticky notes or paper slips (a) one thing they now know using <math>k</math>, and (b) one new question, for example 'What happens to the area when <math>k = 1/4</math>?' or 'How do I enlarge a shape from a point?' They place the notes on the board under headings: 'What we figured out' and 'Questions we still have'. They also tick off any earlier questions that Lesson 4 answered, such as 'What number tells how much bigger or smaller?'</p>                                                                                                              | <p>Say: 'Take 2 minutes to write one thing <math>k</math> helped you explain and one question you still have.' Give WAIT TIME of 10-15 seconds before writing starts. Invite 4 learners to place and read their notes. Ask: 'Which questions can we answer with what we know about <math>k</math>, and which need a new lesson?' Sort the questions together into clusters: area, volume, enlargement with a centre, maps and signs. Point out that area and volume questions will be answered in Lessons 6 and 7 and centre-based enlargement in Lessons 8 to 10. Do not answer them now.</p>                                                                                                                                                                                                                                                                                                                                                                                       | <p>Driving Question Board: learners see their own questions guide the unit. Clustering questions shows how each coming lesson links to the anchoring phenomenon and keeps learners invested in the storyline.</p>                                                                                                         | <p>Check the sticky notes: does each learner name one correct fact about <math>k</math> (for example '<math>k = \text{image} \div \text{object}</math>')? Look for new questions that point forward to area, volume or enlargement. Observe 4 learners sharing aloud and check for accuracy and connection to the Driving Question.</p>                                                  |
| <p><b>Model Building</b></p>                                                                                                                                                                                                                                                                                                                                                        | <p>Groups revise the class model of 'what must stay the same and</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Say: 'Add what Lesson 4 gave us to our model. Where does <math>k</math> fit in?'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Cumulative Model Revision with Gallery Walk: learners add the</p>                                                                                                                                                                                                                                                      | <p>Check each group's model for the formula <math>k = \text{image} \div \text{object}</math>, the</p>                                                                                                                                                                                                                                                                                    |

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| <p><b>Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Dilations: center</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Dilations: center</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>what may change' that they started in Lesson 1. On a poster they add a new section 'Linear scale factor k' with: the formula <math>k = \text{image} \div \text{object}</math>; the three Wanjiru factors (original to passport <math>1/4</math>, original to poster 5, passport to poster 20); the rule that one k must apply to both width and height; and the distorted print counter-example (<math>9/28</math> and <math>1/4</math>). They draw arrows between the <math>14 \times 18</math>, <math>3.5 \times 4.5</math> and <math>70 \times 90</math> cm prints, each labelled with its k and 'enlargement' or 'reduction'. They write a revised model statement: 'Size can change by any k, but the shape is kept only if the same k multiplies every length.' Each group marks one thing they are still unsure about, such as area, and ends with a gallery walk.</p> | <p>Circulate and ask each group: 'How does your model explain why the <math>4.5 \times 4.5</math> cm print looks stretched?' Allow WAIT TIME of 10-15 seconds. Prompt: 'Have you labelled the object and the image on each arrow?' After 8 minutes run a 3-minute gallery walk. Cold-call 3 learners to say one improvement they saw on another group's model. Close with an exit question on paper: 'A <math>14 \times 18</math> cm print is reduced by <math>k = 1/4</math>. Find the new width and height. Is this an enlargement or a reduction?' Collect the slips (answer: <math>3.5 \times 4.5</math> cm; reduction).</p> | <p>new quantitative idea (k) to their existing qualitative model. Comparing models across groups exposes gaps and strengthens a shared explanation, using the NGSS practice of Developing and Using Models.</p> | <p>correct values (<math>1/4</math>, 5, 20, <math>9/28</math> and <math>1/4</math>), and the statement that one k applies to all lengths. Use the exit slip to check that at least 80% of learners give <math>3.5 \times 4.5</math> cm and say 'reduction'. Note learners who need support to name the object and image and revisit them at the start of Lesson 5.</p> |
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## D. TEACHER REFLECTION

After the lesson ask yourself: Did learners name the object and the image before dividing, or did some divide the wrong way round? Which learners still use subtraction ( $14 - 3.5$ ) rather than division to compare sizes? Could they explain why the distorted print ( $9/28$  and  $1/4$ ) has no single k without prompting? Which DQB questions about area and volume will you pick up in Lessons 6 and 7? Which groups need extra practice finding the object length from the image length before moving on to similar triangles with shadows in Lesson 5? What would you change about the pacing of the table activity, and how did the Kisumu photo context support or distract from the mathematics?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| <b>What did I observe?</b>                   | We measured and compared the prints of Wanjiru's photo. Original to passport: width $3.5 \div 14 = 1/4$ and height $4.5 \div 18 = 1/4$ . Original to poster: $70 \div 14 = 5$ and $90 \div 18 = 5$ . Passport to poster: 20 for both. Distorted print: width factor $9/28$ and height factor $1/4$ , which are not equal.                                                                                                         |
| <b>What did I learn?</b>                     | The linear scale factor is $k = \text{image length} \div \text{object length}$ , and we must always say which is the object and which is the image. If $k > 1$ it is an enlargement and if $0 < k < 1$ it is a reduction. We can find a missing length using $\text{image} = k \times \text{object}$ , or $\text{object} = \text{image} \div k$ .                                                                                 |
| <b>How does this explain the phenomenon?</b> | Wanjiru's passport photo ( $k = 1/4$ ) and poster ( $k = 5$ ) keep her likeness because every length, both width and height, is multiplied by the same k, so they are similar (–) to the original. The $4.5 \times 4.5$ cm print changed the width by $9/28$ and the height by $1/4$ . These are different factors, so her face looked stretched. What must stay the same is one scale factor for all lengths. What may change is |

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|  | the size. |
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## LESSON 5 (40 minutes): Similar Triangles in Problem Solving: Measuring Heights with Shadows

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <b>Purpose</b>              | Learners use similar triangles (equal corresponding angles and proportional corresponding sides) to find unknown heights from shadow lengths, so they can see that the same idea that keeps Wanjiru's passport photo undistorted also lets us measure things we cannot reach.                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Knowledge</b>            | Two triangles are similar ( $\sim$ ) when their corresponding angles are equal and the ratios of corresponding sides are equal. Wanjiru (1.5 m tall) casts a 2 m shadow. At the same time the tree in the Nyalenda Secondary School compound casts a 12 m shadow. The sun's rays meet the ground at the same angle, so the two triangles are similar. The scale factor from Wanjiru's triangle (object) to the tree's triangle (image) is $k = 12 \div 2 = 6$ , so the tree height is $1.5 \times 6 = 9$ m. The ratio height : shadow = $1.5 : 2 = 0.75$ holds for both, so $0.75 \times 12 = 9$ m.                                           |
| <b>Skills</b>               | Predicting, measuring and recording lengths with tape measure or ruler; identifying corresponding angles and sides in two triangles; calculating the scale factor $k = \text{image length} \div \text{object length}$ ; setting up and solving proportions for a missing length; constructing and interpreting scale diagrams of triangles; explaining reasoning with evidence (NGSS-aligned practices: using mathematics and computational thinking, analysing and interpreting data, constructing explanations).                                                                                                                            |
| <b>Attitudes</b>            | Communication and Collaboration: discussing and agreeing on measurements in groups. Critical Thinking and Problem Solving: using similar triangles to work out heights that cannot be measured directly. Unity: sharing roles fairly when measuring. Responsibility: taking care of rulers, tape measures and protractors. Appreciating the use of similarity in day-to-day activities.                                                                                                                                                                                                                                                       |
| <b>Key Inquiry Question</b> | 1. Why do we need to enlarge or reduce objects? 2. How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Purpose in Storyline</b> | Lessons 3 and 4 showed that similar figures have equal corresponding angles and equal ratios of corresponding sides, and that a scale factor $k = \text{image} \div \text{object}$ gives missing lengths. Lesson 5 applies both ideas to a new situation: two triangles made by shadows, one small (Wanjiru) and one big (the tree). Learners see that size can change by a large factor ( $k = 6$ here, like $k = 1/4$ and $k = 5$ for the photos) while shape is kept, because the angles stay equal. This adds evidence that similarity is a useful, real-world tool and prepares learners for area and volume factors in Lessons 6 and 7. |
| <b>Safety Notes</b>         | Do the shadow measurement in the school compound in the daytime with the teacher supervising. Do not look directly at the sun. Do not climb the tree or any building. Learners keep to the marked area, carry rulers and tape measures carefully and do not run with them. Where weather or time does not allow going outside, use the pre-measured data on the board and a scale drawing on grid paper.                                                                                                                                                                                                                                      |

### B. LESSON OVERVIEW

Mrs Achieng opens by recalling Wanjiru's passport photo, then asks learners to predict how tall the Nyalenda Secondary School tree is when Wanjiru (1.5 m tall) casts a 2 m shadow and the tree casts a 12 m shadow at the same time. Groups gather evidence by measuring a learner and shadow outdoors (or using the given data) and drawing both triangles to scale on grid paper, measuring their angles with a protractor. In the Explain phase learners show that the two triangles are similar, find  $k = 6$ , and compute the tree height of 9 m by two methods. They then add questions to the Driving Question Board and update the class model: the same rule that stops Wanjiru's face from being distorted (one scale factor in every direction, equal angles) also lets us find heights we cannot measure.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                        | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                      |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Solving similar triangles: same side plays different roles</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Corresponding Sides of Similar Triangles (PLIX)</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | Learners look at a sketch on the board: Wanjiru (1.5 m tall) standing beside the tree in the Nyalenda Secondary School compound. Her shadow is 2 m and the tree's shadow is 12 m at the same time. Individually they write on a scrap of paper: (a) a prediction of the tree's height, (b) one sentence on why they think so. They then share with a partner and decide whether the tree is more or less than 12 m tall. Link to the phenomenon: they recall that the passport photo is a reduction ( $k = 1/4$ ) of the original that keeps Wanjiru's shape, and they predict whether the same idea applies to her shadow and the tree's.                                                                                                                                         | Draw the sketch and label only 1.5 m, 2 m and 12 m, with the tree height as '?'. Say: 'Yesterday we found missing lengths using a scale factor. Today Wanjiru is standing next to a tree. Nobody is climbing it. How tall is it?' Give 1 minute of silent writing. Then say: 'Turn to your partner and explain your number.' Circulate and note different strategies (adding 10 m, multiplying by 6, guessing). WAIT TIME: after asking 'Is the tree taller or shorter than its shadow? How do you know?' wait 10-15 seconds before taking answers. Cold-call 4 learners and record all predictions on the board without saying which is correct. Ask: 'What would have to be true about the sun, Wanjiru and the tree for your method to work?'                             | Predict-Share-Compare (Think-Pair-Share with written predictions). Writing a prediction and a reason before any instruction makes learners' existing ideas (additive versus multiplicative thinking) visible. Comparing them creates a genuine need for evidence in the Observe phase.                                                                                                                                      | Look for: learners who use an additive strategy ( $12 - 2 = 10$ , so $1.5 + 10 = 11.5$ m) versus a multiplicative strategy (about 6 times bigger, so 9 m). Tally on the board how many predict above 12 m, between 5 and 12 m, or below 5 m. Note whether anyone mentions that the sun's angle is the same for both. Use the tally to decide how much to emphasise additive versus multiplicative reasoning later. |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Solving similar triangles</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Use similar triangles</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a>                 | Part A (outdoors, 12 minutes): In groups of four, learners go to a marked area in the school compound. One learner stands upright while the others measure the learner's height and shadow length with a tape measure or ruler and record them. Where possible they also measure the shadow of a tall object such as a flagpole, goalpost or the Nyalenda tree, at the same time. Part B (in class, 10 minutes): Each group draws the two triangles from the data on grid paper using scale 1 cm : 0.5 m: Wanjiru's triangle (vertical side 1.5 m, base 2 m, right angle at the ground) and the tree's triangle (base 12 m, vertical side to be found). They draw the line from the top of the object to the end of the shadow, measure the angle at the end of each shadow with a | Before going out, say: 'Take the measurements at the same time, because the sun moves and the shadows change.' Assign roles: measurer, holder, recorder, reporter. Say: 'Measure to the nearest 0.5 cm and check each other's reading.' Circulate with the questions: 'Which side is the shadow? Where is the right angle? What is the angle between the sun's ray and the ground?' Back indoors, give each group a protractor and say: 'Measure the angle at the end of each shadow. What do you notice?' (Both should be close to $37^\circ$ , since $\tan 37^\circ \approx 0.75$ , but learners just compare them.) Cold-call 3 groups to read out their angle measurements. WAIT TIME: after 'What do you notice about the two angles?' wait 10-15 seconds before taking | Data Table and Pattern Noticing: learners record measurements in a table with columns for object height, shadow length, height $\div$ shadow and angle of the sun's ray. Seeing that the ratio height : shadow is about the same for different objects (0.75 for Wanjiru) and that the angles are equal helps them notice that the triangles have the same shape even though they differ in size, as with the three prints. | Observe: Do learners label the right angle, shadow and height correctly? Do they measure at the same time? Are the measured angles within about $2^\circ$ of each other? Check the table for height $\div$ shadow ratios that agree (Wanjiru $1.5 \div 2 = 0.75$ ). Ask a group: 'What would you expect the ratio to be for the tree?' Note learners who say 0.75 without calculation.                             |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>protractor and record both angles in a table with the height : shadow ratio. If they do not go outside, they use the given data: Wanjiru 1.5 m with 2 m shadow, tree with 12 m shadow, and draw Wanjiru's triangle to scale and the tree's triangle with the vertical side extended until the sun's ray reaches the end of the shadow.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>responses. Remind learners to take care of rulers and protractors.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Solving similar triangles - dubbed(KY)</a><br/> Source: Khan Academy (Kiswahili)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Corresponding Sides of Similar Triangles (PLIX)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners use their evidence to explain why the tree's height can be found from Wanjiru's. Step 1: In groups they list the corresponding angles of the two triangles: the right angle at the ground (90°), the angle at the end of the shadow (equal, because the sun's rays are parallel and strike the ground at the same angle) and the angle at the top (equal because angles in a triangle add up to 180°). Step 2: They state that the triangles are similar (Wanjiru's triangle ~ tree's triangle). Step 3: They find the linear scale factor with Wanjiru's triangle as the object and the tree's triangle as the image: <math>k = 12 \div 2 = 6</math>. Step 4: They find the tree's height: <math>1.5 \times 6 = 9</math> m. Step 5: They check by a second method using the ratio of height to shadow: <math>1.5 \div 2 = 0.75</math> and <math>0.75 \times 12 = 9</math> m. They compare 9 m with their predictions and revise their reasoning. Step 6: In pairs they apply the method to a new problem: 'A school building at Nyalenda casts a shadow of 20 m when Wanjiru's shadow is 2 m. How tall is the building?' (<math>k = 10</math>, so height 15 m). They also explain why the additive method (adding the extra 10 m of shadow to her height) fails, by linking to the distorted print: changing lengths by different amounts does not keep</p> | <p>Say: 'Compare Wanjiru's triangle and the tree's triangle. Which angles are equal, and how do you know?' Give groups 3 minutes. WAIT TIME: after asking 'Why must the angle at the end of the shadow be the same for both?' wait 10-15 seconds. Cold-call 3 groups for angles and 3 learners for the scale factor. Write on the board: 'Object: Wanjiru, shadow 2 m. Image: tree, shadow 12 m. <math>k = \text{image} \div \text{object} = 12 \div 2 = 6</math>.' Say: 'Always say which one is the object and which is the image.' Ask: 'The tree's shadow is 6 times longer. What must be true about its height?' Have learners work out 9 m, then ask for the check using 0.75. Address misconceptions with: 'If we add 10 m to her height, would the angle stay the same? Draw it.' Hand out the building problem and cold-call 2 pairs to explain their set-up (not only the answer). Ask: 'Which earlier print showed us what happens when two lengths change by different factors?' (the <math>4.5 \times 4.5</math> cm distorted print, with factors 9/28 and 1/4).</p> | <p>Claim-Evidence-Reasoning (CER): learners state a claim (the tree is 9 m tall), give evidence (the measurements and equal angles) and reasoning (similar triangles have proportional corresponding sides with one scale factor, <math>k = 6</math>). Comparing with the distorted print makes clear why equal factors in all directions matter. Using a second method (ratio 0.75) strengthens understanding that scale factor and ratio of sides are two views of the same idea.</p> | <p>Check in learners' books or on mini-whiteboards: (1) correct identification of corresponding angles; (2) correct statement <math>k = 12 \div 2 = 6</math> with object and image named; (3) tree height 9 m with units; (4) the second method giving <math>0.75 \times 12 = 9</math>. For the building problem, look for <math>k = 20 \div 2 = 10</math> and <math>1.5 \times 10 = 15</math> m. Indicators of difficulty: dividing the wrong way (<math>2 \div 12</math>), adding instead of multiplying, or giving no units. Ask those learners: 'Is the image bigger or smaller than the object, and should <math>k</math> be more or less than 1?'</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | the shape. Link to the phenomenon: just as the passport photo is similar to the original with $k = 1/4$ , the tree triangle is similar to Wanjiru's triangle with $k = 6$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Solving similar triangles: same side plays different roles</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Use similar triangles</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Learners return to the class Driving Question Board posed by Mrs Achieng: 'How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?' In groups they write on sticky notes (a) what the shadow activity showed that helps answer the driving question, and (b) a new question they now have. Example answers: 'The shape stays the same when all lengths are multiplied by the same $k$ and the angles stay equal'; 'If lengths change by different factors, the shape changes, like the $4.5 \times 4.5$ print'. Example new questions: 'If the tree's shadow is 6 times longer, is the area of its triangle also 6 times bigger?'; 'How do we enlarge a figure from a centre point?'; 'Can we use the same method to find the width of Lake Victoria at a narrow point?'. They place the notes under headings 'What we now know' and 'What we still wonder'. | Say: 'Write one thing today's shadows told us about Wanjiru's photo, and one question we cannot answer yet.' Give 3 minutes. Cold-call 4 groups to read and place a note, asking: 'Where on the board does it belong, and why?' WAIT TIME: after reading a question such as 'Is the area also 6 times bigger?', wait 10-15 seconds and ask: 'What do you predict, and what could we do to test it?' Do not answer; say: 'We will come back to this in Lesson 6.' Group similar notes and star questions leading to the next lessons (area, volume, enlargement from a centre). | Driving Question Board (question tracking): returning to the DQB makes the link between today's evidence and the phenomenon explicit, and turning unanswered questions into the purpose of coming lessons keeps learners' curiosity driving the unit.                                                                                        | Check that notes link the shadow evidence to the photo (equal angles, one scale factor, shape retained) rather than only restating the calculation. Count how many groups raise a question about area or volume. Look for a learner who connects shadows to the distorted print (different factors, different shape). Use these to judge readiness for Lesson 6.                                                                                                               |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">Solving similar triangles</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Dilations: scale factor</a>                                                                                                                                                                                                                                   | Learners revise the class model that explains what must stay the same and what may change when a picture is resized. In groups they update the model drawn in earlier lessons by adding a new panel: 'Shadows and similar triangles'. The panel shows two triangles (Wanjiru: 1.5 m and 2 m; tree: 9 m and 12 m) with equal marked angles and the label ' $k = 6$ '. Beside it they redraw the passport photo ( $3.5 \times 4.5$ cm, $k = 1/4$ from the original $14 \times 18$ cm) and the                                                                                                                                                                                                                                                                                                                                                                                                                      | Say: 'Add today's evidence to our model. What is the same in the shadow triangles and the passport photo?' Give groups 5 minutes. Circulate and ask: 'Where is the scale factor in your drawing? Where are the equal angles?' Cold-call 3 groups to explain their panels. WAIT TIME: after asking 'How does the tree triangle help explain why the passport photo is not distorted?' wait 10-15 seconds, then take 2 answers. Correct vague statements by asking for                                                                                                           | Gotta-Have-It Model Revision: learners add a new component to a cumulative model and justify it with evidence, linking a new context (shadows) to the original phenomenon (the passport photo). Explaining the same rule in two different contexts helps learners transfer the idea of similarity rather than memorising a shadow procedure. | Indicators of success: the revised model shows equal marked angles and one scale factor for both triangles; learners write $k = 6$ with object and image identified; the explanation says that the distorted print used two different factors ( $9/28$ and $1/4$ ). On the exit question look for $k = 16 \div 2 = 8$ and height $1.5 \times 8 = 12$ m with units. Sort exit answers into 'secure', 'nearly' and 'needs support' to plan re-teaching at the start of Lesson 6. |

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| <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>distorted print (4.5 × 4.5 cm, factors 9/28 and 1/4). They add the rule in their own words: 'Size can change by any scale factor, but the shape is kept only when equal angles and one scale factor apply in every direction.' Each learner completes the summary table at the end of the lesson.</p> | <p>numbers: 'Which factor? Which angle?' Ask learners to complete the summary table individually. End with the exit question: 'A flagpole at Nyalenda casts a 16 m shadow when Wanjiru casts 2 m. How high is it?' (k = 8, so 12 m.) Collect the answers.</p> |  |  |
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### D. TEACHER REFLECTION

Which learners used an additive strategy at the start, and did the angle measurements and the distorted print convince them to change to multiplicative reasoning? Could every group explain why the angle at the end of the shadow is equal for Wanjiru and the tree? Was the outdoor measurement manageable in time and safe, and did shadows change during the activity? Which learners named object and image correctly when finding k? What new questions on the DQB (area, volume, enlargement from a centre) should shape the opening of Lesson 6?

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | What did we observe? For example: Wanjiru (1.5 m tall) casts a 2 m shadow and the tree casts a 12 m shadow at the same time; the angles at the end of both shadows measured about the same; height ÷ shadow was 0.75 for Wanjiru.                                                                                                                                                                |
| <b>What did I learn?</b>                     | What did we learn? For example: the two shadow triangles are similar (equal corresponding angles, proportional corresponding sides); with Wanjiru's triangle as the object and the tree's as the image, $k = 12 \div 2 = 6$ , so the tree is $1.5 \times 6 = 9$ m tall.                                                                                                                          |
| <b>How does this explain the phenomenon?</b> | How does this help explain the phenomenon? For example: just as the passport photo ( $k = 1/4$ ) and poster ( $k = 5$ ) are similar to the original because every length is multiplied by one scale factor and the angles stay equal, the tree's triangle keeps Wanjiru's triangle's shape; the distorted $4.5 \times 4.5$ cm print used different factors (9/28 and 1/4), so its shape changed. |

## LESSON 6 (40 minutes (one lesson; extend to 80 minutes if grid-paper cutting takes longer)): Area Scale Factor: Why Doubling Lengths Does Not Double Area

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Purpose</b>              | Learners discover that when every length of a figure is multiplied by the linear scale factor $k$ , the area is multiplied by $k^2$ , using grid paper, cut-outs and Wanjiru's photo prints, so they can explain how much paper and ink Mr Otieno needs for a passport photo and a poster.                                                                                                                                                                                                                                                                                     |
| <b>Knowledge</b>            | Learners know that if lengths are multiplied by $k$ ( $k = \text{image length} \div \text{object length}$ ), the area of the image is $k^2$ times the area of the object. They know the area factors for Wanjiru's prints: original ( $14 \times 18$ , $252 \text{ cm}^2$ ) to passport ( $3.5 \times 4.5$ , $15.75 \text{ cm}^2$ ) is $1/16$ ( $k = 1/4$ ); original to poster ( $70 \times 90$ , $6300 \text{ cm}^2$ ) is $25$ ( $k = 5$ ); passport to poster is $400$ ( $k = 20$ ).                                                                                        |
| <b>Skills</b>               | Learners (a) calculate areas of rectangles from measured sides, (b) calculate the linear scale factor $k$ from corresponding sides, (c) compare areas by division to get the area factor, (d) record results in a table and find the pattern $k \rightarrow k^2$ , (e) use the pattern to predict the area of an enlargement or reduction. NGSS Science and Engineering Practices: Analyzing and Interpreting Data, Using Mathematics and Computational Thinking, Constructing Explanations.                                                                                   |
| <b>Attitudes</b>            | Learners show curiosity about why area grows faster than length, cooperate in groups when cutting and tiling grid paper, take responsibility for accuracy in measuring, and appreciate the use of similarity and enlargement in day-to-day activities such as printing photographs and making posters. Core competencies: Critical Thinking and Problem Solving; Communication and Collaboration. Values: Unity and Responsibility.                                                                                                                                            |
| <b>Key Inquiry Question</b> | Why do we need to enlarge or reduce objects? How do we use similarity and enlargement in day-to-day activities? (Focus today: How much more or less paper does an enlarged or reduced photo need?)                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Purpose in Storyline</b> | Lessons 3 and 4 showed that similar prints keep equal ratios (7:9) and equal angles, and that every length is multiplied by the same linear scale factor $k$ ( $1/4$ , $5$ , $20$ ). Lesson 6 adds the next piece of evidence for the Driving Question: size can change a lot while shape is kept, and while lengths change by $k$ , the area changes by $k^2$ . This explains why the poster ( $k = 5$ ) needs 25 times the paper of the original and why the passport photo ( $k = 1/4$ ) is only $1/16$ of the original's area. Lesson 7 will extend the pattern to volume. |
| <b>Safety Notes</b>         | Scissors: cut away from the body and keep them closed when walking. Collect all grid-paper offcuts in a box so nobody slips on them. Rulers must not be used as swords. No electricity or internet is needed.                                                                                                                                                                                                                                                                                                                                                                  |

### B. LESSON OVERVIEW

Mrs Achieng returns to the Driving Question Board: 'Mr Otieno shrinks Wanjiru's  $14 \times 18 \text{ cm}$  print to  $3.5 \times 4.5 \text{ cm}$ . Every length becomes  $1/4$  as long. Does the paper also become  $1/4$  as big?' Learners predict, then investigate with grid paper in groups. Phase 1: learners predict the area of the passport photo and the poster. Phase 2: groups draw a  $2 \times 3$  unit rectangle on grid paper, enlarge every length by  $k = 2$  and  $k = 3$ , and count unit squares to see that areas become 4 and 9 times as large. They then check the same pattern on Wanjiru's prints by measuring and calculating areas:  $252$ ,  $15.75$  and  $6300 \text{ cm}^2$ . Phase 3: learners explain the pattern  $k \rightarrow k^2$  using tiling, and apply it to find the area of an unseen print. Phase 4: the class adds to the Driving Question Board. Phase 5: learners revise their cumulative model to include 'area changes by  $k^2$ '. The lesson ends with a summary table.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Scale factors and area</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Dilations: scale factor</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar readings | Learners look at the three prints on the board or on paper cut-outs: the original $14 \times 18$ cm, the passport $3.5 \times 4.5$ cm and the poster $70 \times 90$ cm. Individually in their notebooks (2 minutes) they answer: 'Every length of the passport photo is $\frac{1}{4}$ of the original. Predict: is the paper area of the passport photo $\frac{1}{4}$ , $\frac{1}{8}$ , $\frac{1}{16}$ or $\frac{1}{2}$ of the original? And the poster is 5 times as long and wide: is its area 5 times, 10 times, 25 times or 50 times the original?' They write a reason and a confidence rating (1 to 5). Then they discuss with a partner for 2 minutes and mark their prediction on a class tally on the board.                                                                                                                                                                                                                                                                                                               | Say: 'Remember Wanjiru's photo from Lakeview Photo Studio in Kisumu. In Lesson 4 we found $k = \frac{1}{4}$ for the passport photo and $k = 5$ for the poster. Today we ask a new question: what happens to the paper?' Hold up the cut-outs. Ask: 'Mr Otieno uses paper and ink. Who thinks the poster needs 5 times the paper?' Give 3 minutes for writing, then WAIT TIME 10-15 seconds after asking for reasons before taking answers. Cold-call 4 learners (two who chose 5 times and two who chose 25 times) with 'Tell us why you chose that.' Do NOT say which is correct. Record the tally on the board as a bar chart. Say: 'We will test every prediction with evidence.'                                                                                                                                                                  | Predict-Observe-Explain with a class tally: committing to a prediction exposes the common misconception that area scales the same way as length (doubling length doubles area). The tally makes a cognitive conflict visible that learners will want to resolve with evidence in the Observe Phase.                                                                                                                                                                                 | Look for: (1) learners writing a specific number as their prediction and giving a reason; (2) how many choose 'k times' (misconception) versus ' $k \times k$ '; (3) learners who mention 'length and width both change' in their reasoning. Note the group tally results. Do not correct anyone yet.                                                                                                                                                                                                                                                                             |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Scale factors and area</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar videos<br><br> <b>EXERCISE:</b><br><a href="#">Scale drawings</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES</a> for similar readings      | Groups of 4 receive grid paper (1 cm squares), a ruler, scissors and a worksheet. Part A (about 12 minutes): Each group draws a rectangle of $2 \text{ cm} \times 3 \text{ cm}$ on grid paper and counts its area in unit squares (6). They then draw an enlargement with $k = 2$ ( $4 \text{ cm} \times 6 \text{ cm}$ ) and another with $k = 3$ ( $6 \text{ cm} \times 9 \text{ cm}$ ), cut them out, and cover each with unit squares or count the grid squares. They fill a table: k, new width, new height, area, area $\div 6$ . Results: $k = 2$ gives area 24 and factor 4; $k = 3$ gives area 54 and factor 9. They also try a reduction by cutting a $4 \times 6$ rectangle in half in both directions to see that a rectangle with $k = \frac{1}{2}$ holds $\frac{1}{4}$ of the squares. Part B (about 10 minutes): Learners use the prints data: original $14 \times 18 = 252 \text{ cm}^2$ ; passport $3.5 \times 4.5 = 15.75 \text{ cm}^2$ ; poster $70 \times 90 = 6300 \text{ cm}^2$ . For each pair they calculate | Distribute materials and say: 'Be careful with scissors, cut away from your body. Each of you has a role: measurer, cutter, recorder, reporter.' Circulate and ask: 'How many unit squares fit along the width now? Along the height? So how many squares in total?' If a group says the area of the $k = 2$ rectangle is 12, ask: 'Count the squares again, row by row. How many rows? How many in each row?' For Part B, say: 'Use the sizes from Mr Otieno's printing: 14 by 18, 3.5 by 4.5 and 70 by 90.' Check that learners state which print is the object and which is the image. After each part, WAIT TIME 10-15 seconds before asking: 'What do you notice when you compare k with the area factor?' Cold-call 3 groups to report one row of their table each. Put the class table on the board and ask: 'Is the area factor ever the same | Data table with pattern-finding: learners collect paired values of k and area factor from concrete grid cut-outs and from real print data, then compare them side by side. Seeing $2 \rightarrow 4$ , $3 \rightarrow 9$ , $\frac{1}{4} \rightarrow \frac{1}{16}$ , $5 \rightarrow 25$ and $20 \rightarrow 400$ lets them generalise the rule themselves instead of memorising it. Using both small whole numbers and Wanjiru's prints connects the abstract rule to the phenomenon. | Observable indicators: (1) learners count or calculate areas correctly ( $24$ and $54 \text{ cm}^2$ for $k = 2$ and $k = 3$ ); (2) learners state object and image when computing k ( $k = \text{image} \div \text{object}$ ); (3) correct table entries 4, 9, $\frac{1}{16}$ , 25, 400; (4) learners spontaneously say 'k times k' or 'squared'. Check in each group that the recorder's table matches the measurements. Note any group that divides the wrong way ( $252 \div 15.75$ gives 16, which is the inverse factor) and ask them to re-state which print is the object. |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>the linear scale factor <math>k</math> and the area factor (area of image <math>\div</math> area of object): original to passport <math>k = 1/4</math>, area factor <math>15.75 \div 252 = 1/16</math>; original to poster <math>k = 5</math>, area factor <math>6300 \div 252 = 25</math>; passport to poster <math>k = 20</math>, area factor <math>6300 \div 15.75 = 400</math>. They add these rows to their table and compare area factor with <math>k</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>number as <math>k</math>? What is the connection?' Do not give <math>k^2</math> yet; let learners say 'k times k'.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <p><b>Explain Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Dilations: scale factor</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Corresponding sides and points</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/>           Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar readings</p> | <p>Learners return to their predictions and revise them in a different colour. In the same groups they write a claim-evidence-reasoning explanation (CER): Claim: 'When every length is multiplied by <math>k</math>, the area is multiplied by ____.' Evidence: two rows from their table (for example <math>k = 3</math> gives 9 and <math>k = 1/4</math> gives <math>1/16</math>). Reasoning: 'Length is multiplied by <math>k</math> and width is multiplied by <math>k</math>, so area = <math>(k \times \text{width}) \times (k \times \text{height}) = k \times k \times \text{width} \times \text{height} = k^2 \times \text{the old area}</math>.' They use the <math>2 \times 3</math> grid rectangle drawing to show it, with <math>k = 2</math> showing 2 rows of 2 blocks. Then they apply the rule: 'Mr Otieno wants a second poster with <math>k = 10</math> from the original. What is its area?' (<math>10^2 \times 252 = 25\,200 \text{ cm}^2</math>) and 'A passport print is <math>3.5 \times 4.5 \text{ cm}</math>. What area does <math>k = 20</math> give?' (<math>400 \times 15.75 = 6300 \text{ cm}^2</math>). They discuss why the answer is NOT 5 times for the poster and finally explain why the distorted <math>4.5 \times 4.5 \text{ cm}</math> print (area <math>20.25 \text{ cm}^2</math>) cannot be found with one <math>k^2</math> rule because its width factor <math>9/28</math> and height factor <math>1/4</math> are different.</p> | <p>Say: 'Take out the predictions you made at the start. Use a different colour pen to fix anything that evidence changed.' Ask: 'Why does the poster need 25 times the paper and not 5 times?' WAIT TIME 10-15 seconds. Cold-call 3 learners. Draw a <math>2 \times 3</math> rectangle on the board and show <math>k = 2</math> with the rows and columns: 'Two times as many rows and two times as many in each row. So <math>2 \times 2 = 4</math> times the squares.' Ask: 'How would you say that for any <math>k</math>?' Cold-call 2 more learners to state '<math>k^2</math>'. Then give the application problems and ask groups to write answers on mini-whiteboards or paper to hold up. Finally ask: 'The distorted <math>4.5 \times 4.5</math> print has width factor <math>9/28</math> and height factor <math>1/4</math>. Can we use <math>k^2</math> for it? Why not?' Use WAIT TIME 10-15 seconds, then cold-call 2 learners. Reinforce: 'Area factor <math>k^2</math> works because the same <math>k</math> is used in every direction. That is exactly what similarity means.' Do not re-teach scale factor calculation; refer to Lesson 4 only for <math>k = \text{image} \div \text{object}</math>.</p> | <p>Claim-Evidence-Reasoning (CER) with revision of predictions: learners must justify the rule <math>k^2</math> with their own data and an algebraic/geometric argument (rows <math>\times</math> columns), which turns pattern recognition into explanation. Linking back to the distorted print shows the limits of the rule and ties it to the Driving Question (one scale factor in all directions).</p> | <p>Look for: (1) CER statements that name <math>k^2</math> with two correct pieces of evidence; (2) correct application values <math>25\,200 \text{ cm}^2</math> and <math>6300 \text{ cm}^2</math>; (3) the reasoning 'both length and width are multiplied by <math>k</math>'; (4) learners recognising that the distorted print has two different factors so <math>k^2</math> does not apply. Use thumbs-up/sideways/down after the explanation: 'How confident are you in explaining <math>k^2</math> to a friend?' Pull aside learners who still say 'area is multiplied by <math>k</math>' for a quick re-count with the grid cut-outs.</p> |
| <p><b>Driving Question Board (DQB) Creation</b><br/>  <b>VIDEO:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>Learners gather around the Driving Question Board with Mrs Achieng's question: 'How can a full-size photograph of a person be resized to fit a small passport photo box</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <p>Say: 'Our Driving Question is still on the board. What did today add?' Give each group 2 minutes to write a note, then WAIT TIME 10-15 seconds before sharing. Cold-call</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Driving Question Board as a public record of growing understanding: moving learner questions into answered, partly answered and future categories shows how each</p>                                                                                                                                                                                                                                      | <p>Indicators: (1) questions that build on <math>k^2</math> (for example about volume, triangles or cost) rather than repeating earlier ones; (2) learners placing notes in the right category;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                               |

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| <p><a href="#">Side lengths after dilation</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Dilations: scale factor</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar readings</p>                               | <p>without distortion?' Each group writes one new question or finding on a sticky note or strip of paper, for example: 'Why does area grow faster than length?', 'Does the area rule work for triangles?', 'How much would a poster with <math>k = 5</math> cost compared with the original?', 'Why does the distorted print not follow <math>k^2</math>?'. They place their notes under headings: 'What we found' and 'What we still wonder'. They add the finding card: 'Lengths <math>\times k \rightarrow</math> area <math>\times k^2</math> (1/16, 25, 400 for Wanjiru's prints).'</p>                                                                                                                                                                                                                                                                                                                                                           | <p>4 groups to read a note and place it. Ask: 'Which of today's findings helps explain why the passport photo is not distorted? Which one helps explain why it costs different amounts of paper?' Tell learners that tomorrow we will see whether the same pattern happens in three dimensions, to prepare for Lesson 7. Sort questions: 'This is about area (answered), this is about volume (coming), this is about centres of enlargement (coming later).'</p>                                                                                                                                                                                                                                                                    | <p>lesson's evidence contributes to the Driving Question and keeps learners invested in the sequence.</p>                                                                                                                                                                                                     | <p>(3) learners referring to evidence ('we found 25 for the poster'). Record which groups only ask surface questions and prompt them with: 'What would you like to know about a box instead of a flat print?'</p>                                                                                                                                                                                                                                |
| <p><b>Model Building Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Scale drawings</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Scale drawings</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar readings</p> | <p>Learners open their cumulative model page (started in Lesson 1 and revised in Lessons 3 to 5). Individually, for 8 minutes, they add a new section titled 'Area' to their model of how Wanjiru's photo can be resized without distortion. They draw the original (<math>14 \times 18</math>), passport (<math>3.5 \times 4.5</math>) and poster (<math>70 \times 90</math>) with labels: <math>k = 1/4</math>, 5, 20; area factors 1/16, 25, 400; and areas 252, 15.75, 6300 <math>\text{cm}^2</math>. They add the rule: 'Same <math>k</math> in every direction <math>\rightarrow</math> ratios 7:9 and angles equal <math>\rightarrow</math> area <math>\times k^2</math>.' They also add a note on the distorted print: 'Different factors 9/28 and 1/4, so ratio becomes 1:1 and the chin angle changes; <math>k^2</math> rule does not apply.' Finally they complete the summary table (Observed, Learned, Explained) in their notebooks.</p> | <p>Say: 'Open your model page. Today's new piece is area. Add it in a different colour so we can see how your thinking is growing.' Walk around and ask: 'Where does <math>k^2</math> appear in your model? How does it connect to the ratio 7:9 and the angles we found earlier?' WAIT TIME 10-15 seconds before each response. Cold-call 3 learners to share their revised model: one on the passport (reduction), one on the poster (enlargement), one on the distorted print. Ask: 'Why does a reduction by <math>k = 1/4</math> make the area 1/16?' Close with an exit question: 'If Mr Otieno doubled every length of the passport photo, what would happen to its area?' (4 times). Collect exit answers on paper slips.</p> | <p>Cumulative model revision: learners integrate the new area idea with earlier ideas (ratios, angles, scale factor) in one visual model, which strengthens connections and lets them show how the explanation for the phenomenon has become more complete. Colour-coded additions make progress visible.</p> | <p>Observable indicators: (1) model includes <math>k</math>, <math>k^2</math> and the correct three area factors; (2) labelled diagram with correct values 252, 15.75, 6300; (3) the model links the rule to using the same <math>k</math> for all lengths; (4) correct exit answer 4 times. Sort exit slips into correct, 'doubles' misconception and blank; plan a short revisit at the start of Lesson 7 for the second and third groups.</p> |

## D. TEACHER REFLECTION

After the lesson ask yourself: (1) How many learners moved from 'area  $\times k$ ' to 'area  $\times k^2$ ', and what evidence convinced them (grid counts, table, or the rows-and-columns argument)? (2) Did any group divide the areas in the wrong direction ( $252 \div 15.75 = 16$  instead of  $15.75 \div 252 = 1/16$ )? How did I help them state object and image? (3) Were cold-calls evenly spread, and did the WAIT TIME produce deeper answers? (4) Did learners link  $k^2$  back to the Driving Question, and to the distorted print having two different factors? (5) Which learners need a quick revisit before volume in Lesson 7, and what concrete materials (small blocks or cubes of cut paper) will I prepare?

### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>What did I observe?</b>                   | We measured and counted unit squares for rectangles enlarged by $k = 2$ and $k = 3$ (areas 24 and 54 $\text{cm}^2$ from a $2 \times 3$ rectangle) and calculated the areas of Wanjiru's prints: original $14 \times 18 = 252 \text{ cm}^2$ , passport $3.5 \times 4.5 = 15.75 \text{ cm}^2$ , poster $70 \times 90 = 6300 \text{ cm}^2$ .                                                                                                  |
| <b>What did I learn?</b>                     | When every length is multiplied by the linear scale factor $k$ , the area is multiplied by $k^2$ . The area factors for Wanjiru's prints are $1/16$ (original to passport, $k = 1/4$ ), $25$ (original to poster, $k = 5$ ) and $400$ (passport to poster, $k = 20$ ).                                                                                                                                                                     |
| <b>How does this explain the phenomenon?</b> | Length and width are both multiplied by $k$ , so area = $(k \times \text{width}) \times (k \times \text{height}) = k^2 \times \text{old area}$ . Wanjiru's prints keep her likeness because one scale factor is used in every direction (similar figures), so area changes by $k^2$ . The distorted $4.5 \times 4.5 \text{ cm}$ print used different factors ( $9/28$ and $1/4$ ), so it is not similar and the $k^2$ rule does not apply. |

## LESSON 7 (40 minutes): Volume Scale Factor: From Small Cartons to Large Cartons ( $k^3$ )

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Purpose</b>              | Learners discover that when every length of a solid is multiplied by the scale factor $k$ , its volume is multiplied by $k^3$ , using Mr Otieno's small and large similar cartons for packing prints.                                                                                                                                                                                                                                                                          |
| <b>Knowledge</b>            | Similar solids have all corresponding edges in the same ratio $k$ . Volume of a cuboid = length $\times$ width $\times$ height. For the small carton $10 \times 6 \times 4 = 240 \text{ cm}^3$ and the large carton $20 \times 12 \times 8 = 1920 \text{ cm}^3$ ; $k = 2$ (large $\div$ small, object = small carton, image = large carton) and the volume factor is $1920 \div 240 = 8 = 2^3$ . Volume scale factor = $k^3$ , extending the area pattern $k^2$ from Lesson 6. |
| <b>Skills</b>               | Calculating volumes of cuboids; calculating $k = \text{image} \div \text{object}$ for edges; spotting patterns across area and volume factors; building a $2 \times 2 \times 2$ arrangement from unit cubes; using the $k^3$ rule to find an unknown volume; communicating reasoning (NGSS practices: using mathematics and computational thinking, analysing and interpreting data, constructing explanations).                                                               |
| <b>Attitudes</b>            | Curiosity about why volume grows so much faster than length; collaboration and unity in groups; responsibility in handling cubes and rulers; appreciation of similarity in everyday packaging.                                                                                                                                                                                                                                                                                 |
| <b>Key Inquiry Question</b> | 1. Why do we need to enlarge or reduce objects? 2. How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                                                                                                          |
| <b>Purpose in Storyline</b> | Lesson 4 gave us $k = \text{image} \div \text{object}$ and Lesson 6 showed that areas are multiplied by $k^2$ . This lesson adds the third dimension: volume is multiplied by $k^3$ . It completes the idea for the Driving Question that size can change by any scale factor, but shape is kept only if one $k$ is applied to all lengths in every direction, with area changing by $k^2$ and volume by $k^3$ . Lesson 11 will use $k$ , $k^2$ and $k^3$ in applications.     |
| <b>Safety Notes</b>         | Cubes and boxes must be kept on desks, not thrown. Scissors are not needed. Keep walkways clear of bags and boxes. Learners with sight difficulties should sit near the front and use larger-print worksheets.                                                                                                                                                                                                                                                                 |

### B. LESSON OVERVIEW

Mrs Achieng brings in two cartons that Mr Otieno of Lakeview Photo Studio in Kisumu uses to pack prints: a small carton of  $10 \times 6 \times 4 \text{ cm}$  and a large similar carton of  $20 \times 12 \times 8 \text{ cm}$ . Learners first predict how many small cartons would fill the large one, with many guessing 2 because every length doubled. They then gather evidence by calculating edge ratios and volumes ( $240 \text{ cm}^3$  and  $1920 \text{ cm}^3$ ) and building a  $2 \times 2 \times 2$  block from unit cubes to see why the factor is 8. In the Explain phase they compare the three factors ( $k$ ,  $k^2$ ,  $k^3$ ) and link the pattern to the passport photo (area  $1/16$  from  $k = 1/4$ ). The DQB and the model are updated: similarity keeps shape by using one  $k$  in every direction, and volume changes by  $k^3$ .

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                     | Learner Experience                                                                                                                                                                                                   | Teacher Moves                                                                                                                                                       | Sensemaking Strategy                                                                                                                                        | Formative Assessment Strategy                                                                                                                                    |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Area and Volume of Similar Solids Principles</a> | Learners sit in groups of four. Mrs Achieng holds up a small carton ( $10 \times 6 \times 4 \text{ cm}$ ) and a large carton ( $20 \times 12 \times 8 \text{ cm}$ ) that Mr Otieno uses to pack Wanjiru's prints and | Say: 'Mr Otieno packs Wanjiru's prints in this small carton. He also has a larger carton with every edge doubled. How many small cartons could fit inside the large | Predict-Observe-Explain (POE): Committing to a written prediction with a reason before any calculation exposes learners' intuition that volume doubles when | Observe and note: (1) How many learners predict 2 (linear thinking), 4 (area thinking), or 8 (volume thinking). (2) Whether reasons mention edges, area or space |

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| <p>Source: CK-12<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Area and Volume of Similar Solids (Flexbook)</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                       | <p>posters. Each learner silently writes a prediction: every length of the large carton is double the small one, so how many small cartons of prints would fit inside the large carton? They also write one sentence of reasoning and a confidence rating from 1 to 3. Groups then share predictions. Many will say 2 or 4 based on the earlier area idea. Learners link back to the phenomenon: when the passport photo was reduced with <math>k = 1/4</math>, the area did not change by <math>1/4</math>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>one? Write your number and your reason.' Show the cartons but do NOT reveal the dimensions of volume yet. WAIT TIME: 15 seconds of silent thinking before writing. Give 2 minutes for individual writing. Then ask groups to share. Cold-call 4 learners: 'What is your prediction and why?' Record all predictions on the board as a tally (2, 4, 6, 8, other) without saying which is right. Say: 'We will test these predictions with evidence. Keep your paper, we shall return to it.'</p>                                                                                                                                                                                                                                                                                        | <p>length doubles. This sets up cognitive conflict when the data show a different pattern, so learners revise their own ideas rather than just receiving a rule.</p>                                                                                                                                                                                                                       | <p>inside. (3) Whether learners connect to Lesson 4 by stating <math>k = 2</math> for the edges. Use the tally on the board as a baseline. Learners who write no reason are prompted individually by the teacher with 'What did you notice about the edges?'</p>                                                                                                                                                                             |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Area and Volume of Similar Solids Principles</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Area and Volume of Similar Solids Practice</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Learners collect evidence in a table. Step 1: measure the edges of both cartons with a ruler and confirm <math>10 \times 6 \times 4</math> cm and <math>20 \times 12 \times 8</math> cm. Step 2: calculate <math>k</math> for each pair of corresponding edges (object = small carton, image = large carton): <math>20 \div 10 = 2</math>, <math>12 \div 6 = 2</math>, <math>8 \div 4 = 2</math>, so all edges have the same <math>k = 2</math> and the cartons are similar (<math>\sim</math>). Step 3: calculate volumes: <math>10 \times 6 \times 4 = 240</math> cm<sup>3</sup> and <math>20 \times 12 \times 8 = 1920</math> cm<sup>3</sup>. Step 4: divide <math>1920 \div 240 = 8</math>. Step 5: using 8 unit cubes (1 cm edges) per group, build a <math>1 \times 1 \times 1</math> cube and then a <math>2 \times 2 \times 2</math> cube, counting that 8 small cubes fill the enlarged cube. Step 6: complete a table of <math>k = 1, 2, 3</math> for a unit cube: volumes 1, 8, 27 (<math>3 \times 3 \times 3</math>). Learners record the area factors (from Lesson 6: <math>k^2</math>) beside the volume factors.</p> | <p>Circulate with the question: 'Are all three pairs of edges in the same ratio? How do you know?' Ask groups to say which carton is the object and which is the image before dividing. Say: 'Calculate the volume of each carton and then divide the larger by the smaller.' WAIT TIME: 10 seconds after asking 'Is 8 the same as your prediction?' Distribute 8 cubes per group (and extra cubes for <math>k = 3</math> for fast groups, 27 total, if available; otherwise use drawings). Ask: 'How many layers does the <math>2 \times 2 \times 2</math> block have? How many cubes in each layer?' Cold-call 3 groups to report their volume factor. Challenge groups finished early: 'What would the volume factor be if <math>k = 3</math>? Test with your cubes or a drawing.'</p> | <p>Data table and pattern-finding: Putting <math>k</math>, area factor and volume factor side by side (<math>1, 2, 3 \rightarrow 1, 4, 9 \rightarrow 1, 8, 27</math>) lets learners see structure in data. The concrete cube build gives a physical reason (length <math>\times</math> width <math>\times</math> height each doubled) so the pattern is explained, not just memorised.</p> | <p>Look for: correct edge ratios all equal to 2; correct volumes 240 and 1920 cm<sup>3</sup>; the division <math>1920 \div 240 = 8</math>; the cube build showing 2 layers of 4 cubes. Indicators of struggle: using different <math>k</math> for different edges, or adding instead of multiplying. Ask a struggling learner to point to each edge doubling in the cube model. Tick a checklist for each group on completing the table.</p> |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Dilations: scale factor</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p>                                                                                                                                                                                                  | <p>Learners explain why volume is multiplied by 8 when <math>k = 2</math>. In groups they write: 'Length is multiplied by <math>k</math>, width by <math>k</math>, and height by <math>k</math>, so volume = <math>(k \times \text{length}) \times (k \times \text{width}) \times (k \times \text{height}) = k \times k \times k \times \text{original volume} = k^3 \times</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>Ask: 'Why 8 and not 2?' WAIT TIME: 15 seconds. Use a think-pair-share: 1 minute individual, 2 minutes in pairs, then cold-call 4 pairs. Press for reasoning: 'Which of the three edges is doubled? So how many times does the volume</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>Claim-Evidence-Reasoning (CER): Learners state a claim (volume factor = <math>k^3</math>), cite evidence (240, 1920, the cube build) and give reasoning (three dimensions each multiplied by <math>k</math>). This builds mathematical argumentation and</p>                                                                                                                            | <p>Listen for correct use of vocabulary: 'corresponding edges', 'scale factor <math>k</math>', 'cube of <math>k</math>'. Indicators of mastery: learners compute <math>240 \times 27 = 6480</math> and <math>240 \times 1/8 = 30</math> correctly, and explain with all three dimensions.</p>                                                                                                                                                |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>HTML:</b><br/> <a href="#">Area and Volume of Similar Solids (Flexbook)</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                        | <p>original volume.' They check with the cartons: <math>2^3 = 8</math>, and <math>240 \times 8 = 1920 \text{ cm}^3</math>. They then apply the rule: a similar carton with <math>k = 3</math> from the small carton would have volume <math>240 \times 27 = 6480 \text{ cm}^3</math> (check edges <math>30 \times 18 \times 12 = 6480</math>). They compare the patterns: <math>k</math> gives length, <math>k^2</math> gives area (Lesson 6), <math>k^3</math> gives volume. Linking to the phenomenon: for Wanjiru's passport photo <math>k = 1/4</math> gives area <math>1/16</math> of the original, and the same thinking shows why only one <math>k</math> applied in every direction keeps the shape. They revisit their predictions and rewrite them using the evidence.</p>                                                                                                                                 | <p>grow in each direction?' Say: 'Last lesson we saw <math>k^2</math> for area. What do you think is new here, and why?' Write the generalisation on the board: 'Volume factor = <math>k^3</math>'. Pose a transfer question: 'If Mr Otieno made a carton with <math>k = 1/2</math> from the small carton, what is its volume?' (Answer: <math>240 \times 1/8 = 30 \text{ cm}^3</math>; edges <math>5 \times 3 \times 2 = 30</math>.) Cold-call 3 learners. Address misconception: 'Doubling the edges does NOT double the volume.' Link back: 'In our photo, one <math>k</math> for all lengths kept her face the same shape. Is that true for the cartons? What changed, and what stayed the same?'</p> | <p>ties the rule to why shape and size behave differently.</p>                                                                                                                                                                                           | <p>Misconception to flag: writing volume factor = <math>2k</math> or <math>k^2</math>. Use a quick whiteboard check: 'Volume factor when <math>k = 3</math>?' (27) and show answers at the same time. Record the number of correct answers.</p>                                                                                                                                               |
| <p><b>Driving Question Board (DQB) Creation</b></p> <p><b>VIDEO:</b><br/> <a href="#">Volume of triangular prism &amp; cube</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Area and Volume of Similar Solids Practice</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to the Driving Question Board posed by Mrs Achieng: 'How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?' Each group writes one new sticky note: either an answer they now have (for example 'Volume changes by <math>k^3</math>, area by <math>k^2</math>, length by <math>k</math>') or a new question (for example 'Does a flat photo have volume?' 'Do all shapes follow <math>k^3</math>?' 'How do we use <math>k^3</math> for real containers in Kisumu such as water tanks?'). They place their notes on the board and group them under headings such as 'What stays the same', 'What changes', and 'Still wondering'. They also note which of the KICD key inquiry questions they can now answer: why we enlarge or reduce objects (packing, storage, models) and how similarity is used in daily life (cartons, containers).</p> | <p>Say: 'Write one thing this lesson added to our answer to the Driving Question, or one new question.' Give 3 minutes. WAIT TIME: 10 seconds before the first group places a note. Cold-call 3 groups to read their notes aloud and say where it belongs on the board. Ask: 'Does this note answer an old question? Which one?' Move answered questions to the 'Answered' column. Highlight a question about real containers, such as a small and large jerrycan or a mandazi box, and say: 'We shall use <math>k</math>, <math>k^2</math> and <math>k^3</math> in Lesson 11.' Photograph or copy the board for the next lesson.</p>                                                                     | <p>Driving Question Board as a public record of learning: Placing and sorting questions makes learners track how their understanding grows lesson by lesson and shows that new questions arise from evidence. It keeps the phenomenon at the centre.</p> | <p>Check the quality of sticky notes: do they mention <math>k^3</math> with a reason, or only copy a number? Count the notes that connect volume to the photo phenomenon. Look for learners who ask extension questions (a sign of deeper thinking). Note any group that records a misconception such as 'volume doubles when <math>k = 2</math>' for follow-up at the start of Lesson 8.</p> |
| <p><b>Model Building Phase</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                | <p>Learners revise the class model that explains Wanjiru's photo. In groups, they update a poster with</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>Say: 'Add to our model: what happens to length, area and volume when we multiply every</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Cumulative model revision: Extending the class model from ratios, angles and <math>k</math> (earlier</p>                                                                                                                                              | <p>Check posters for: three correct rows (<math>1/4 \rightarrow 1/16</math>; <math>5 \rightarrow 25</math>; <math>2 \rightarrow 8</math>), the statement about one <math>k</math> in every</p>                                                                                                                                                                                                |

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| <p> <b>VIDEO:</b><br/> <a href="#">Dimensions of a cube from its volume</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Dilations: scale factor</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>three columns: 'Length (k)', 'Area (k<sup>2</sup>)', 'Volume (k<sup>3</sup>)'. Using the photo data they fill in: original to passport k = 1/4 and area factor 1/16 (252 to 15.75 cm<sup>2</sup>), original to poster k = 5 and area factor 25 (6300 cm<sup>2</sup>), and cartons k = 2 and volume factor 8 (240 to 1920 cm<sup>3</sup>). They add a sentence: 'Shape is kept only if the same k is applied to all lengths in every direction; then area changes by k<sup>2</sup> and volume by k<sup>3</sup>.' They sketch the two cartons and the 2 × 2 × 2 cube beside the photo diagrams, and mark the distorted 4.5 × 4.5 cm print as 'width factor 9/28 and height factor 1/4: not equal, so shape changes'. Each learner then completes a short exit task: a similar carton has k = 3 from the small carton; give its edges and its volume (30 × 18 × 12 = 6480 cm<sup>3</sup>).</p> | <p>length by k?' Give 6 minutes. Circulate and ask: 'Where on your model do you see the same k in every direction? Where is it different for the distorted print?' WAIT TIME: 10 seconds after each question. Cold-call 3 groups to explain one column of their model. Then say: 'Show me with your fingers: what is the volume factor when k = 2?' (8) Collect exit task answers. Preview Lesson 8: 'Next we shall construct enlargements from a centre using rays.' Remind learners to bring rulers and grid paper.</p> | <p>lessons) to include k<sup>2</sup> and k<sup>3</sup> helps learners combine separate ideas into one coherent explanation of the phenomenon. Comparing the similar prints and cartons with the distorted print makes the necessary condition (one k in every direction) explicit.</p> | <p>direction, and the distorted print marked with unequal factors. Exit task success criteria: edges 30 × 18 × 12 cm, volume 6480 cm<sup>3</sup> (or 240 × 27). Learners with fewer than 2 of 3 correct get a short re-teach with cubes at the start of Lesson 8.</p> |
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## D. TEACHER REFLECTION

Which learners moved from the linear prediction (2) to the k<sup>3</sup> reasoning, and what evidence convinced them: the numbers or the cube build? Did any group confuse k<sup>2</sup>, and k<sup>3</sup> or apply k to only one edge, just as the distorted print used unequal factors? Were the WAIT TIMES of 10 to 15 seconds enough for quieter learners to contribute? What would I change about the cube materials if some groups lacked enough cubes (for example using drawings or maize grains arranged in layers)? Which DQB questions should I use to open Lesson 8?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                        |
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| <b>What did I observe?</b>                   | Small carton 10 × 6 × 4 cm has volume 240 cm <sup>3</sup> ; large similar carton 20 × 12 × 8 cm has volume 1920 cm <sup>3</sup> ; all edges have k = 2 (large ÷ small); 1920 ÷ 240 = 8; a 2 × 2 × 2 block uses 8 unit cubes.                                                                                           |
| <b>What did I learn?</b>                     | When every length of a solid is multiplied by k, the volume is multiplied by k <sup>3</sup> (and area by k <sup>2</sup> ), so k = 2 gives volume factor 8 and k = 3 gives 27.                                                                                                                                          |
| <b>How does this explain the phenomenon?</b> | Volume depends on three lengths (length × width × height), each multiplied by the same k, so volume is multiplied by k × k × k = k <sup>3</sup> . This extends the idea from Wanjiru's photo: one scale factor in every direction keeps the shape, while area changes by k <sup>2</sup> and volume by k <sup>3</sup> . |



## LESSON 8 (40 minutes): Enlargement with a Centre and a Positive Scale Factor: Building Wanjiru's Poster from Rays

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners construct an enlargement of a triangle using a centre of enlargement, rays and a positive scale factor greater than 1, and explain why the image is similar to the object. This shows how a picture such as Wanjiru's can be made bigger from one fixed point without being distorted.                                                                                                                                                                                                                                     |
| <b>Knowledge</b>            | An enlargement needs a centre O and a scale factor k, with $k = \text{image length} \div \text{object length}$ (object = ABC, image = A'B'C'). Each image point lies on the ray from O through its object point, with $OA' = k \times OA$ . For enlargement, centre O(0,0), scale factor 2, triangle ABC with A(1,1), B(3,1), C(1,2) maps to A'(2,2), B'(6,2), C'(2,4). Object and image are similar: corresponding angles are equal and corresponding sides are in the ratio 1 : 2.                                                |
| <b>Skills</b>               | Draw rays from a centre through object points on grid paper; measure and mark distances using a ruler; read and write coordinates; compare corresponding sides and angles of object and image (NGSS SEP: Developing and Using Models; Analyzing and Interpreting Data; Constructing Explanations).                                                                                                                                                                                                                                  |
| <b>Attitudes</b>            | Appreciate the precision needed in construction; cooperate in groups and value each other's checking; show curiosity about how enlargement is used in photo studios such as Lakeview Photo Studio in Kisumu.                                                                                                                                                                                                                                                                                                                        |
| <b>Key Inquiry Question</b> | 1. Why do we need to enlarge or reduce objects? 2. How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Purpose in Storyline</b> | Lessons 3 and 4 taught that similar figures have equal corresponding angles and equal ratios of corresponding sides, and how to calculate a scale factor $k = \text{image} \div \text{object}$ . Lesson 8 shows HOW an image is constructed from one centre using a single k, so that every point is moved in the same proportion. This answers part of the driving question: Mr Otieno's poster stays like Wanjiru because every point is moved from one centre by the same factor. Lesson 9 extends to fractional and negative k. |
| <b>Safety Notes</b>         | Use sharp pencils and rulers safely. Do not point pencils or compasses at others. Scissors, if used to cut out triangles, must be used under teacher supervision and returned to the table after use. Keep walkways clear when groups move to display work.                                                                                                                                                                                                                                                                         |

### B. LESSON OVERVIEW

Learners begin by predicting how Mr Otieno could make Wanjiru's photo bigger for the prize-giving poster while keeping her likeness, then predict the position of the enlarged triangle ABC. In the Observe Phase, groups work on grid paper with triangle ABC, A(1,1), B(3,1), C(1,2), and centre O(0,0). They draw rays from O through A, B and C, mark A', B' and C' at twice the distance from O, join the points, and measure sides and angles. In the Explain Phase, they connect the construction to the scale factor and similarity from Lessons 3 and 4 and explain why a single k for all points keeps the shape, unlike the distorted  $4.5 \times 4.5$  cm print, which used different factors. The DQB and model phases record what is now known and what remains, such as  $k < 1$ , negative k, and finding an unknown centre.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                | Learner Experience                                                 | Teacher Moves                                                    | Sensemaking Strategy                                        | Formative Assessment Strategy                               |
|----------------------|--------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------|
| <b>Predict Phase</b> | Learners recall the anchoring phenomenon: Wanjiru's $14 \times 18$ | Display the four print sizes and triangle ABC on the board. Say: | Predict-Observe-Explain (Prediction Gallery): committing to | Observe whether learners (a) place the image farther from O |

|                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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| <p><b>VIDEO:</b><br/> <a href="#">Dilations: scale factor</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar videos</p> <p><b>EXERCISE:</b><br/> <a href="#">Dilations: scale factor</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar readings</p>               | <p>cm original, the <math>3.5 \times 4.5</math> cm passport photo, the <math>70 \times 90</math> cm poster and the distorted <math>4.5 \times 4.5</math> cm print. Mr Otieno at Lakeview Photo Studio in Kisumu must make the poster from the original. Each learner silently sketches in their notebook where they think the enlarged triangle ABC will land if every point is pushed away from O(0,0) so that all lengths double. Triangle ABC has A(1,1), B(3,1), C(1,2). They write one sentence beginning 'I think the image will be...' and give a reason. Pairs compare predictions for 2 minutes.</p>                                                                                                                                                                                                           | <p>'Mr Otieno has one fixed point, the centre, and one number, the scale factor. Triangle ABC is a simple stand-in for Wanjiru's photo. If we double every length from O, where will A, B and C go? Sketch it silently.' Give 2 minutes of silent writing. Then say: 'Turn to your partner and compare.' Circulate and note predictions. Cold-call 3 pairs: 'What did you predict for point A, and why?' After each answer, WAIT TIME of 10 seconds before calling on the next group. Do NOT confirm or correct. Record 3 different predictions on the board, labelled P1, P2 and P3.</p>                                                                                                                                                                                 | <p>a prediction creates a cognitive stake. When learners later compare the construction with their sketch, the difference between prediction and evidence drives sensemaking and exposes ideas such as 'add 2 to each coordinate' versus 'multiply by 2'.</p>                | <p>than the object, (b) keep the same shape, (c) use multiplication rather than addition. Indicators: at least 3 of 5 learners in a sampled group attempt a prediction with a reason; note learners who say 'add 2' for later targeted discussion. Tally the P1/P2/P3 types on a clipboard.</p>                                                                                                                                                                                          |
| <p><b>Observe Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Dilations: center</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar videos</p> <p><b>EXERCISE:</b><br/> <a href="#">Dilations: center</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES</a> for similar readings</p> | <p>In groups of 4 with grid paper, rulers and protractors, learners plot O(0,0), A(1,1), B(3,1) and C(1,2) and draw triangle ABC. Step 1: draw a straight ray from O through A and extend it. Step 2: measure OA, and mark A' so that <math>OA' = 2 \times OA</math>; A' should fall at (2,2). Step 3: repeat for B to get B'(6,2) and for C to get C'(2,4). Step 4: join A', B', C'. Step 5: complete a table of coordinates and then measure AB, A'B', AC, A'C', BC and B'C' and the angles at A, B and C and at A', B', C'. They record that A'B' = 4 units and AB = 2 units, A'C' = 2 and AC = 1 give ratio 2, and the angles are equal. They also compute x- and y-coordinates of each image as 2 times the object (A(1,1) to A'(2,2)). Groups must have each member construct one vertex and check another's.</p> | <p>Distribute materials: 'Take out your grid paper, ruler and protractor. Plot O at the origin and triangle ABC. Today, our rule is that nobody moves on until the whole group agrees on the position of A'.' Model on the board only Step 1 for point A. Circulate with these prompts: 'Which ray does B' lie on? How do you know?' 'How far is A' from O compared with A from O?' 'What do you notice about the coordinates of A and A'?' Ask groups who finish early: 'Does C' lie on the line through O and C?' Pause the class at the halfway point: 'Stop and look at your ray for C. WAIT TIME 10 seconds. Is C' where you expected?' Check that rays are extended, not just segments. Cold-call 2 groups to report the length ratios A'B' : AB and B'C' : BC.</p> | <p>Data Table and Construction Modelling: building the image step by step with rays makes the abstract rule '<math>OA' = k \times OA</math>' visible. The measurement table makes the pattern (all ratios = 2, all angles equal) emerge from the learners' own evidence.</p> | <p>Check, using a rubric: (1) rays pass through O and the object points (within 2 mm); (2) A', B', C' have coordinates (2,2), (6,2), (2,4); (3) measured ratios of sides are 2 (<math>\pm 0.1</math>) and angles match the original (<math>\pm 2^\circ</math>). Indicators of difficulty: A' placed by adding 2 units to OA rather than doubling; rays not drawn through O; image drawn but not joined. Provide a hint card ('Measure OA, then double it') to groups that are stuck.</p> |
| <p><b>Explain Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Dilating shapes: expanding</a><br/> Source: Khan</p>                                                                                                                                                                                                                                                       | <p>Groups use their evidence to answer in writing: (a) What stays the same between triangle ABC and triangle A'B'C'? (b) What changes? (c) What is the scale</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>Say: 'Use your table. Claim first, then evidence, then reasoning.' Give 3 minutes of writing with WAIT TIME of at least 10 to 15 seconds after each question you</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Claim-Evidence-Reasoning with Revisiting Predictions: learners compare their initial predictions (P1, P2, P3) with their constructed evidence, which moves them from</p>                                                                                                  | <p>Look for explanations that include (1) equal scale factor in every direction, (2) equal corresponding angles, (3) sides in ratio 1 : 2, (4) reference to the distorted print</p>                                                                                                                                                                                                                                                                                                      |

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| <p>Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Dilations: scale factor</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                                                          | <p>factor, and how do you calculate it using <math>k = \text{image length} \div \text{object length}</math>? (d) Why does a single centre and a single <math>k</math> keep the shape, and how does this connect to Mr Otieno's passport photo (<math>k = 1/4</math>) and poster (<math>k = 5</math>)? (e) In the distorted <math>4.5 \times 4.5</math> cm print, the width factor was <math>9/28</math> and the height factor <math>1/4</math>, so what went wrong? Learners write one claim, two pieces of evidence (measured ratio, measured angle) and one reasoning sentence (Claim-Evidence-Reasoning). They also state the formal description: 'enlargement, centre <math>O(0,0)</math>, scale factor 2'.</p> | <p>pose to the whole class before taking answers. Ask: 'If we had used a scale factor of 2 for the width but 3 for the height, what would happen to the angle at A?' Cold-call 4 learners from different groups to share a claim and push with 'Why does that follow from your measurements?' and 'Who can add to or challenge that?' Connect: 'Mr Otieno applies one <math>k</math> in every direction, like our rays from one centre. The distorted print used <math>9/28</math> and <math>1/4</math>, so it was not an enlargement.' Do not re-teach area or volume factors.</p> | <p>procedure to a causal explanation: one centre plus one <math>k</math> means that every point is stretched from <math>O</math> in the same proportion, so ratios and angles are preserved (similar figures from Lessons 3 and 4).</p>               | <p>having two different factors. Success criterion: at least 75% of groups write a complete CER with measured data. Misconception check: 'enlargement means adding the same length to each side' should be corrected through the evidence <math>AB = 2</math> to <math>A'B' = 4</math>.</p>                                                                            |
| <p><b>Driving Question Board (DQB) Creation</b><br/> <b>VIDEO:</b><br/> <a href="#">Dilations: scale factor</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Dilations: center</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to the class Driving Question Board posed by Mrs Achieng: 'How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?' On sticky notes, each learner writes (a) one thing they can now answer using Lesson 8 (for example 'An enlargement from <math>O</math> with <math>k = 2</math> moves <math>A(1,1)</math> to <math>A'(2,2)</math>') and (b) one new question (for example 'What if <math>k</math> is less than 1, like the passport photo <math>k = 1/4</math>?' or 'What if we only see the image and need to find the centre?'). They place notes on the DQB under 'Answered' and 'New Questions'.</p>                                  | <p>Say: 'Take one sticky note for what we can now explain and one for a question we still have. Think for 10 to 15 seconds before you write.' Walk to the DQB with the class. Cold-call 3 learners to read and place their notes, asking 'Which part of the driving question does this help to answer?' Group related questions (<math>k &lt; 1</math>, negative <math>k</math>, finding the centre) and tell learners: 'Lessons 9 and 10 will help us explore these.' Praise questions that link to Wanjiru's passport photo.</p>                                                  | <p>Driving Question Board (Question Sorting): organising questions shows learners that their curiosity steers the unit. It makes the learning trajectory visible and links today's construction to future investigations.</p>                         | <p>Count the notes: at least 80% of learners add a note. Check the 'Answered' notes for accuracy (<math>k</math>, centre, ray, similarity) and the 'New Questions' for relevance. Indicators: questions that link to fractional <math>k</math>, negative <math>k</math>, finding the centre, or areas of images show good progress.</p>                                |
| <p><b>Model Building Phase</b><br/> <b>VIDEO:</b><br/> <a href="#">Dilations: center</a><br/> Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p>                                                                                                                                                                                                  | <p>Learners add a new section to their class model titled 'How to make a similar image': (1) choose a centre <math>O</math> and a scale factor <math>k</math>, (2) draw rays from <math>O</math> through every object point, (3) mark each image point at <math>k</math> times the distance from <math>O</math>, (4) join the image points. They draw a labelled diagram of triangle</p>                                                                                                                                                                                                                                                                                                                            | <p>Say: 'Revise the model we started in Lesson 1. Add the steps you used today, and show how they explain why Wanjiru's poster keeps her likeness.' Provide 5 minutes. Circulate and ask: 'Where on your model do you show the centre? Where is the scale factor?' Select 2 groups to present using</p>                                                                                                                                                                                                                                                                             | <p>Cumulative Model Revision: updating one shared model each lesson helps learners integrate new ideas (centre, rays, <math>k</math>) with earlier ones (equal angles, equal ratios, scale factor) into a coherent explanation of the phenomenon.</p> | <p>Check that models include the centre, the rays, the scale factor, the coordinates <math>A'(2,2)</math>, <math>B'(6,2)</math>, <math>C'(2,4)</math>, and a statement that corresponding angles are equal and corresponding sides are in the ratio <math>1 : 2</math>. Indicators of mastery: the learner can explain the model to a peer without notes. Note any</p> |

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| <p> <b>HTML:</b><br/> <a href="#">Scale Factor of Similar Polygons (Flexbook)</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>ABC and A'B'C' with O(0,0), <math>k = 2</math>, and annotate the 1 : 2 side ratio and equal angles. They then add a note on the poster: 'Wanjiru's poster (<math>k = 5</math>) can be built the same way from the original <math>14 \times 18</math> cm print.' They also complete the summary table.</p> | <p>the board (cold-call 2 groups); WAIT TIME 10 seconds after each presentation before inviting 'one question or one addition' from the class. Close with: 'Today we answered the how. In Lesson 9 we ask what happens when <math>k</math> is a fraction like the passport photo's <math>\frac{1}{4}</math>, or negative.'</p> | <p>learner who says the centre must be inside the triangle, and address this in Lesson 9 or 10.</p> |
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#### D. TEACHER REFLECTION

Did every group place A', B' and C' using multiplication from O rather than adding a fixed length? Which learners could explain why a single  $k$  from one centre keeps the shape, and link this to Wanjiru's photo and the distorted  $4.5 \times 4.5$  cm print? Where did measurement errors arise (rays not through O, imprecise ruler reading), and how could next lesson's grouping help? Which questions on the DQB show readiness for  $k < 1$  and negative  $k$  in Lesson 9?

#### E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | We drew rays from the centre O(0,0) through A(1,1), B(3,1) and C(1,2) and marked A'(2,2), B'(6,2) and C'(2,4) at twice the distance from O. We measured the sides and angles of triangle ABC and triangle A'B'C'.                                                                                                                              |
| <b>What did I learn?</b>                     | An enlargement needs a centre and a scale factor $k$ , where $k = \text{image length} \div \text{object length}$ . With $k = 2$ , every image point lies on the ray from O through its object point at twice the distance. The image is similar to the object: corresponding angles are equal and corresponding sides are in the ratio 1 : 2.  |
| <b>How does this explain the phenomenon?</b> | Wanjiru's poster keeps her likeness because every point is moved from one centre by the same scale factor ( $k = 5$ ), so ratios and angles stay the same. The distorted $4.5 \times 4.5$ cm print used different factors ( $\frac{9}{28}$ for width and $\frac{1}{4}$ for height), so it is not an enlargement and her face looked stretched. |

## LESSON 9 (40 minutes): Fractional and Negative Scale Factors: Reduction and Inverted Images

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners extend enlargement with a centre from $k > 1$ (Lesson 8) to fractional and negative scale factors, so they can explain how Wanjiru's passport print ( $k = 1/4$ ) is a reduction that keeps her likeness, and why a projector image can appear upside down yet remain similar.                                                                                                                                                                                                    |
| <b>Knowledge</b>            | Enlargement with centre $O$ and $0 < k < 1$ gives a smaller image (a reduction) on the same side of $O$ as the object. Enlargement with negative $k$ gives an image on the opposite side of $O$ , inverted, with lengths multiplied by $ k $ . In both cases the image is similar to the object: corresponding angles are equal and corresponding sides are in the same ratio.                                                                                                             |
| <b>Skills</b>               | Constructing images by drawing rays from the centre through each vertex; calculating image coordinates by multiplying object coordinates by $k$ ; measuring lengths and angles to test similarity; (NGSS-style practices) making predictions, using mathematical models, analysing data, constructing explanations from evidence.                                                                                                                                                          |
| <b>Attitudes</b>            | Curiosity about how a picture can be turned upside down yet keep its shape; precision and care when plotting and measuring; collaboration and respect when groups compare constructions; appreciation of the use of enlargement and reduction in day-to-day activities.                                                                                                                                                                                                                    |
| <b>Key Inquiry Question</b> | 1. Why do we need to enlarge or reduce objects? 2. How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Purpose in Storyline</b> | Lesson 8 taught construction of an enlargement from a centre using rays and $k > 1$ . Lesson 9 adds the two missing cases: reduction ( $0 < k < 1$ ), which links directly to the passport print ( $k = 1/4$ ), and negative $k$ , which links to the inverted projector image. Learners find that every type of enlargement keeps similarity, which prepares Lesson 10 (finding the centre and $k$ from an object and image) and Lesson 12 (the full resolution of the Driving Question). |
| <b>Safety Notes</b>         | Use rulers, protractors and scissors with care; scissors are used only to cut grid paper and are passed handle-first. Keep bags and feet out of aisles. If a sunny window or a torch is used for the projector demonstration, never look directly at the sun or the torch bulb. Keep the demonstration area clear of desks and cords. No electricity or internet is required.                                                                                                              |

### B. LESSON OVERVIEW

Mrs Achieng returns to the Driving Question Board: How can a full-size photograph be resized to fit a small passport box without distortion? Learners recall that Wanjiru's original  $14 \times 18$  cm print became a  $3.5 \times 4.5$  cm passport print, a scale factor of  $1/4$  (image  $\div$  object), and that Lesson 8 showed how to build an enlargement from a centre using rays with  $k = 2$ . In this lesson learners predict what happens when  $k$  is a fraction or negative, then construct on grid paper the images of triangle ABC with  $A(1,1)$ ,  $B(3,1)$ ,  $C(1,2)$  and centre  $O(0,0)$ : first  $k = 1/2$  giving  $A'(0.5,0.5)$ ,  $B'(1.5,0.5)$ ,  $C'(0.5,1)$ , then  $k = -2$  giving  $A'(-2,-2)$ ,  $B'(-6,-2)$ ,  $C'(-2,-4)$ . They measure sides and angles, compare ratios and angles with the object, and connect the reduction to the passport print and the inverted image to a projector (a pinhole or torch-and-cutout demonstration). They conclude that size can shrink, grow or flip, but shape is kept because one scale factor applies to all lengths. Learners add to the DQB and revise the class model.

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sensemaking Strategy                                                                                                                                                                                                                                                                                                                                                                                  | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Dilating shapes: shrinking</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Dilations: scale factor</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a>        | Learners sit in groups of four with grid paper, rulers and pencils. Mrs Achieng displays triangle ABC with A(1,1), B(3,1), C(1,2) and centre O(0,0) on the board. Each learner silently writes a prediction in their notebook (3 minutes): (1) 'If I enlarge ABC from O with $k = 1/2$ , where will A' be and will the triangle be bigger or smaller than ABC?' (2) 'What might happen if $k = -2$ ?' They sketch their predicted position of the image on grid paper, labelling it 'My prediction'. They then link the idea to Wanjiro's prints: the passport print is $1/4$ the length of the original, so is that an enlargement too? Groups share predictions in turn.                                                                           | Say: 'Last lesson we built an enlargement with $k = 2$ using rays from the centre O. Today the scale factor changes. Before we draw anything, I want your own prediction.' Recall from Lesson 8 only the ray method and $k = 2$ giving A'(2,2), B'(6,2), C'(2,4). Point to the DQB and say: 'Wanjiro's original print was $14 \times 18$ cm and her passport print is $3.5 \times 4.5$ cm. Image length divided by object length is $1/4$ . Is that a scale factor?' WAIT TIME: wait 10-15 seconds after each question before taking answers. Cold-call 4 learners from different groups: 'What do you think $k = 1/2$ does to the triangle?' and 'What might a negative $k$ do?' Record every prediction on the board without judging. Walk around and note learners who place A' beyond A or on the opposite side of O. | Predict-Observe-Explain with a public prediction record: committing to a written and sketched prediction first creates a cognitive 'stake', so that the later construction produces surprise or confirmation. Recording the class's predictions makes competing ideas visible and gives a reference to return to in the Explain phase.                                                                | Look for: (a) learners who write that $k < 1$ gives a smaller image (naive but correct idea) versus those who say 'enlargement always makes it bigger'; (b) learners who predict the negative- $k$ image will be on the other side of O or turned around; (c) learners who connect $1/4$ with the passport print. Tally on the board how many of the 4 cold-called learners predicted smaller, bigger, or flipped. Note groups needing support with plotting coordinates.                                            |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Dilating shapes: shrinking by 1/2</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Dilations: center</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Part A (15 min total for Observe): Using grid paper (axes from -7 to 7), each group plots triangle ABC with A(1,1), B(3,1), C(1,2) and centre O(0,0). Construction 1, $k = 1/2$ : they draw rays from O through A, B and C, measure OA, OB, OC, mark A', B', C' at half the distance from O, and read the coordinates, obtaining A'(0.5,0.5), B'(1.5,0.5), C'(0.5,1). They check by multiplying each coordinate by $1/2$ . Construction 2, $k = -2$ : they draw the rays from O through A, B, C extended backwards past O, mark A', B', C' on the opposite side at twice the distance, obtaining A'(-2,-2), B'(-6,-2), C'(-2,-4). They measure AB, A'B' and angles at A, A' ( $90^\circ$ ) and compare. Part B: a short projector demonstration: Mrs | Distribute grid paper and say: 'Use the same ray method from Lesson 8. For $k = 1/2$ the image point is half as far from O as the object point. Measure it, do not guess.' Circulate and ask: 'Which side of O is A' on? How far is OA' compared with OA?' For $k = -2$ , say: 'Negative means go through O and out the other side. Extend your ray backwards.' Cold-call 3 groups to read their image coordinates aloud and write them on the board for checking against A'(0.5,0.5), B'(1.5,0.5), C'(0.5,1) and A'(-2,-2), B'(-6,-2), C'(-2,-4). WAIT TIME: give 10-15 seconds of silent measuring before asking 'What do you notice about AB and A'B'?' For the projector demonstration, stand to the side, switch on the torch,                                                                                       | Structured data table with a gallery of evidence: recording the same measurements (coordinates, side of O, orientation, length ratios) for each scale factor lets learners spot patterns across cases rather than seeing each construction as an isolated drawing. Mathematical modelling with a physical analogue (torch and cut-out) connects the abstract rays to a real inverted projected image. | Check that: image points lie on the ray through O and the object point; A'(0.5,0.5) etc. are within a half-square of accuracy; negative- $k$ points lie on the opposite side of O; learners measured A'B' = 1 (half of $AB = 2$ ) for $k = 1/2$ and A'B' = 4 (twice $AB = 2$ ) for $k = -2$ ; angles at A' measured as $90^\circ$ . Ask 3 learners to point to the exact ray used for point B. Note learners who place the negative- $k$ image on the same side of O and re-direct them to extend the ray backwards. |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Achieng holds a cut-out triangle in front of a torch (or a pinhole in a card on a sunny wall) and learners see the inverted, enlarged image on the wall. Learners record observations in a table: object, k, image coordinates, side of O, upright or inverted, size compared with object.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | and ask: 'What do you see? Is the picture upside down? Is it still the same shape?' Remind learners not to look into the torch.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Explain Phase</b><br> <b>VIDEO:</b><br><a href="#">Dilating shapes: shrinking</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Dilations: scale factor</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Groups use their table to answer in writing (8 minutes): (1) What happens to lengths when $0 < k < 1$ ? (2) What does the negative sign in $k = -2$ do, and what does the number 2 do? (3) Is each image similar to ABC? Give evidence using corresponding angles and ratios of corresponding sides (e.g. $AB : BC$ ratio unchanged; right angle at A stays $90^\circ$ ). (4) Link to Wanjiru: the passport print is $k = 1/4$ (image $\div$ object, $3.5/14 = 4.5/18 = 1/4$ ), so it is a reduction with both width and height multiplied by $1/4$ ; the width : height ratio stays 7:9. Explain why the $4.5 \times 4.5$ cm distorted print is not an enlargement of the original (factors $9/28$ and $1/4$ are unequal). (5) Link to the projector: why does the image appear upside down yet keep the shape? Learners share explanations in a 'talk-move' pair-share, then present to the class. | Pose: 'Wanjiru's passport print is a reduction. Which scale factor describes it, and which of today's constructions looks like it?' WAIT TIME: 10-15 seconds before calling on anyone. Cold-call 5 learners across the room for the following: one to state what k between 0 and 1 does; one to state what a negative k does; one to compare $ k  = 2$ with the size of the image; one to use the ratio 7:9 for the passport print; one to explain why $9/28$ and $1/4$ differ for the distorted print. Use talk moves: 'Who can add to that?', 'Can you say why using the angle at A?', 'Do you agree or disagree, and why?' Correct misconceptions firmly: 'Enlargement is the name of the transformation even when the image is smaller; negative k flips the image through O but the shape is kept.' Do not re-teach area or volume factors, which come from Lessons 6 and 7. | Claim-Evidence-Reasoning with talk moves: learners must state a claim (the image is similar), support it with measured evidence (equal angles, equal ratios), and give reasoning (one scale factor multiplies every length). Revoicing and 'agree or disagree' prompts push the whole class to reason with each other's ideas and connect the abstract rule to the passport and projector situations. | Listen for correct statements: ' $k = 1/2$ halves every distance from O'; 'negative k puts the image on the opposite side of O and inverts it'; 'the lengths are multiplied by the size of k'; 'corresponding angles stay $90^\circ$ , ratios stay equal, so the images are similar'; ' $k = 1/4$ for the passport print, equal for width and height'. Indicators of misunderstanding: saying the image is distorted because it is upside down; saying $k = -2$ makes the image smaller; confusing object and image in $k = \text{image} \div \text{object}$ . Record on a class checklist and revisit with a quick re-measurement for those learners. |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Dilating shapes: shrinking by 1/2</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a>                                                                                                                                                                                                                                                                                                                                                                        | Learners return to the DQB, which holds the question from Lesson 1: How can a full-size photograph be resized to fit a small passport box without distortion? In groups they write one sticky-note answer to a fresh question (for example, 'Can a picture be shrunk and still look like Wanjiru?' or 'Why does a projector flip the picture?') and one new question for the board (for                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Say: 'Mrs Achieng's board must show our learning. Write one thing we can now explain and one question we still have.' Give 3 minutes. Read 3 notes aloud (cold-call 3 groups to read their own) and sort them under the headings. Praise precise language ('image', 'object', 'centre', 'k'). WAIT TIME: wait 10-15 seconds after asking 'Which of these new questions will                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Driving Question Board as a public record of evolving understanding: moving questions between 'new' and 'answered' shows learners that their evidence is accumulating, motivates the next lesson, and gives ownership of the inquiry direction to learners.                                                                                                                                           | Check that notes use correct vocabulary (k, centre, image, reduction, inverted), that at least one note per group links to the passport photo or the projector, and that questions are genuine (e.g. about locating the centre) rather than off-topic. Count the notes moved to 'Answered today' and identify groups with no connection to the phenomenon for                                                                                                                                                                                                                                                                                          |

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| <p> <b>EXERCISE:</b><br/> <a href="#">Dilations: center</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>example, 'How do we find the centre if we only have the object and the image?'). They place notes under headings: Answered today / Partly answered / New questions. They also add a progress note: 'Today we learned that <math>k = 1/4</math> reduces Wanjiru's print and that negative <math>k</math> flips it but keeps the shape.'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Lesson 10 help us answer?' Point out that the question about finding the centre and <math>k</math> will be addressed in Lesson 10. Link back to the Key Inquiry questions: 'Why might a photographer, a map maker, or a projector use enlargement or reduction?'</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                            | <p>follow-up.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Model Building Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Dilations: scale factor</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Dilations: scale factor</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a><br/> for similar readings</p> | <p>Learners update the class model of 'what must stay the same and what may change' (started in Lesson 1 and revised since). In groups they add a new panel with a labelled diagram: object ABC with centre O; the image for <math>k = 1/2</math> (smaller, same side) and <math>k = -2</math> (bigger, opposite side, inverted); the passport photo as <math>k = 1/4</math> (<math>3.5 \times 4.5</math> from <math>14 \times 18</math>). Under it they write a short rule: 'Lengths change by the factor <math> k </math>; position is on the same side of O if <math>k &gt; 0</math> and the opposite side if <math>k &lt; 0</math>; angles stay equal and the ratios of corresponding sides stay equal, so the shape stays the same.' They then fill the summary table (what we observed, what we learned, how it explains the Driving Question) and revise their personal model sentence for Wanjiru's photo.</p> | <p>Say: 'Revise your model. Include one diagram and one rule that explains why a reduction by <math>k = 1/4</math> keeps Wanjiru's face looking like her, and why the distorted <math>4.5 \times 4.5</math> print does not.' Give 8 minutes. Circulate and ask each group: 'Where is the scale factor in your diagram? What stays equal?' WAIT TIME: wait 10-15 seconds after asking 'What would change if <math>k</math> were <math>-1/4</math>?' and cold-call 2 learners (prediction only: a smaller inverted image on the opposite side of O, not constructed today). Select 2 groups to display their models, then ask the class to point out what has been added since Lesson 8. Close by stating that Lesson 10 will work backwards: given an object and image, find the centre and <math>k</math>. Collect exit tickets with one sentence: 'The passport print is similar to the original because...'</p> | <p>Cumulative model revision: adding fractional and negative <math>k</math> to the existing model forces learners to reconcile new evidence with earlier ideas (similarity, scale factor, rays from a centre) and produce a single coherent explanation. Using multiple representations (diagram, rule, numbers) strengthens transfer.</p> | <p>Observable indicators: the model shows image points on rays through O; labels <math>k = 1/2</math> as reduction and <math>k = -2</math> as inverted; includes <math>k = 1/4</math> for the passport print with <math>14 \times 18</math> to <math>3.5 \times 4.5</math>; states that angles and ratios are equal; uses 'image <math>\div</math> object' correctly. Exit ticket: a correct answer mentions one scale factor for all lengths, equal corresponding angles and equal ratios (7:9). Identify learners who still say 'inverted means distorted' and plan a short follow-up.</p> |

## D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners correctly place negative- $k$  images on the opposite side of O, and how many needed the ray extended backwards? (2) Did they understand that 'enlargement' with  $0 < k < 1$  produces a reduction, and could they connect it to the  $1/4$  passport print? (3) Which prediction misconceptions (e.g. 'negative means smaller' or 'inverted means distorted') were most common, and what evidence changed learners' minds? (4) Was WAIT TIME of 10-15 seconds used consistently, and did cold-calling reach quieter learners? (5) Were object and image used consistently in  $k = \text{image} \div \text{object}$ ? (6) What adjustments to grid paper, torch or sunlight availability would help next time? Use these notes to prepare Lesson 10, where learners find the centre and scale factor from a given object and image.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
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| What did I observe?                                          | We observed that when triangle ABC with A(1,1), B(3,1), C(1,2) is enlarged from centre O(0,0) with $k = 1/2$ , the image A'(0.5,0.5), B'(1.5,0.5), C'(0.5,1) is smaller and on the same side of O; with $k = -2$ , the image A'(-2,-2), B'(-6,-2), C'(-2,-4) is larger, inverted and on the opposite side of O. A projector shows an inverted image of a slide.                                                                                                                                                                                 |
| What did I learn?                                            | We learned that a scale factor between 0 and 1 gives a reduction, a negative scale factor gives an inverted image on the opposite side of the centre, and lengths are multiplied by the size of k. In every case the corresponding angles stay equal (for example $90^\circ$ ) and the ratios of corresponding sides stay equal, so the image is similar to the object.                                                                                                                                                                         |
| How does this explain the phenomenon?                        | Wanjiru's passport print ( $3.5 \times 4.5$ cm) is a reduction of her $14 \times 18$ cm original with $k = 1/4$ , since both width and height are multiplied by the same factor and the width : height ratio stays 7:9, so her face looks like her. The $4.5 \times 4.5$ cm print used different factors (9/28 for width and 1/4 for height), so the shape changed and her face looked stretched. Size can change by any scale factor, even a fraction or a negative one, but shape is kept only if one scale factor is applied to all lengths. |

## LESSON 10 (40 minutes): Finding the Centre and Scale Factor of an Enlargement

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                         |
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| <b>Purpose</b>              | Learners work backwards from an object and its image to find the centre of enlargement and the scale factor $k$ . This links to the Driving Question: if Mr Otieno resizes Wanjiru's photo, a single centre and a single $k$ describe how every point moves, which is why the shape is kept.                                                                                                                       |
| <b>Knowledge</b>            | By the end of the lesson, the learner should be able to: (1) explain that lines joining each object point to its image point (P to P', Q to Q', R to R') meet at the centre of enlargement; (2) find $k = \text{image length} \div \text{object length}$ , stating which is object and which is image; (3) check the centre by showing that image point minus centre equals $k$ times (object point minus centre). |
| <b>Skills</b>               | Plotting points on grid paper; drawing straight lines with a ruler; extending lines to find where they meet; measuring and calculating ratios of corresponding lengths; constructing arguments from evidence; checking results with a second method. NGSS-aligned practices: analysing and interpreting data, using mathematics and computational thinking, constructing explanations.                             |
| <b>Attitudes</b>            | Learners appreciate the use of enlargement in day-to-day activities such as photo printing and posters; they show unity by working in groups, responsibility by taking care of rulers and grid paper, and perseverance in checking their answers.                                                                                                                                                                  |
| <b>Key Inquiry Question</b> | 1. Why do we need to enlarge or reduce objects? 2. How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                                              |
| <b>Purpose in Storyline</b> | Lessons 8 and 9 taught how to construct an image from a given centre and scale factor. Lesson 10 reverses this: given the object and the image, learners find the centre and $k$ . This shows that the passport photo and poster are exact enlargements of the original, each with one centre and one scale factor, and it prepares learners for the applications in Lesson 11.                                    |
| <b>Safety Notes</b>         | Scissors and rulers are used with care. Learners keep pencils pointed down when moving about. Rulers are long, so keep the aisles clear and do not swing them. Grid paper is shared and neatly returned. No electricity or internet is needed.                                                                                                                                                                     |

### B. LESSON OVERVIEW

Mrs Achieng opens by recalling Wanjiru's photos at Lakeview Photo Studio in Kisumu: the  $14 \times 18$  cm original, the  $3.5 \times 4.5$  cm passport photo and the  $70 \times 90$  cm poster all keep her likeness. Learners predict how to find the centre when only the object and image are given. In groups they plot triangle PQR with P(2,2), Q(3,2), R(2,3) and its image P'(4,4), Q'(7,4), R'(4,7) on grid paper, join corresponding points with a ruler, extend the lines until they meet at (1,1), and calculate  $k = 3$  from lengths (for example  $PQ = 1$  and  $P'Q' = 3$ ). They explain why the rays meet at a single point, check the centre with coordinates, update the DQB and revise the class model of what must stay the same (one scale factor for all lengths) and what may change (size, position).

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                | Learner Experience                                                                                                          | Teacher Moves                                                                                                                   | Sensemaking Strategy                                                                                                   | Formative Assessment Strategy                                                                                                      |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Dilations: scale factor</a> | Learners recall Lessons 8 and 9, where an image was built from a centre and $k$ . Mrs Achieng sketches triangle PQR and its | Draw the two triangles on the board. Say: 'In Lesson 8 we started with the centre. Today we start with the answer. Where did it | Predict-Observe-Explain (prediction before evidence): committing to an idea beforehand makes learners notice where the | Look for: learners who suggest joining object points to image points or extending lines; learners who guess a point by eye without |

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| <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Dilations: scale factor</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p>                                                                                   | <p>image P'Q'R' on the board without a centre and asks: 'Mr Otieno has the small print and the big print but forgot which point he enlarged from. How could we find that point?' Learners write a prediction in their notebooks, with a sketch, then share in pairs. Prediction prompts: (a) Where might the centre be? (b) What could we draw to find it? (c) How might we find k? Link to the phenomenon: the original, passport photo and poster must all come from one enlargement rule.</p>                                                                                                                                                                                                                                                                                                                                                               | <p>come from?' Give 10 seconds of silent thinking time, then WAIT TIME of 10-15 seconds before taking any answers. Cold-call 4 learners for their predictions and record them on the board without judging. Ask: 'What did you use from Lesson 8 to make that prediction?' Circulate during pair talk and listen for the idea of joining corresponding points. Say: 'Hold on to your prediction; the evidence today will test it.'</p>                                                                                                                                                                                                                                          | <p>evidence supports or contradicts them, so the new method (joining corresponding points) is built on their own reasoning from Lessons 8 and 9.</p>                                                                                                                                                                                                                                                      | <p>a method; learners who confuse centre and scale factor. Tally on a class list how many predictions mention drawing lines. Note which groups need support with plotting.</p>                                                                                                                                                                                                                             |
| <p><b>Observe Phase</b></p> <p><b>VIDEO:</b><br/> <a href="#">Dilations: center</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p> <p><b>EXERCISE:</b><br/> <a href="#">Dilations: center</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar readings</a></p> | <p>In groups of 4, learners use grid paper (axes from 0 to 8), a ruler and pencil. Step 1: plot P(2,2), Q(3,2), R(2,3) and join to form triangle PQR. Step 2: plot P'(4,4), Q'(7,4), R'(4,7) and join. Step 3: with a ruler, draw a straight line through P and P', another through Q and Q', and another through R and R', extending each backwards past the object. Step 4: mark the point where all three lines meet and read its coordinates (1,1). Step 5: measure PQ = 1 and P'Q' = 3, PR = 1 and P'R' = 3, and calculate <math>k = \text{image length} \div \text{object length} = 3 \div 1 = 3</math>. Step 6: check the centre numerically: P'-centre = (4,4) - (1,1) = (3,3) = <math>3 \times (1,1) = 3 \times (\text{P} - \text{centre})</math>. Learners record observations in a table: point pair, line drawn, where lines meet, lengths, k.</p> | <p>Distribute the grid paper and rulers, one set per group, and assign roles: plotter, ruler-holder, measurer, recorder. Say: 'Draw your lines thin and long, and extend them backwards beyond the object.' Circulate and ask: 'What do you notice about where your three lines cross?' Ask groups that have lines not meeting: 'Check your coordinates for Q'. Is it (7,4)?' WAIT TIME of 10-15 seconds after each question before giving hints. When most groups have the meeting point, cold-call 3 groups to report the coordinates. Remind learners to state which length is the object and which is the image when computing k. Remind them to look after the rulers.</p> | <p>Structured data collection and pattern-seeking with a recording table: using a table makes learners compare the three lines and the three length ratios side by side and notice that the lines meet at one point and the ratios are equal. This is the same pattern of 'equal ratios of corresponding sides' met in Lesson 3 and '<math>k = \text{image} \div \text{object}</math>' from Lesson 4.</p> | <p>Observable indicators: plots P, Q, R, P', Q', R' accurately; draws lines with a ruler through corresponding points (P to P', not P to Q'); extends the lines until they meet; reads (1,1); writes <math>k = 3</math> with object and image stated. Quick check: ask a learner to point to a corresponding pair on the grid. Note groups with lines that do not meet and prompt a check of plotting.</p> |
| <p><b>Explain Phase</b></p> <p><b>VIDEO:</b><br/> <a href="#">Dilations: scale factor</a></p> <p>Source: Khan Academy (English - US curriculum)<br/> <a href="#">Search ARES for similar videos</a></p>                                                                                                                                                                      | <p>Groups explain, in words and in a diagram, why the three lines meet at one point and what <math>k = 3</math> means. They answer: (1) Why does the centre (1,1) lie on every line joining a point to its image? (2) How does the centre check P'-centre = <math>3 \times (\text{P} - \text{centre})</math> show that</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Say: 'Turn to your group and explain to each other why all three lines meet at one place.' Give WAIT TIME of 10-15 seconds before calling on groups. Cold-call 3 different groups to explain, and then ask a fourth learner to restate: 'Who can say that in their</p>                                                                                                                                                                                                                                                                                                                                                                                                       | <p>Claim-Evidence-Reasoning with sentence frames: learners make a claim (the centre is at (1,1) and <math>k = 3</math>), support it with evidence (the lines, the lengths, the coordinate check) and give reasoning (every image point lies on a ray from the centre and is k times as far). This</p>                                                                                                     | <p>Listen for: lines meet because each image point lies on a ray from the centre; k is found from image length <math>\div</math> object length; the coordinate check is used to verify the centre; (0,0) is rejected using evidence. Use a thumbs-up/middle/down check on</p>                                                                                                                              |

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| <p> <b>EXERCISE:</b><br/> <a href="#">Dilations: scale factor</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p>                                                                                                                                                                                                                                                                                                                                                                                        | <p><math>k = 3</math>? (3) Find <math>k</math> a second way using the centre: distance from centre to <math>P'</math> is 3 times distance from centre to <math>P</math>. (4) Link to the phenomenon: if the original print is the object and the passport is the image with <math>k = 1/4</math>, would the lines still meet at one point? What is different for the poster with <math>k = 5</math>? Learners use the sentence frame: 'The centre is where ____ because every image point is ____ times as far from the centre as its object point.' They then try a quick extension: if the image had been <math>P''(7,7)</math>, what would <math>k</math> be with the same centre <math>(1,1)</math>? (Answer: <math>P''\text{-centre} = (6,6) = 6 \times (1,1) \dots k = 6</math>.)</p> | <p>own words?' Press for reasoning: 'How do you know the centre is not <math>(0,0)</math>? Test it with <math>P</math> and <math>P'</math>.' (From <math>(0,0)</math>, <math>P'(4,4) = 2 \times P(2,2)</math> but <math>Q'(7,4)</math> is not <math>2 \times Q(3,2)</math>, so <math>(0,0)</math> fails.) Correct only through questioning, for example: 'Check your subtraction again.' Connect to the phenomenon: 'One centre and one scale factor for every point is why Wanjiru's face is not stretched.' Do not re-teach construction from Lessons 8 and 9; only refer to it.</p> | <p>makes the link between the construction from Lesson 8 and its reverse explicit.</p>                                                                                                                                                             | <p>'I can find a centre from an object and image'. Look for misconceptions: using differences <math>(4 - 2 = 2)</math> instead of ratios for <math>k</math>; joining the wrong pairs; taking <math>k = 4</math> by counting from the wrong point.</p>                                                                          |
| <p><b>Driving Question Board (DQB) Creation</b><br/>  <b>VIDEO:</b><br/> <a href="#">Dilations: center</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Dilations: center</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners return to the Driving Question Board: 'How can a full-size photograph of a person be resized to fit a small passport photo box without distortion?' Groups write on sticky notes or strips: one thing learned today (e.g. 'Every point of an enlargement lies on a line through the centre, and one <math>k</math> describes all lengths') and one new question (e.g. 'Does the centre work for a reduction like <math>k = 1/4</math>?', 'What if the lines never meet?'). They place notes under the headings 'What we have learned' and 'What we still wonder'. They also link today's finding to earlier ideas: equal ratios (Lesson 3), <math>k</math> (Lesson 4), construction from a centre (Lessons 8 and 9).</p>                                                        | <p>Say: 'Mrs Achieng's DQB is growing. What can we now add that answers part of our question?' Give WAIT TIME of 10-15 seconds before groups write. Cold-call 3 groups to read their notes aloud and place them on the board. Cluster notes with the class, for example 'one scale factor for all lengths' and 'how to use this in real life'. Point to the questions about maps and signs and say: 'Lesson 11 will help with these.' Do not answer new questions yet; ask learners to keep them for later.</p>                                                                        | <p>Driving Question Board as a public record of understanding: seeing learned ideas and open questions side by side helps learners track how each lesson contributes to the Driving Question and keeps their curiosity directing the sequence.</p> | <p>Check that notes are accurate (e.g. mention a single centre, <math>k</math> from image <math>\div</math> object) and that new questions are specific. Observe whether learners connect the notes to the phenomenon (Wanjiru's photos, Mr Otieno's printing). Record which questions are unanswered to use in Lesson 11.</p> |
| <p><b>Model Building Phase</b><br/>  <b>VIDEO:</b><br/> <a href="#">Dilating shapes: shrinking</a><br/> <i>Related topic: not</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Learners revise the class model of 'What must stay the same and what may change when a photo is resized'. They add a new section: 'An enlargement has one centre and one scale factor. Joining each object point to its image point</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Say: 'Add today's evidence to our model. Draw it, label it and write one sentence about what stays the same.' Provide chart paper and markers. Give WAIT TIME of 10-15 seconds before groups begin drawing. Cold-call 3 learners to</p>                                                                                                                                                                                                                                                                                                                                             | <p>Cumulative model revision (modelling): adding each new piece of evidence to a shared diagram makes learners integrate ideas about ratios, angles, scale factor and centre into one explanation of why the passport</p>                          | <p>Look at diagrams and sentences for: centre <math>(1,1)</math> and <math>k = 3</math> stated correctly; object and image identified; the link to <math>1/4</math>, 5 and <math>9/28</math> versus <math>1/4</math> correct; an explanation that one <math>k</math> applies to all lengths. Exit check: each</p>              |

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| <p><i>an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b></p> <p><a href="#">Dilations and properties</a></p> <p><i>Related topic: not an exact match for this lesson.</i></p> <p>Source: Khan Academy (English - US curriculum)</p> <p><a href="#">Search ARES for similar readings</a></p> | <p>locates the centre; <math>k = \text{image length} \div \text{object length}</math>.' They draw a labelled diagram of PQR, P'Q'R', centre (1,1) and <math>k = 3</math>. They then add a line connecting the model to the photos: the original <math>14 \times 18</math>, passport <math>3.5 \times 4.5</math> (<math>k = 1/4</math>) and poster <math>70 \times 90</math> (<math>k = 5</math>) are all enlargements of the original with a single <math>k</math>, while the distorted <math>4.5 \times 4.5</math> print used two different factors (<math>9/28</math> and <math>1/4</math>), so it is not an enlargement from any centre. Learners write a two-sentence explanation in their notebooks.</p> | <p>read their sentences and ask the class to add or correct. Prompt: 'Why can the distorted print not have a centre and a single <math>k</math>?' Accept: because the width factor is <math>9/28</math> and the height factor is <math>1/4</math>, which are not equal. Ask: 'What would happen if we tried joining corresponding points on the distorted print?' (The lines would not meet at one point.) Close by saying: 'Next lesson we use scale factors, area factors and volume factors on maps and signs.' Ask learners to return rulers and grid paper neatly.</p> | <p>photo keeps Wanjiru's likeness.</p> | <p>learner writes how to find the centre and <math>k</math> from an object and image. Success: mentions joining corresponding points and <math>\text{image} \div \text{object}</math>. Follow up with learners who cannot do this in Lesson 11.</p> |
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## D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners join the correct corresponding points (P to P', Q to Q', R to R')? (2) Which groups used differences instead of ratios for  $k$ , and what questioning moved their thinking? (3) Did learners reason about why the lines meet, or only follow the steps? (4) Were the WAIT TIME periods and cold-calls enough for quieter learners such as those who rarely speak? (5) How well did learners connect today's work to Wanjiru's photos and the distorted print? (6) Which DQB questions should be addressed in Lesson 11 (maps and painted signs)? Note any learners needing extra practice with plotting coordinates or using a ruler.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>What did I observe?</b>                   | We plotted P(2,2), Q(3,2), R(2,3) and their image P'(4,4), Q'(7,4), R'(4,7), joined P to P', Q to Q' and R to R', extended the lines, and saw that all three meet at one point, (1,1). We measured that each image length is 3 times its object length (for example PQ = 1 and P'Q' = 3).                                                                                                         |
| <b>What did I learn?</b>                     | The centre of an enlargement is the point where the lines joining corresponding object and image points meet. The scale factor is $k = \text{image length} \div \text{object length} = 3$ . We checked the result with coordinates: P' minus centre = $(3,3) = 3 \times (1,1)$ , which is $3 \times (\text{P minus centre})$ .                                                                    |
| <b>How does this explain the phenomenon?</b> | Wanjiru's passport photo ( $k = 1/4$ ) and poster ( $k = 5$ ) are enlargements of the original because one centre and one scale factor apply to every point and length, so her face keeps its shape. The distorted $4.5 \times 4.5$ cm print used two different factors ( $9/28$ for width and $1/4$ for height), so it is not an enlargement from a single centre and her face looked stretched. |

## LESSON 11 (40 minutes): Applications of Scale: Maps, Models and Costing Painted Signs

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Purpose</b>              | Learners apply the linear scale factor $k$ , the area factor $k^2$ and the volume factor $k^3$ to scale drawings, maps, models and the costing of painted signs, and use these applications to explain why a resized photograph stays undistorted only when one scale factor is applied to all lengths.                                                                                                                                                                                                                                             |
| <b>Knowledge</b>            | By the end of the lesson the learner should be able to: (a) use a map scale such as 1:5 000 000 to convert between real distances and map distances; (b) use $k$ to find lengths in scale drawings and models; (c) use $k^2$ to find how area (and so paint cost) changes; (d) use $k^3$ to find how volume changes in a scale model.                                                                                                                                                                                                               |
| <b>Skills</b>               | Calculating and interpreting scale factors (always stating object and image); converting units (km to cm, $\text{cm}^2$ to $\text{m}^2$ ); using mathematics to model real situations (NGSS SEP: Using Mathematics and Computational Thinking); constructing explanations from evidence; measuring with a ruler; working collaboratively.                                                                                                                                                                                                           |
| <b>Attitudes</b>            | Appreciate the use of similarity and enlargement in day-to-day activities such as maps, signwriting and model making. Core competencies: Critical Thinking and Problem Solving, Communication and Collaboration. Values: Responsibility (taking care of rulers and grid paper) and Unity (working together in groups). PCI: Life Skills (decision making: costing a sign wisely).                                                                                                                                                                   |
| <b>Key Inquiry Question</b> | Why do we need to enlarge or reduce objects? How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Purpose in Storyline</b> | Lessons 4, 6 and 7 gave learners $k$ , $k^2$ and $k^3$ using the photo and carton data. Lesson 11 shows that the same rules work outside the photo studio: on a map of Kenya, on a signboard and on a model. This advances the Driving Question by showing that real users (map makers, signwriters) also apply ONE factor to every length, which is why the passport photo ( $k = 1/4$ ) is not distorted, while the 4.5 cm $\times$ 4.5 cm print (width factor $9/28$ , height factor $1/4$ ) is. Lesson 12 will bring all the evidence together. |
| <b>Safety Notes</b>         | Learners use rulers, scissors and grid paper only. Keep walkways clear while groups move to the gallery wall. Scissors are handled with the points down and are used only for cutting grid paper. No electricity or internet is needed.                                                                                                                                                                                                                                                                                                             |

### B. LESSON OVERVIEW

Mr Otieno at Lakeview Photo Studio resizes Wanjiru Atieno's photograph with one scale factor:  $1/4$  for the 3.5 cm  $\times$  4.5 cm passport print, 5 for the 70 cm  $\times$  90 cm prize-giving poster. In this lesson learners discover that other people who resize things do the same. Learners first predict how far apart Kisumu and Nairobi would be on a 1:5 000 000 map and how much more paint an enlarged school sign needs. They then measure and calculate: the Kisumu to Nairobi straight-line distance of about 265 km becomes 5.3 cm on the map ( $26\ 500\ 000\text{ cm} \div 5\ 000\ 000$ ); a 70 cm  $\times$  90 cm sign ( $0.63\text{ m}^2$ ) enlarged with  $k = 3$  becomes 210 cm  $\times$  270 cm ( $5.67\text{ m}^2$ ), with area factor 9, so paint at KES 400 per  $\text{m}^2$  costs KES 252 for the small sign and KES 2268 for the large one. A short scale-model task using  $k = 2$  cartons (volume factor 8) recalls Lesson 7. Learners then explain how the applications connect to the photograph, add to the Driving Question Board (DQB), and update the class model: size changes by any  $k$ , shape stays only if one  $k$  is applied to all lengths, area changes by  $k^2$  and volume by  $k^3$ .

### C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
|-------|--------------------|---------------|----------------------|-------------------------------|
|-------|--------------------|---------------|----------------------|-------------------------------|

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| <p><b>Predict Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Scale drawing: centimeters to kilometers</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>EXERCISE:</b><br/> <a href="#">Reading Maps, Legends/Scale Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p>          | <p>Learners sit in groups of four. Mrs Achieng displays two prompts on the board. (1) Kisumu to Nairobi is about 265 km in a straight line. Wanjiru's school atlas map uses the scale 1:5 000 000. Predict how many centimetres apart the two towns will be on the map. (2) The school sign painted by a local signwriter is 70 cm × 90 cm, the same size as the poster. The head teacher wants a sign three times as long and three times as high (<math>k = 3</math>). Predict: will it need 3 times the paint, 6 times, 9 times or 27 times? Each learner writes a prediction and a one-sentence reason in the notebook, then compares with a partner. Groups record their predictions on a half sheet of paper that they will check at the end.</p>                                                                                                                                                                                                                                                                                                    | <p>Say: "Mr Otieno changed the photo size by a huge amount and Wanjiru still looks like herself. Today we ask: do map makers and signwriters follow the same rule?" Write the two prompts on the board. Give 2 minutes of silent individual writing. Say: "Write a number, not just a guess. Give me a reason." Then give pair talk of 1 minute. WAIT TIME: after asking "Which choice for the paint do you expect and why?" wait 10-15 seconds before taking answers. Cold-call 4 learners (2 for the map, 2 for the sign) and record every answer on the board without saying right or wrong. Ask: "Who thought differently? What made you choose that?" Collect the group prediction sheets.</p>                                                                                                                                                                                                                                                     | <p>Prediction with reasoning (Predict-Observe-Explain). Learners commit to a number and a reason before calculating, which exposes common ideas (for example that 3 times as long means 3 times as much paint). The later conflict between prediction and evidence drives conceptual change and links to the area factor <math>k^2</math> from Lesson 6.</p>                                          | <p>Look and listen for: (a) how many learners write a map distance in cm and not km; (b) how many choose 3 times the paint (a linear-thinking misconception) versus 9 times; (c) whether reasons mention 'scale', 'same ratio' or 'area'. Tally predictions on the board (e.g. 3 times: __, 9 times: __). A learner who says "the sign grows in two directions" shows readiness for <math>k^2</math>. Note the learners to support during the Observe Phase.</p>                                                                                                                                                                                                                                                                                 |
| <p><b>Observe Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Interpreting scale factors in drawings</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES for similar videos</a></p> <p> <b>HTML:</b><br/> <a href="#">Similar Polygons and Scale Factors (Flexbook)</a><br/> Source: CK-12<br/>  <a href="#">Search ARES for similar readings</a></p> | <p>Learners rotate through three 6-minute stations at group tables and record all results in a table. Station A, Map of Kenya: Using the 1:5 000 000 scale, groups change 265 km into cm (<math>265 \text{ km} \times 100\,000 = 26\,500\,000 \text{ cm}</math>), divide by 5 000 000 to get 5.3 cm, then measure 5.3 cm on a ruler and mark two dots labelled Kisumu and Nairobi. They state object (real distance) and image (map distance), and write <math>k = 5.3 \div 26\,500\,000 = 1/5\,000\,000</math>. They then find the real distance for a second map length of 2 cm (<math>2 \times 5\,000\,000 = 10\,000\,000 \text{ cm} = 100 \text{ km}</math>). Station B, Painted sign: Groups draw, on grid paper (1 square = 10 cm), the small 70 cm × 90 cm sign and the enlarged sign with <math>k = 3</math> (210 cm × 270 cm). They work out the area of each in <math>\text{m}^2</math>: <math>0.7 \times 0.9 = 0.63 \text{ m}^2</math> and <math>2.1 \times 2.7 = 5.67 \text{ m}^2</math>. They find <math>5.67 \div 0.63 = 9 = 3^2</math>.</p> | <p>Before groups move, model the first line of Station A on the board: "Object: the real distance 26 500 000 cm. Image: the map distance. <math>k = \text{image} \div \text{object}</math>." Circulate with a checklist. Ask at Station A: "Why must we change kilometres to centimetres before dividing?" At Station B ask: "The sides tripled. Did the area triple? Show me on your grid." At Station C ask: "Which number do you cube, the sides or the scale factor?" Use WAIT TIME of 10-15 seconds after each question before prompting. Cold-call 3 groups to report one result from each station while others check. Insist on units in every line (cm, <math>\text{cm}^2</math>, <math>\text{m}^2</math>, <math>\text{cm}^3</math>, KES). Prompt groups who finish early: "Does the 5.3 cm map keep the same shape as the real road? Why does every length on the map use the same factor?" Give a time call at the 6-minute mark: "Rotate</p> | <p>Data table with pattern finding (Using Mathematics and Computational Thinking). Learners record the linear factor, area factor, volume factor and cost in one table so the <math>k</math>, <math>k^2</math>, <math>k^3</math> pattern emerges from their own calculations across three contexts. This connects the map, the sign and the model to the same rule that governs the photo prints.</p> | <p>Observable indicators: (a) learners write <math>k = \text{image} \div \text{object}</math> with object and image named; (b) they convert 265 km to 26 500 000 cm correctly; (c) they get 5.3 cm on the map; (d) they get 5.67 <math>\text{m}^2</math> and 9 as the area factor; (e) they get KES 252 and KES 2268; (f) they get 8 as the volume factor. Use a checklist per group. Common errors to catch: dividing the area by 3 instead of 9; mixing <math>\text{cm}^2</math> and <math>\text{m}^2</math> (for example 6300 <math>\text{cm}^2</math> read as 6300 <math>\text{m}^2</math>); costing the large sign as <math>3 \times \text{KES } 252 = \text{KES } 756</math>. Ask those learners to redraw the grid and count squares.</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>They cost both at KES 400 per m<sup>2</sup>: small sign <math>0.63 \times 400 = \text{KES } 252</math>, large sign <math>5.67 \times 400 = \text{KES } 2268</math> (which is <math>9 \times 252</math>). Station C, Scale model: Groups use the cartons for packing prints: small carton <math>10 \times 6 \times 4 \text{ cm} = 240 \text{ cm}^3</math> and large similar carton <math>20 \times 12 \times 8 \text{ cm} = 1920 \text{ cm}^3</math>. They state <math>k = 2</math> and check that the volume factor is <math>1920 \div 240 = 8 = 2^3</math>. They then say how much more material is needed for the model's surfaces and how much more it holds, noting that volume, not length, is what grows by 8.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | now."                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <p><b>Explain Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Making a scale drawing</a><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Similar Polygons and Scale Factors Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>Learners return to their predictions and check them against the evidence: Kisumu and Nairobi are 5.3 cm apart; the large sign needs 9 times the paint. Each group completes the claim-evidence-reasoning frame in the notebook: Claim: "When a drawing, map or sign is made with scale factor <math>k</math>, lengths change by <math>\_\_\_\_\_\_</math>, areas by <math>\_\_\_\_\_\_</math> and volumes by <math>\_\_\_\_\_\_</math>." Evidence: figures from the three stations. Reasoning: why the paint cost rose by 9 and not by 3 (the sign grew in height as well as in width, so each of the two directions multiplies by 3). Then learners link back to Wanjiru's photograph: Mr Otieno's passport print has <math>k = 1/4</math> for both width and height (<math>3.5 \div 14 = 4.5 \div 18 = 1/4</math>), so its area is <math>(1/4)^2 = 1/16</math> of the original (<math>15.75 \text{ cm}^2</math> compared with <math>252 \text{ cm}^2</math>). The poster has <math>k = 5</math> and area factor 25. Learners also explain why the map maker and the signwriter, like Mr Otieno, must apply the same <math>k</math> to width and height: if only one direction were changed (like the <math>4.5 \text{ cm} \times 4.5 \text{ cm}</math> print, with width factor <math>9/28</math> and</p> | <p>Ask: "Who can compare our predictions with the evidence? What changed in your thinking?" WAIT TIME 10-15 seconds. Then say: "Turn and tell your partner why the sign cost 9 times and not 3 times." After 1 minute cold-call 4 pairs and press for reasons: "Which two lengths were both tripled?" Draw a <math>1 \times 1</math> square and a <math>3 \times 3</math> square on the board so learners see 9 small squares. Ask: "Mr Otieno used <math>k = 1/4</math> for the passport photo. What was the area factor? Why is it smaller than <math>1/4</math>?" Take 3 responses. Then ask: "If the signwriter tripled only the width, would the sign look like the original? Compare this with the <math>4.5 \text{ cm} \times 4.5 \text{ cm}</math> print." Ask learners to answer the budget question: KES 2268 is less than KES 2500, so the sign is affordable. Say: "Check your claims against the table, not against what I say."</p> | <p>Claim-Evidence-Reasoning with revisiting of predictions. Learners justify area and volume factors using their own measurements, then transfer the reasoning to the photograph. Comparing the correct scaling with the single-direction scaling of the distorted print connects the applications to the Driving Question.</p> | <p>Listen for statements such as "area changes by <math>k^2</math> because both length and width are multiplied by <math>k</math>" and "the same <math>k</math> must be used for every length or the shape changes". Check notebooks for: (a) a complete CER with numbers; (b) area factor 9 linked to <math>k = 3</math>; (c) passport area = <math>1/16</math> of the original; (d) a correct comparison with the <math>4.5 \text{ cm} \times 4.5 \text{ cm}</math> print (unequal factors <math>9/28</math> and <math>1/4</math>). Probe learners who say "area is 3 times" with a grid drawing. Use a thumbs-up check: "Show me if you can explain <math>k^2</math> to a partner."</p> |

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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | height factor 1/4), the town shapes or the sign letters would look stretched. Pairs finish with a real-life decision: "The school can pay KES 2500. Can the head teacher afford the $k = 3$ sign?"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Driving Question Board (DQB) Creation</b><br> <b>VIDEO:</b><br><a href="#">Identifying scale factor in drawings</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Identify scale factor in scale drawings</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar readings</a> | Mrs Achieng's DQB stays on the wall with the Driving Question: How can a full-size photograph of a person be resized to fit a small passport photo box without distortion? Groups write new questions and findings on sticky notes or paper strips. Examples: "Do all maps of Kenya use 1:5 000 000?", "What scale would fit Kenya on one page?", "Why do signwriters charge by area?", "Does a model of a building need $k^3$ for the cement?" They also add an answer note: "Map and sign makers use ONE $k$ for all lengths; area is $k^2$ and volume is $k^3$ ." Learners place each note under a heading: Answered, Partly answered, New question. They tick questions from earlier lessons that the map, sign or model now help to answer, and mark with a star the ones that Lesson 12 should resolve. | Say: "What new questions do you have after costing the sign and measuring the map?" Give 2 minutes for writing and WAIT TIME 10-15 seconds before sharing. Cold-call 3 groups to read one note aloud and justify where it goes. Ask: "Which question on our board can we answer today? Which evidence from Lesson 4, 6 or 7 helped?" Move one or two answered notes to the Answered column and read the evidence aloud. Point to the Driving Question and ask: "What did the map and the sign teach us about 'without distortion'?" Record one sentence from a learner beside the question. | Driving Question Board as a public record of growing understanding. Posting questions and answers makes learners' thinking visible, shows how each lesson adds evidence, and gives them ownership of what is still unknown for the final lesson.                                                             | Check that notes are specific and mathematical (not just 'what is a map?'). Count how many learners add an answer with evidence (for example ' $k = 3$ , area $\times 9$ , cost KES 2268'). Observe whether new questions point to ideas that need the one- $k$ rule or to unrelated topics. Note groups that cannot relate a note to the Driving Question and revisit them in Lesson 12.                                                                                                                              |
| <b>Model Building Phase</b><br> <b>VIDEO:</b><br><a href="#">Scale drawing: centimeters to kilometers</a><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Reading Maps, Legends/Scale</a>                                                                                                                                                                                                            | Learners revise the class model, a poster showing three prints of Wanjiru's photograph ( $3.5 \times 4.5$ cm, $14 \times 18$ cm, $70 \times 90$ cm) and the distorted print ( $4.5 \times 4.5$ cm). Groups add a new panel called Applications. It has a map strip (Kisumu to Nairobi: 265 km becomes 5.3 cm, $k = 1/5\ 000\ 000$ ), the sign ( $70 \times 90$ cm to $210 \times 270$ cm, $k = 3$ , area $\times 9$ , cost KES 252 to KES 2268) and the carton ( $k = 2$ , volume $\times 8$ ). Each group updates the model rule in their own words: "Size can change by                                                                                                                                                                                                                                     | Say: "Our model must now explain photos, maps, signs and cartons. What do we add?" Give groups 6 minutes and circulate. Ask: "Where in your model do you show that the same $k$ is used for width and height?" WAIT TIME 10-15 seconds. Cold-call 3 groups to explain one new link. Challenge: "What would your model say about the $4.5$ cm $\times$ $4.5$ cm print?" Expect: unequal factors $9/28$ and $1/4$ , so not similar. Collect exit tickets. Close by saying: "Next lesson, we use everything to give a full                                                                     | Cumulative model revision with transfer. Learners extend the class model from the photograph to new contexts and show relationships with arrows and rules. Applying the model to a new costing problem tests whether the idea (one $k$ ; $k^2$ for area; $k^3$ for volume) can be used, not just remembered. | Exit ticket success criteria: (a) $140$ cm $\times$ $180$ cm; (b) area factor 4; (c) cost KES 1008; (d) units given. Model criteria: map, sign and carton panels appear with correct figures; the rule states that one $k$ is used for all lengths; area is linked to $k^2$ and volume to $k^3$ . Sort tickets into three piles: secure, nearly there (arithmetic or unit slips), needs reteaching (uses 2 instead of 4 for area). Plan a short revisit using grid paper for the third pile at the start of Lesson 12. |

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| <p><a href="#">Practice</a></p> <p>Source: CK-12<br/> <a href="#">Search ARES</a><br/> for similar<br/> readings</p> | <p>any scale factor <math>k</math>. Shape stays the same only if the same <math>k</math> is applied to all lengths. Area changes by <math>k^2</math> and volume by <math>k^3</math>." They draw arrows linking the passport print (<math>k = 1/4</math>, area <math>1/16</math>) to the map (a big reduction) and the poster (<math>k = 5</math>, area 25) to the sign (an enlargement). Then they complete the exit ticket: "The signwriter makes a sign with <math>k = 2</math> from a <math>70 \times 90</math> cm sign. Give the new size, the area factor and the cost if the original sign costs KES 252." (Answer: <math>140 \text{ cm} \times 180 \text{ cm}</math>; area factor 4; KES 1008.)</p> | <p>answer to the Driving Question."</p> |  |  |
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## D. TEACHER REFLECTION

After the lesson, consider: Did learners predict 3 times the paint, and how did their thinking change once they counted 9 squares on the grid? Which learners confused  $\text{cm}^2$ ,  $\text{m}^2$  and  $\text{km}$  when working with the map and sign, and what quick support (such as a conversion strip on the board) would help them? Could groups explain why the same  $k$  must be used for width and height by connecting the sign and the map to the  $4.5 \text{ cm} \times 4.5 \text{ cm}$  print? Did every group finish all three stations, or should the carton station be shortened next time? Which questions on the DQB still need evidence, and what should Lesson 12 prioritise so that every learner can give a complete explanation to the Driving Question?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
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| <b>What did I observe?</b>                   | We observed that Kisumu and Nairobi, about 265 km apart, are 5.3 cm apart on a 1:5 000 000 map ( $26\,500\,000 \text{ cm} \div 5\,000\,000$ ). We observed that the $70 \text{ cm} \times 90 \text{ cm}$ sign ( $0.63 \text{ m}^2$ ) enlarged with $k = 3$ becomes $210 \text{ cm} \times 270 \text{ cm}$ ( $5.67 \text{ m}^2$ ), 9 times the area, so paint at KES 400 per $\text{m}^2$ costs KES 252 for the small sign and KES 2268 for the large one. We also saw the $k = 2$ carton hold 8 times as much ( $240 \text{ cm}^3$ to $1920 \text{ cm}^3$ ). |
| <b>What did I learn?</b>                     | We learned that map makers, signwriters and model makers use scale factor $k$ , area factor $k^2$ and volume factor $k^3$ to resize objects. Costs that depend on area (such as paint) change by $k^2$ , not $k$ . The same $k$ must be applied to all lengths in every direction.                                                                                                                                                                                                                                                                           |
| <b>How does this explain the phenomenon?</b> | This helps explain the photograph: Mr Otieno applied one factor ( $k = 1/4$ for the passport print, $k = 5$ for the poster) to both width and height, so Wanjiru's face keeps its shape, just as a map keeps the shape of Kenya. The $4.5 \text{ cm} \times 4.5 \text{ cm}$ print used different factors ( $9/28$ for width and $1/4$ for height), so it was distorted.                                                                                                                                                                                      |

## LESSON 12 (40 minutes (one double period may be used if available)): Consolidation: Resolving the Passport Photo Puzzle

### A. SPECIFIC LEARNING OUTCOMES

| Category                    | Specific Learning Outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>Purpose</b>              | To bring together everything learned in Lessons 1 to 11 into one complete, evidence-based Final Explanation of why Wanjiru's passport photo (3.5 cm × 4.5 cm) and prize-giving poster (70 cm × 90 cm) keep her likeness while the 4.5 cm × 4.5 cm print looks distorted, and to resolve the Driving Question Board.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Knowledge</b>            | Learners will state that: (1) similar figures have equal corresponding angles and equal ratios of corresponding sides (width : height = 7:9 for the original, passport and poster; 1:1 for the distorted print); (2) one linear scale factor $k$ is applied to all lengths (original to passport $k = 1/4$ ; original to poster $k = 5$ ; passport to poster $k = 20$ ), whereas the distorted print used $9/28$ for width and $1/4$ for height; (3) the chin angle is about $42.5^\circ$ in the original, passport and poster but about $53^\circ$ in the distorted print, and all corners stay $90^\circ$ ; (4) area changes by $k^2$ (passport is $1/16$ of the original: $252 \div 15.75 = 16$ ; poster is 25 times the original; poster is 400 times the passport), so size can change greatly while shape is kept. Learners also apply this to the wider uses seen earlier: shadows, enlargement from a centre, maps and signs. |
| <b>Skills</b>               | Developing and using models (revising a cumulative model); Constructing explanations (claim-evidence-reasoning); Analysing and interpreting data (comparing ratios, angles and scale factors); Engaging in argument from evidence; Communicating (using the terms corresponding sides, corresponding angles, scale factor $k$ , $\sim$ ). Measuring with a ruler and protractor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Attitudes</b>            | Appreciate the use of similarity and enlargement in day-to-day activities. Communication and Collaboration: share and defend ideas in groups. Critical Thinking and Problem Solving: use ratios, angles and scale factors to justify a conclusion. Values: Unity (working together to build one group explanation); Responsibility (taking care of rulers, protractors and shared materials). PCI, Life Skills (self-esteem): confidently presenting an explanation to peers.                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Key Inquiry Question</b> | 1. Why do we need to enlarge or reduce objects? 2. How do we use similarity and enlargement in day-to-day activities?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Purpose in Storyline</b> | This is the final lesson. In Lesson 1 learners met the puzzle: the size of Wanjiru's photo changes hugely, yet her shape is kept, and they recorded initial ideas about what must stay the same and what may change. Lessons 2 to 11 supplied the evidence: sorting similar figures (L2), equal angles and equal ratios (L3), scale factor $k$ (L4), similar triangles with shadows (L5), area factor $k^2$ (L6), volume factor $k^3$ (L7), enlargement from a centre (L8 to L10) and applications to maps and signs (L11). Today learners synthesise this into one Final Explanation, compare their final model with their Lesson 1 model, and resolve every question on the Driving Question Board.                                                                                                                                                                                                                                 |
| <b>Safety Notes</b>         | Scissors and protractors have sharp points: carry them point down and keep them on desks when not in use. Keep walkways clear when groups display posters. No electricity or internet is needed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### B. LESSON OVERVIEW

Mrs Achieng opens by returning to Wanjiru Atieno's photographs from Lakeview Photo Studio in Kisumu. Learners first predict individually how they would convince Mr Otieno, the photographer, why the  $3.5 \times 4.5$  cm passport photo is faithful but the  $4.5 \times 4.5$  cm print is not. In the Observe phase, groups re-examine the four prints ( $14 \times 18$ ,  $3.5 \times 4.5$ ,  $70 \times 90$ ,  $4.5 \times 4.5$  cm) and complete a one-page evidence table of width : height ratios, scale factors, chin angles and area factors using their own earlier measurements. In the Explain phase, groups write a claim-evidence-reasoning Final Explanation and test it against a new challenge. Then the class resolves the Driving Question Board, and finally learners revise their cumulative model and compare it with their Lesson 1 model, noting what changed in their thinking. The lesson ends by connecting the idea to the two KICD Key Inquiry Questions.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Learner Experience                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Teacher Moves                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Sensemaking Strategy                                                                                                                                                                                                                                                                                              | Formative Assessment Strategy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>Predict Phase</b><br> <b>VIDEO:</b><br><a href="#">Using similarity to estimate ratio between side lengths</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Corresponding Parts of Similar Figures (Flexbook)</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | Learners look at the four prints (original $14 \times 18$ cm, passport $3.5 \times 4.5$ cm, poster $70 \times 90$ cm, distorted $4.5 \times 4.5$ cm) pinned at the front. Individually, in 3 minutes, they write on a scrap of paper: 'If Mr Otieno asks me what he must keep the same and what he may change when resizing Wanjiru's photo, I will tell him...' They must name at least two things (e.g., ratio of width to height, angles, one scale factor) and one thing that may change (e.g., length, area). They then pair up and compare predictions with a partner.                                                                                                                                                                                                  | Display the four prints and say: 'Remember Wanjiru at Lakeview Photo Studio in Kisumu. Her photo shrinks to 3.5 by 4.5 centimetres and grows to 70 by 90 on the school hall poster, yet she still looks like herself. Today you will give Mr Otieno a complete answer.' Pose: 'What must stay the same, and what is allowed to change?' Give WAIT TIME of 10-15 seconds of silent thinking before anyone writes. Walk around and note which learners mention 'same ratio' versus only 'same shape'. Cold-call 3 learners to share partner ideas: 'Achieng's group partner, what did your partner say?' Record all ideas on the left side of the board under 'Our predictions today'. Do not correct yet. Say: 'Hold these ideas. We will test them with evidence.' | Think-Pair-Share with a written commitment: writing before talking makes each learner's current thinking visible and gives a baseline to compare with the final explanation, so learners can see how their understanding has grown since Lesson 1.                                                                | Observe whether learners (a) mention ratios or angles rather than only 'it looks the same'; (b) separate what stays the same (shape: ratio of sides, angles) from what changes (size). Tally on a class list: how many mention ratios, angles, scale factor, or area. Learners who say only 'same picture' are noted for support during group work.                                                                                                                                                                                               |
| <b>Observe Phase</b><br> <b>VIDEO:</b><br><a href="#">Using similarity to estimate ratio between side lengths</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>EXERCISE:</b><br><a href="#">Corresponding Parts of Similar Figures Practice</a><br>Source: CK-12<br> <a href="#">Search ARES</a>              | In groups of 4, with rulers, protractors and grid paper, learners re-examine the four prints (or the group's record from Lessons 3, 4 and 6) and complete a one-page Evidence Table. Rows: original $14 \times 18$ , passport $3.5 \times 4.5$ , poster $70 \times 90$ , distorted $4.5 \times 4.5$ . Columns: width : height ratio (7:9, 7:9, 7:9, 1:1); scale factor from original (width and height separately: $1/4$ and $1/4$ ; 5 and 5; $9/28$ and $1/4$ ); chin angle from the face triangle (about $42.5^\circ$ , $42.5^\circ$ , $42.5^\circ$ , about $53^\circ$ ); corner angles (all $90^\circ$ ); area in $\text{cm}^2$ (252, 15.75, 6300, 20.25). They verify at least two entries by measuring with a ruler or protractor, e.g., check that the passport eye gap | Distribute a printed or board-copied Evidence Table and say: 'You are detectives for Mr Otieno. Fill every cell with evidence, not opinion. Two cells must be checked by measuring.' Circulate and ask each group: 'What scale factor did the width use? What did the height use? Are they equal?' and 'Which angle changed, and by how much?' Prompt with: 'Show me where $k = 1/4$ appears in your table.' Give WAIT TIME of 10-15 seconds after each question before offering hints. Ask groups that finish early: 'Why is the area factor 16 and not 4 for the passport photo?' (linking to Lesson 6: area changes by $k^2$ ). Remind learners of conventions: $k = \text{image}$                                                                              | Data table synthesis (evidence organiser): placing all measurements for the four prints side by side lets learners see patterns across rows (equal ratios and angles for the three similar prints; unequal factors for the distorted print), turning scattered lesson evidence into a single comparative picture. | Check that each group's table has: ratios 7:9, 7:9, 7:9 and 1:1 correct; the distorted print's two factors $9/28$ and $1/4$ identified as unequal; chin angles about $42.5^\circ$ versus about $53^\circ$ ; area factors 16, 25 and 400. Indicator of mastery: the learner states the scale factor as $\text{image} \div \text{object}$ and can say which print is the object and which is the image. Note groups that confuse 16 with $1/16$ (original is 16 times the passport; passport is $1/16$ of the original) and address it immediately. |

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| for similar readings                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1.4 cm and chin distance 1.8 cm have ratio 7:9, while the distorted print's 1.8 and 1.8 have ratio 1:1. They also compute area factors: $252 \div 15.75 = 16$ (passport is 1/16 of the original), $6300 \div 252 = 25$ ( $5^2$ ), $6300 \div 15.75 = 400$ ( $20^2$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | length $\div$ object length; original is the $14 \times 18$ print. Cold-call 2 groups to report one verified measurement each. Collect any measurement differences (e.g., a protractor reading of $43^\circ$ instead of $42.5^\circ$ ) and discuss that small measuring error is expected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Explain Phase</b><br> <b>VIDEO:</b><br><a href="#">Using similarity to estimate ratio between side lengths</a><br><i>Related topic: not an exact match for this lesson.</i><br>Source: Khan Academy (English - US curriculum)<br> <a href="#">Search ARES for similar videos</a><br><br> <b>HTML:</b><br><a href="#">Unknown Measures of Similar Figures (Flexbook)</a><br>Source: CK-12<br> <a href="#">Search ARES for similar readings</a> | Each group writes a Final Explanation using Claim-Evidence-Reasoning on a sheet of manila paper. CLAIM: The passport photo and poster keep Wanjiru's likeness, but the $4.5 \times 4.5$ cm print is distorted. EVIDENCE: use at least four numbers from the table (ratios 7:9 versus 1:1; factors 1/4 and 5 versus 9/28 and 1/4; angles about $42.5^\circ$ versus about $53^\circ$ ). REASONING: a photo keeps its shape only if one scale factor $k$ multiplies every length, so corresponding sides stay in equal ratios and corresponding angles stay equal (the prints are similar, written original ~ passport ~ poster). Size can change by any $k$ (1/4, 5 or 20) and area then changes by $k^2$ (1/16, 25, 400). Groups then do a Transfer Challenge: 'Mr Otieno wants a 7 cm $\times$ 9 cm print. Is it similar to the original? What is $k$ ? What is the area factor?' ( $k = 7/14 = 9/18 = 1/2$ ; area factor 1/4; area $63 \text{ cm}^2$ compared with $252 \text{ cm}^2$ : $252 \times 1/4 = 63$ ). They also decide whether a $10 \times 12$ cm print would be similar (10:12 is 5:6, not 7:9, so no). | Say: 'A good explanation has a claim, evidence and reasoning. Your reasoning must explain WHY, using the words corresponding sides, corresponding angles and scale factor $k$ .' Provide a sentence frame: 'The passport photo is similar to the original because ____; the distorted print is not similar because ____.' Give WAIT TIME of 10-15 seconds, then circulate asking: 'You say the shape is kept. What exactly is kept?' and 'If I only changed the width, which number in your evidence changes?' Pose the Transfer Challenge and give 4 minutes. Select 3 groups to present (cold-call: 'Group 2, please read your reasoning'). After each, ask a different group: 'Do you agree? Which piece of evidence convinces you?' Allow challenge from the class. Then synthesise on the board: 'What stays the same: ratios of corresponding sides, corresponding angles. What changes: lengths (by $k$ ), area (by $k^2$ ), volume (by $k^3$ ).' Briefly link Lesson 5 (shadow triangles: Wanjiru 1.5 m, shadow 2 m; tree, shadow 12 m, height 9 m) as another place where equal angles give proportional sides, without re-teaching. | Claim-Evidence-Reasoning (CER) with a transfer task and peer argument: structuring the explanation forces learners to connect the evidence to a general principle, and applying it to a new print size (7 $\times$ 9 cm, 10 $\times$ 12 cm) tests whether the principle really explains and predicts, not just describes. | Use a 4-point checklist on each group's CER: (1) claim is clear; (2) at least four correct numbers cited; (3) reasoning mentions one scale factor for all lengths, equal ratios and equal angles; (4) area factor $k^2$ linked correctly. Observable indicator for the transfer task: learner computes $k = 1/2$ from both width and height and states area factor 1/4 ( $63 \text{ cm}^2$ ). Learners who answer ' $k = 1/2$ , so area is half' have the $k^2$ misconception: ask them to recompute using 252 and 63. |
| <b>Driving Question Board (DQB)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | The class returns to the Driving Question Board begun in Lesson 1, headed 'How can a full-size                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Say: 'Mrs Achieng's board has been waiting since Lesson 1. Let us see which questions we can                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Driving Question Board resolution: matching questions to evidence shows learners how the sequence                                                                                                                                                                                                                         | Look for answers that cite specific evidence (number, angle or factor) and the lesson of origin, rather                                                                                                                                                                                                                                                                                                                                                                                                                |

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| <p><b>Creation</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Using similarity to estimate ratio between side lengths</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>EXERCISE:</b><br/> <a href="#">Unknown Measures of Similar Figures Practice</a><br/> Source: CK-12<br/>  <a href="#">Search ARES</a> for similar readings</p> | <p>photograph of a person be resized to fit a small passport photo box without distortion?' Each group takes 2 or 3 question cards from the board (for example, 'Why does the <math>4.5 \times 4.5</math> print look stretched?', 'Does a big change in size change the shape?', 'Why do we need to enlarge or reduce objects?', 'How are maps and signs related?'). They write a short evidence-based answer on a sticky note, naming the lesson where they found the evidence (e.g., Lesson 3 for angles and ratios, Lesson 4 for k, Lesson 6 for <math>k^2</math>, Lesson 11 for the Kisumu-Nairobi map <math>265 \text{ km} = 5.3 \text{ cm}</math> at 1:5 000 000 and the sign costing KES 252 and KES 2268). They stick the answer beside the question and mark it Answered (green), Partly answered (yellow) or New question (red). Learners add at least one new question to carry forward (for example, 'What happens to similarity when k is negative?' or 'How is this different from congruence?').</p> | <p>now answer with evidence.' Hand each group 2 or 3 cards. Give WAIT TIME of 10-15 seconds when a group reads a question aloud before they respond. Ask: 'Which lesson gave you the evidence?' and 'Can you answer using k, ratios or angles?' Cold-call 4 groups to read and defend one answer each. For the KICD Key Inquiry Questions, pose them directly: 'Why do we need to enlarge or reduce objects?' (passport photos, posters, maps like Kisumu to Nairobi, sign painting) and 'How do we use similarity and enlargement in day-to-day activities?' (shadows to find the height of a tree: 9 m; scale drawings; costing a sign: area factor 9 gives KES 2268). Mark any questions not yet answered as 'to be explored in 2.2 Reflection and Congruence' (note that <math>k = 1</math> gives an identical size, which is studied later).</p> | <p>of lessons built toward the answer, makes the learning visible and gives ownership of the questions; leaving some questions open models how science and mathematics generate new inquiries.</p>                                                                                                      | <p>than generic statements. Check that the Answered/Partly/New colour coding is justified. Indicator: at least 80% of cards have an evidence-based answer and each learner contributes to at least one note. Note vague answers such as 'because it is similar' and ask 'similar in what way, and how do you know?'</p>                                                                                                                                                                                                                                                                                                                                                                                    |
| <p><b>Model Building Phase</b></p> <p> <b>VIDEO:</b><br/> <a href="#">Using similarity to estimate ratio between side lengths</a><br/> <i>Related topic: not an exact match for this lesson.</i><br/> Source: Khan Academy (English - US curriculum)<br/>  <a href="#">Search ARES</a> for similar videos</p> <p> <b>HTML:</b><br/> <a href="#">Corresponding Parts of Similar</a></p>                                                                                                                                                            | <p>Individually, learners complete the cumulative model they have developed through the unit: a labelled diagram of Wanjiru's original print, passport print and poster, joined by arrows labelled with scale factors <math>k = 1/4</math>, <math>k = 5</math> and <math>k = 20</math>, with the face triangle drawn on each (eye-gap and chin distance: 5.6 and 7.2; 1.4 and 1.8; 28 and 36), angles marked (<math>90^\circ</math> corners, about <math>42.5^\circ</math> at the chin), and the distorted print beside it with unequal factors <math>9/28</math> and <math>1/4</math> and angle about <math>53^\circ</math>. They add a rule box: 'Shape kept <math>\Rightarrow</math> one k for all lengths. Lengths <math>\times k</math>, area <math>\times k^2</math>, volume <math>\times k^3</math>.' Then they place their Lesson 1 model next to the final</p>                                                                                                                                             | <p>Say: 'Take out your Lesson 1 model. Now complete your final model so that it explains what Mr Otieno did right and what went wrong with the <math>4.5</math> by <math>4.5</math> print.' Insist that every arrow has a scale factor and the units are cm. Give WAIT TIME of 10-15 seconds after the prompt 'What is the biggest change between your two models?' before taking answers. Cold-call 3 learners to read their 'In Lesson 1 I thought... Now I know...' sentences. Use probing questions: 'Where does your model show why the chin angle changed?' and 'Where does your model show area changing by <math>k^2</math>?' Conclude with a 2-minute exit slip: 'In one</p>                                                                                                                                                                 | <p>Model revision and comparison (then versus now): placing the Lesson 1 model beside the final model makes conceptual change explicit, strengthens metacognition, and consolidates the evidence into a single representation that learners can use to explain and predict new resizing situations.</p> | <p>Rubric for the final model: (1) all three similar prints labelled with correct sizes and k values (<math>1/4</math>, 5, 20); (2) distorted print labelled with unequal factors <math>9/28</math> and <math>1/4</math> and angle about <math>53^\circ</math>; (3) equal ratios (<math>7:9</math>) and angles (<math>90^\circ</math>, about <math>42.5^\circ</math>) shown; (4) area factor <math>k^2</math> shown (16, 25, 400) and volume factor <math>k^3</math> mentioned; (5) written comparison shows a clear change in thinking. Exit slip success indicator: states 'one scale factor for all lengths, so ratios and angles stay equal; size can change'. Collect the slips to plan revision.</p> |

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| <a href="#">Figures (Flexbook)</a><br>Source: CK-12<br><a href="#">Search ARES for similar readings</a> | model and write three sentences: 'In Lesson 1 I thought _____. Now I know _____ because _____.'<br>Volunteers share how their thinking changed (for example, from 'it just looks the same' to 'ratios and angles are equal'). | sentence, what must stay the same and what is allowed to change when a photo is resized?'<br>Expected: the ratios of corresponding sides and the angles must stay the same (one scale factor for all lengths); size, lengths and area may change. |  |  |
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## D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners explain WHY the passport photo is not distorted using ratios, angles and one scale factor, or did they only say 'it looks the same'? (2) Which learners still confuse the area factor  $k^2$  with  $k$  (for example, saying a  $k = 1/4$  passport is  $1/4$  of the area instead of  $1/16$ ), and what extra practice with  $252 \div 15.75 = 16$  is needed? (3) Were measurement differences (such as  $43^\circ$  instead of  $42.5^\circ$ ) handled as normal measuring error? (4) Which DQB questions remain open, and can they be linked to 2.2 Reflection and Congruence? (5) How well did learners link to the Key Inquiry Questions with local examples (photos at Lakeview Photo Studio, the Kisumu-Nairobi map, the tree shadow at Nyalenda Secondary School, the painted sign)? (6) How did the quality of Final Explanations compare with the Lesson 1 initial ideas, and what should be revisited in revision sessions?

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

|                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <b>What did I observe?</b>                   | We observed that the original $14 \times 18$ cm print, the passport photo $3.5 \times 4.5$ cm and the poster $70 \times 90$ cm all have a width : height ratio of 7:9 and the same angles ( $90^\circ$ corners, about $42.5^\circ$ at the chin), while the $4.5 \times 4.5$ cm distorted print has ratio 1:1 and a chin angle of about $53^\circ$ .                                                                                                                        |
| <b>What did I learn?</b>                     | We learned that similar figures have equal corresponding angles and equal ratios of corresponding sides, that the linear scale factor $k = \text{image length} \div \text{object length}$ ( $1/4$ , 5 and 20 for the passport, poster and passport-to-poster), and that area changes by $k^2$ ( $1/16$ , 25 and 400) and volume by $k^3$ .                                                                                                                                 |
| <b>How does this explain the phenomenon?</b> | Wanjiru's photo keeps her likeness because every length was multiplied by the same scale factor ( $k = 1/4$ for the passport photo, $k = 5$ for the poster), so ratios and angles stay equal; the $4.5 \times 4.5$ cm print used different factors ( $9/28$ for width and $1/4$ for height), so the ratio and the chin angle changed and her face looked stretched. Size can change by any amount, but shape is kept only when one scale factor is applied to all lengths. |

| DIFFERENTIATION AND INCLUSION       |                                                                                       |                                                                                  |
|-------------------------------------|---------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Learner Need                        | Support Strategies                                                                    | Extension Strategies                                                             |
| English Language Learners           | Provide bilingual glossaries; allow use of first language for initial model drawing.  | Write extended explanations of observations; lead group discussions in English.  |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts.               |
| Gifted and advanced learners        | Access to additional reading materials; challenge questions during DQB.               | Lead model-building discussions; research and present extension topics to class. |